

Relational Frictions Along the Supply Chain: Evidence from Senegalese Traders*

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Abstract

Search and trust frictions have historically made it hard for small firms in lower-income countries to buy inputs from foreign markets. The growth in smartphone ownership and social media usage has the potential to alleviate these barriers. Informed by a dynamic model of relational contracting, we run a field experiment leveraging these technological tools to provide exogenous variation in (1) search frictions and (2) trust frictions (adverse selection and moral hazard) in a large international import market. In the search treatment, we connect a randomly selected 80% of 1,862 small garment firms in Senegal to new suppliers in Türkiye. We then cross-randomize two trust treatments that provide additional information about the types (adverse selection) and incentives (moral hazard) of these new suppliers. Reducing search frictions is sufficient to increase access to foreign markets: in all treated groups, firms are 26% more likely to have the varieties a mystery shopper requests, and the goods sold are 30% more likely to be high quality. However, the trust treatments are necessary for longer-term effects: using both transaction-level mobile payments data and a follow-up survey, we show that these groups are significantly more likely to develop the connections into relationships that persist beyond the endline survey. These new relationships lead to increases in medium-run profit and sales. Finally, we use the treatment effects to estimate the model and counterfactually lower the trust frictions among the whole supplier pool for a given firm, finding that the largest gains come from alleviating adverse selection.

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1 Introduction

Search and trust frictions make it hard for small firms to access foreign inputs. Consider a clothing wholesaler in Senegal who wants to start selling high quality European-made jeans. First, they must find a supplier, most of whom are in Europe, as well as some way of seeing what this supplier sells. Second, even if they manage to do this and decide to make an order, they often have to pay before observing quality—exposing them to both adverse selection and moral hazard. These issues may be particularly severe in lower-income settings, whose vast informal sectors complicate information aggregation and lower state capacities make contracts hard to enforce. Yet, while there is a long theoretical tradition studying these frictions—dating back to [Stigler \(1961\)](#) on search and [Shapiro \(1983\)](#) on trust—it has proven harder to analyze them both empirically as researchers rarely observe variation that is both exogenous and a close map onto the specific theoretical objects.

In recent years, there has been substantial growth in smartphone ownership and social media usage in lower-income countries, which has the potential to fundamentally change how small firms approach buying and selling decisions. Not only are firms increasingly selling online—with e-commerce revenue in Africa estimated to have doubled between 2019 and 2024—but they are primarily doing so through social media rather than traditional platforms.¹ This growth in “social commerce” may reflect the fact that social media could meaningfully alleviate both search and trust frictions in supply chains. For search, the supplier in Europe can send photos of their wares at minimal cost. For trust, domestic firms may be able to much more easily share information and coordinate action to discipline suppliers that cheat—a modern day version of the mechanism in [Greif \(1993\)](#). However, whether it actually does alleviate these frictions is ultimately an empirical question.

In this paper, we provide the first experimental evidence on the extent to which search and trust frictions limit access to foreign input markets and whether information transmitted via social media can alleviate them. We designed a field experiment leveraging key features of social media to alleviate these frictions in the context of a large international import market. Specifically, we randomly allocated 1,862 small garment firms in Dakar across treatment arms that connected them to new suppliers in Türkiye and varied the information available about the types and incentives of these suppliers. We then measured how these interventions affected their access to foreign goods, supplier relationships, and profits and sales, using data from a mystery shopping exercise, real-time transactions from Senegal’s largest mobile payments provider, and a follow-up survey.

¹In a survey in six African countries, a large non-profit found that among firms that use some e-commerce, 60% use social media exclusively ([GSMA, 2023](#)). Total B2C e-commerce revenue in Africa increased from 18 to 34 billion USD between 2019 and 2024 ([Statista, 2023](#)).

We first provide novel descriptive evidence from our baseline survey showing that firms use social media extensively in ways consistent with it alleviating search and trust frictions. For search, 86% of firms are in WhatsApp groups managed by suppliers (both domestic and international), with the median firm in four. These groups help firms see what suppliers sell, but it is unclear how much of the friction it alleviates as firms must still find the suppliers—and thus join the groups—in the first place. For trust, one-quarter of firms are in WhatsApp groups with other firms for the purpose of sharing business information, and two-thirds of firms have recommended or warned against particular suppliers to other firms in the past year. However, these networks are often highly fragmented, so it is unclear how relevant the information that they provide is to the average firm. While we focus on WhatsApp, other apps, such as TikTok, Instagram, and Facebook, are also popular, each used by around a third of firms. Motivated by the widespread use of “social commerce” among businesses, we design an experiment that leverages social media to create exogenous variation in search and trust frictions in international trade.

To make precise what we mean by search and trust frictions and to discuss what types of variation might separate them, we develop a model of relational contracting featuring sequential search for suppliers and both adverse selection and moral hazard, which we refer to as trust frictions. A firm can either buy inputs from a local supplier without frictions, or can pay a search cost to match with a random foreign supplier. Foreign suppliers may be thought of as selling newer varieties, higher quality varieties, or the same varieties at a lower price. Foreign suppliers must take a costly but unobservable action to ensure that the goods are high quality. Bad-type suppliers will never do this, while good-type suppliers will only do so if the future value of the relationship exceeds the current cost. We characterize the optimal contract and derive a wedge equation showing that adverse selection and moral hazard distort quantity downwards. We also show that interventions that alleviate both of these trust frictions are complements: if the firm’s beliefs about a supplier’s type exogenously improve, then they buy more today, which creates greater incentives for the supplier to shirk, which increases the return to alleviating moral hazard.

The model highlights the types of variation that the ideal experiment aiming to isolate the three frictions would generate. To isolate search frictions, treatments should either create matches or lower the cost of finding a random new supplier. To isolate adverse selection, treatments should either directly give information about a particular supplier or improve the ability to learn this information over time, but should not affect suppliers’ incentives. To isolate moral hazard, treatments should strengthen the incentives of the supplier or firms’ perceptions of these incentives, but should not provide any other information. Understanding the relative importance of each friction is not only of theoretical interest but

also has direct policy implications, since they require different solutions.

The experiment comprises three treatments. Each treatment is designed to be of real-world interest in its own right, while also mapping onto one of the three frictions. In the search treatment, we add treated firms to the supplier WhatsApp groups of three different suppliers in Türkiye, the second-largest source of ready-to-wear garments in Dakar. We inform firms that the suppliers were recruited by a local team in Türkiye and export to Senegal, but do not provide any further details. Thus, the treatment creates new matches between firms in Senegal and suppliers abroad.

We then cross-randomise the adverse selection and moral hazard treatments among the firms in the search treatment. In the adverse selection treatment, we add treated firms to a fourth WhatsApp group containing other firms matched with the same suppliers. The purpose of this group is for them to meet other firms with whom they can privately share information about whether these suppliers are good or bad. We also seed these groups with initial information: treated firms receive a recommendation for one of the suppliers, based on real mystery orders that we commissioned prior to the study. Thus, the treatment directly provides (positive) information about a supplier's type and improves learning.

In the moral hazard treatment, we inform firms that we will ask them to rate the study suppliers, and that we will remove any supplier receiving consistently negative feedback from the study, thereby losing access to a large number of potential clients. We emphasize that the suppliers know this and that they therefore have strong incentives to exert effort. Thus, the treatment aims to shift firms' perceptions about the suppliers' incentives.

Altogether, we have five equally sized groups: Pure Control, Search Only, Search + Adverse Selection, Search + Moral Hazard, and Search + Adverse Selection + Moral Hazard.

We measure outcomes using three independent data sources. The first is a revealed preference measure of access to foreign goods. We designed a mystery shopping exercise in which trained surveyors, acting as real customers, buy goods from the study firms. This allows us to measure access to foreign goods using real behavior, and to separately measure gains along a horizontal dimension (access to a wide set of differentiated varieties), a vertical dimension (access to high quality varieties), and a price dimension (cheaper access to otherwise available varieties). On the horizontal dimension, each good that we try to buy is defined by 5 criteria, such as color and sleeve style, and our outcome is an indicator for whether the firm has a good matching at least 3 criteria. On the vertical dimension, the outcome is an index that aggregates three measures, including a detailed quality scorecard that we designed together with hired experts. We pre-specified these outcomes and the regression specification that we use throughout the paper in our Pre-Analysis Plan (PAP).

We find that the treatments have a large and significant effect on access to foreign goods. Pooling the four treated groups (all of whom received the search treatment), treated firms are 9.3 percentage points more likely to have a suitable good: a 26.1% increase ($p < 0.01$). Conditional on having a good, the vertical index increases by 0.412 standard deviations ($p < 0.01$). We find no significant effect on price, suggesting that gains stem from access to more or higher quality varieties rather than lower prices. Disaggregating across the four groups, we do not find significantly larger effects in the trust-treated arms relative to Search Only. This does not necessarily imply the absence of trust frictions, but rather that these mystery transactions involved small orders (around USD 20-40), and are potentially below the threshold at which trust becomes binding. Overall, the results suggest that (1) firms face constraints in accessing foreign goods, (2) alleviating search frictions improves this access, (3) social media can be an effective tool to do so.

The extent to which firms are able to realise these gains in practice depends on whether these connections develop into lasting relationships. We measure this using data from our two other sources: a follow-up survey that we conducted after 3 months, and real-time, transaction-level administrative data from the largest mobile money provider in Senegal, tracking a large share of transactions for up to 18 months after the study started.

From the survey data, we find that (pooled) treatment increases the likelihood of having a regular supplier in Türkiye by 3.7 percentage points ($p = 0.088$), a 22.2% increase relative to control. Disaggregating the treatments matters: the effect comes primarily from the groups with the trust treatments, and is largest in the Search + Adverse Selection + Moral Hazard group at 7.5 percentage points ($p < 0.01$). We find no effect on the total number of suppliers, suggesting that firms have substituted away from a local wholesaler and towards importing directly. From the mobile money data, we find that all treated groups order similar amounts from study suppliers in the first few months. However, this changes in the medium- to long-run: firms in the trust treatments order 209.8% ($p = 0.067$) more than Search Only in the 15 months after the study finishes. When we disaggregate this effect into the separate trust treatments, the coefficients are all positive, with Search + Adverse Selection + Moral Hazard again the largest. We thus conclude that the trust treatments caused these connections to develop into more meaningful relationships.

To understand whether alleviating these frictions ultimately flows through to producer surplus, we collect standard summary measures of monthly profit and sales in our follow-up survey. Pooling the treatments together, we find increases of 82.4 USD ($p = 0.025$) in profits and 245.2 USD ($p = 0.051$) in sales. These are 43.8% and 40.2% increases relative to control. The coefficient in the Search + Adverse Selection + Moral Hazard is, again, much larger than the others, which is consistent with the result that trust treatments are comple-

ments in the model. The implied increases are very large, and the magnitudes reduce by around half (but remain significant) when we winsorize at the 1% level. When we look at distributional treatment effects, we find that these average results come primarily from the upper tail of the profit and sales distributions: we see large and significant increases starting at around the 75th percentile for the Search + Adverse Selection + Moral Hazard group. Overall, we conclude that firms are able to realise meaningful gains from accessing a new foreign supplier by using social media to overcome search and trust frictions.

Finally, we use the experimental treatment effects as moments to estimate the model. We solve the model numerically using the recursive Lagrangian formulation of [Marcet and Marimon \(2019\)](#), and then calculate theoretical treatment effects, which we match to the experimental ones using Simulated Method of Moments. The estimates imply that firms behave in line with just over half of suppliers in the marketplace being bad types, and that the moral hazard parameter is close to the unity assumed by many models of moral hazard. The estimated search costs are large but noisy. With the estimated parameters, we evaluate counterfactuals where we alleviate the frictions among all potential suppliers that a firm faces (i.e., not just the three in the experiment). Allowing the firm to screen out half of the bad type suppliers (a substantial reduction in adverse selection) has a large effect on lifetime discounted profit. Lowering moral hazard on its own has a much smaller effect: for most firms, this is not the binding constraint, as the adverse selection problem is sufficiently severe that they do not want to order large quantities anyway. However, as in the experiment, there is a strong complementarity between interventions that lower adverse selection and moral hazard, and the gains are very large to alleviating both simultaneously.

Overall, our results show that both search and trust frictions limit the ability of small firms to buy inputs from foreign markets, and social media can reduce these barriers. We provide new experimental evidence on the nature of information frictions in international trade, and show how the rapidly evolving digital landscape can change how firms find, learn about, and develop relationships with suppliers.

This paper contributes to several strands of literature. First, we add to research on information frictions within buyer-seller relationships. A rich theoretical tradition has emphasized the importance of these frictions in shaping firm behavior, and recent empirical work has leveraged novel data and experiments to study these frictions. For example, [Antràs and Foley \(2015\)](#) shows that a large US exporter requires buyers from low-enforcement countries to pay in advance, [Macchiavello and Morjaria \(2015, 2021\)](#) highlight the implications of relational contracting as a second-best solution, and [Startz \(2024\)](#) uses data on travel and orders to quantify search frictions and moral hazard. Recent experimental work includes [Bergquist et al. \(2024\)](#), which studies a platform lowering search frictions by match-

ing buyers and sellers, and [Cai et al. \(2024\)](#), which finds that spatially targeted matches increase trade when subsidized. We advance this emerging literature in two key ways. First, to our knowledge, this is the first field experiment to directly vary and measure trust frictions—and to evaluate them jointly with search frictions. We do so in a setting (international trade) where spatial distance makes such frictions both more severe and typically harder to vary experimentally. Second, we show that in low-trust, high-friction environments, small firms actively use new technologies—particularly social media—to reduce these frictions and engage in international trade. In doing so, we provide causal evidence on how trust can be built between distant and previously unconnected firms.

Second, we contribute to the literature on digital trade. Prior work has documented how the internet ([Freund and Weinhold, 2004](#); [Fernandes et al., 2019](#); [Akerman et al., 2022](#)) and formal B2B platforms like Alibaba ([Lendle et al., 2016](#); [Chen and Wu, 2021](#); [Carballo et al., 2022](#)) reduce the impact of distance and facilitate cross-border trade. However, most of this evidence comes from formal platforms in high- or middle-income settings. In contrast, we show that, in a lower-income setting, the vast majority of firms operate outside formal platforms, relying instead on informal, flexible, and highly relational tools like WhatsApp and Facebook—a form of “social commerce” rather than traditional “e-commerce.” We experimentally demonstrate that, in such environments, social media serves as a way to mitigate the same information frictions that formal platforms aim to alleviate.

Third, we contribute to a literature in trade and development that emphasises the importance of networks when formal institutions to solve information frictions and enforce contracts are inadequate ([Rauch, 1999](#); [McMillan and Woodruff, 1999](#); [Karlan et al., 2009](#); [Fisman et al., 2017](#); [Cai and Szeidl, 2018](#); [Boken et al., 2024](#)).² In a classic article, [Greif \(1993\)](#) highlights how 11th-century Maghribi traders sustained a multilateral punishment system for overseas agents through informal information flows over social networks. While our study does not focus on the role of networks *per se*, one of the main channels through which social media may alleviate trust frictions is exactly the mechanism in [Greif \(1993\)](#). Our study thus highlights how social media facilitates a modern manifestation of this idea.

The rest of the paper is organised as follows. Section 2 presents the setting and descriptive evidence on how firms use social media in supply chains. Section 3 describes the model. Section 4 describes the experimental design. Section 5 describes the data and methods. Section 6 presents the results. Section 7 presents the model estimation. Section 8 concludes.

²There is also a wide literature outside of economics that analyzes how shared ties substitute for formal institutions among particular groups or geographies. For example, [Cohen \(1969\)](#) studies the Hausa trade diaspora in Yoruba towns in 1960s Nigeria, [Weidenbaum and Hughes \(1996\)](#) studies the “bamboo network” of Chinese entrepreneurs across Southeast Asia in the latter half of the 20th century, and [Chin et al. \(1996\)](#) studies the role of Korean immigrants in Los Angeles in 1968-1977 in facilitating trade in the wig industry.

2 Setting

2.1 Ready-to-Wear Garments in Dakar

Our study focuses on the ready-to-wear garment industry in Dakar, the capital city and economic hub of Senegal. The ready-to-wear garment industry exhibits substantial horizontal and vertical differentiation, making it ideal for our study: horizontal differentiation (a wide range of varieties) is well-suited for studying search frictions, and vertical differentiation (the presence of high and low qualities) is well-suited for studying trust frictions. It is also a large and important industry in its own right: in a consumer survey that we conducted with 400 households in Dakar, ready-to-wear garments represented an average of 48% of clothing expenditure and 6% of total household expenditure.

Within the ready-to-wear garment industry, our study places particular emphasis on goods made in Türkiye. We chose Turkish-made goods for two reasons. First, Türkiye is the second largest source of ready-to-wear garments in Dakar (after China). Second, Turkish-made goods have a reputation for being higher quality than goods made in China, which is ideal for studying trust frictions. In fact, in this setting, requesting a good “Made in Türkiye” is a common way to signal willingness to pay for quality.³

2.2 Sample

Firms in Senegal The main subjects of the study are 1,862 small firms in the ready-to-wear garments industry in Dakar. These firms are typical of small, informal, owner-operated businesses in many large cities in lower- and middle-income countries. Our sample includes both firms that operate a physical store in a market (33%) and online-only firms that sell exclusively through platforms like WhatsApp and Facebook (67%). Firms with a physical store were recruited through a census in selected markets that sell both high and low quality goods while online-only firms were identified through a combination of advertisements on Facebook and snowball sampling.

Importantly, firms operating exclusively online are typically not captured in existing datasets, despite their growing prevalence in African cities. Our study offers an opportunity to document and understand the behavior of these firms. 93% of online firms and 83% of firms with a physical store agreed to participate when contacted. To ensure the relevance of the study, we conducted a short screening survey with some eligibility criteria (see Ap-

³To quantify this, in our consumer survey we showed households an image of a product and randomised whether we said the good was made in Türkiye or made in China. We then asked for their willingness to pay. We plot the CDF of willingness to pay in Figure 4, Panel (a). The Türkiye CDF is shifted uniformly rightward relative to the China CDF, with an average premium of 34% ($p < 0.01$).

pendix F.1), which were met by 85% of online firms and 89% of physical-store firms.

At baseline, 91% of eligible firms sell Turkish-made goods, and Turkish-made goods represent 44% of sales on average. Average monthly profit is 227 USD and 33% of firms sell wholesale. 10% have traveled internationally for business in the past 5 years, so most buy their goods from other firms in Dakar or through e-commerce. 22% of firms have at least one regular supplier based in Türkiye. As we show in Section 2.3, these firms ubiquitously use social media, particularly WhatsApp, to receive information from suppliers.

Firms have pessimistic beliefs about unknown foreign suppliers. 60% know multiple other firms that have had bad experiences ordering from a supplier online. To measure firms' priors, we asked them to consider a scenario in which they made orders from 10 unknown foreign suppliers, and to opine as to how many such orders would arrive with the anticipated quality. The median firm's view was that this would happen only 50% of the time.

Suppliers in Türkiye The study involves connecting firms in Senegal with suppliers in Türkiye. We work with 30 suppliers, all of whom are based in Istanbul and exporters of ready-to-wear garments to West Africa. We conducted a census of two neighborhoods of Istanbul that are textile wholesale and export hubs to many parts of the world, including to several countries in West Africa. Among the suppliers that met our inclusion criteria—including being based in Istanbul, regularly exporting ready-to-wear garments to West Africa, of Senegalese nationality, and using WhatsApp as a channel for commerce—we then conducted a mystery shopping exercise to identify the most active. We focused on suppliers of Senegalese nationality for three main reasons. First, a sizable Senegalese diaspora operates in the Türkiye–West Africa export industry, alongside Turkish exporters. This reflects general patterns documented by Greif (1993) and Rauch (2001), where shared ties have played a key role in reducing search and contracting frictions in long-distance trade. These shared ties may facilitate reputation-based mechanisms and reduce—but certainly do not eliminate—trust frictions. Second, Senegalese suppliers help reduce logistical barriers, such as language differences and incompatible payment technologies, which ensures we can focus on the search and trust frictions of interest. Third, most Senegalese suppliers in Türkiye rely on the same mobile money system used by firms in Senegal. This provides a unique measurement advantage, as we can track transactions directly through the platform's administrative data.

2.3 Social Media and e-Commerce in Supply Chains

In this section, we describe two types of WhatsApp groups that firms use that appear closely related to search and trust frictions, and that we will leverage in our experimental

design. We also present statistics on how firms use social media and formal e-commerce platforms for business.

Supplier Groups Many suppliers in our setting—both domestically and internationally—operate WhatsApp groups with their clients to advertise their goods, post prices, and highlight new items in stock. We refer to these as “supplier groups” (see Figure 2 for examples). A typical group has one supplier and 50-100 clients, most of whom are regular or repeat customers. These are not discussion groups: the purpose is for the supplier to regularly post photos and videos of their goods (typically only the supplier has permission to post). Buyers can send a private message to negotiate prices or make enquiries. These groups may be thought of as virtual storefronts where clients can see what the supplier is selling.

Supplier groups primarily reduce search frictions by allowing firms to observe a wide range of products directly using their mobile phone, with minimal effort required to negotiate and follow up. These groups also facilitate resale: firms often forward supplier images to their own customers via social media. To a lesser extent, supplier groups may also mitigate trust frictions: a large, active group could raise the reputational cost of dishonest behavior, and suppliers can use these groups to build a brand over time.

Firm Discussion Groups Firms also participate in WhatsApp groups aimed at sharing information—for example, about suppliers—with other firms, which we refer to as “discussion groups”. These groups primarily address trust frictions by allowing firms to exchange experiences and warnings. In our sample, 23% of firms are members of at least one group whose explicit purpose is information sharing between businesses. More generally, firms exchange information about suppliers through their networks: 67% have recommended or recommended against a particular supplier to another firm in the past year.

Social Media Usage In Figure 3, we present statistics from the 1,862 firms in our baseline survey. We present sample-wide averages here, but we also show in Appendix Figure A1 that the results are almost identical among firms with and without a physical store. In Panel (a), we plot the share of firms that reported using different types of social media to obtain information about suppliers, such as learning about new varieties or price information. WhatsApp is ubiquitous: 92% of firms use WhatsApp Status (a feature, used less often in the United States, that broadcasts content to all contacts for 24 hours), and 86% use supplier groups. TikTok, Instagram, and Facebook are also popular, each used by about a third of firms. Panel (b) shows the distribution of the number of unique supplier WhatsApp groups that firms belong to. Firms are in many supplier groups, with almost half of firms in 5 or more, allowing them to continuously observe information about different

varieties and prices. Finally, firms are not simply passive observers: the distribution of the number of groups that they have bought goods from in the past year is almost identical.

To understand why such groups are so widely used, in Panel (c) we present the responses to a question asking about the main advantages of supplier groups. Firms highlight that it allows them to see more varieties (both a wider set and higher quality) and to compare prices across suppliers. Finally, in Panel (d), we show the location of the suppliers running these groups. The majority of firms (81%) are in a group with a supplier in Senegal, while a large minority (27%) are in at least one group with a supplier abroad. 21% are in a group with a supplier in Türkiye, 12% are in a group with a supplier in China, and 6% are in a group with a supplier in Dubai. Since this focuses only on WhatsApp, this is a lower bound on the share of firms using social media more generally for international trade.

Firms are therefore familiar with the concept of using supplier WhatsApp group to transact with suppliers, but, since only 21% are in a group with a supplier in Türkiye, our experiment is still able to generate meaningful variation.

e-Commerce Platforms Formal B2B e-commerce platforms, such as Alibaba, also aim to alleviate search and trust frictions. Yet, in this setting, they are seldom used: 85% of firms have never bought from these platforms, and, of those who have, half have done so only very rarely. This is not because they have not heard of them (88% have), and reflects a broader trend of large e-commerce companies having limited success at penetrating African markets. In response to a our survey question asking why they do not use B2B platforms, apart from the 39% who gave no specific reason, the two most common answers were that firms find them too complicated to use (40%)—often due to language barriers and bank access requirements—and that firms do not trust them (33%).

These statistics confirm the observations that led us to run this study: that small firms use social media extensively in ways that appear related to search and trust frictions, including a sizable share that use it as a means of doing international trade.

3 Theory

In this section, we describe a model of relational contracting featuring adverse selection and moral hazard, embedded within a sequential search framework. The goal is to make precise the role of the frictions and to highlight the types of variation necessary to identify them. In Section 7, we use the experimental treatment effects to estimate the parameters governing these frictions. While the forces in the model are all canonical, we are not aware

of existing work that combines them in this way.⁴

3.1 Setup

Time is discrete and infinite horizon. All players are risk neutral, with common discount factor δ . Every period, a firm with productivity z buys high-quality inputs q_t^j from one or more suppliers $j = 1, 2, \dots$ and re-sells them according to a downward-sloping demand curve that generates revenue $zr(q_t^1, q_t^2, \dots)$, where r satisfies gross substitutes and is strictly increasing, concave, and with $\lim_{q \rightarrow \infty} \partial r / \partial q = 0$ in each argument. Inputs may also be of low quality, and for simplicity we set the productive value of low-quality inputs to zero.

3.2 Sequential Search

The firm can always buy high quality inputs without frictions from local suppliers at constant price c_l , which can be thought of as visiting the wholesale market in Dakar and directly inspecting quality. The firm can also engage in international trade. To do so, the firm must pay a search cost s to match with a random foreign supplier. Upon matching, the firm immediately observes a realization of a fixed match-specific productivity shifter, $\psi \sim G$, that transforms inputs q_t into effective inputs ψq_t . The rationale for engaging in international trade may be horizontal (if the revenue function r features love-of-variety or if ψ is interpreted as representing newer styles), vertical (if ψ is interpreted as a quality shifter), or to obtain lower prices (if ψ is interpreted as a price shifter). We describe the details of the relationship with the foreign supplier in the next section, but, for now, denote the period-0 value of a relationship with match-specific productivity ψ to a firm with productivity z as $U_0(\psi, z)$. Search is sequential, meaning that if a firm's current best option delivers discounted utility $\hat{U}$, then the firm will search if $E_\psi[\max\{U_0(\psi, z), \hat{U}\}] - s \geq \hat{U}$. Standard arguments then imply that there exists a cutoff value $\bar{U}$ such that the firm will search if and only if their current best option is less than $\bar{U}$, and that the expected utility of doing so is exactly $\bar{U}$.⁵ Moreover, with heterogeneous z , we obtain an importing equivalent of the cutoff result in Melitz (2003): since s is fixed, there exists a cutoff $\bar{z}$ such that only firms with $z > \bar{z}$ ever engage in search and thus import directly.

⁴The dynamic contracting part of our model differs from much of the relational contracts literature in two main ways. First, we have both limited liability and on-path learning, studied separately in Fong and Li (2017) and Yang (2013), either of which can lead to non-stationary optimal contracts. Second, and perhaps more importantly, we allow for the contractible quantity to be endogenous, so the principal chooses not only how to incentivize the agent but also the stakes of the incentives. Martimort et al. (2017) take a mechanism design approach to study the quantity issue, though with a number of differences (including no limited liability or on-path learning). Moreover, unlike all of these papers, we (1) embed the dynamic contracting problem within a sequential search environment, and (2) structurally estimate the model.

⁵Unlike standard sequential search, U_0 also depends directly on $\bar{U}$ because the contract is relational (discussed in the next section). However, we prove in Propositions 5 and 6 that the usual arguments go through.

3.3 Relational Contracting with Foreign Suppliers

If the firm has paid the search cost s and obtained a match, then they play a repeated game. For ease of notation, we set both ψ and z to 1 in this section as they are fixed within any firm-supplier pair. The stage game has three sub-periods. First, the firm specifies a quantity q_t and makes a transfer τ_t . Second, the supplier chooses whether to take a costly but unobservable action $a_t \in \{0, 1\}$ that influences the probability that the goods are high quality. If they choose $a_t = 1$, then the goods are high quality with probability 1 and the supplier pays cost cq_t , where $c > 0$ is a constant marginal cost. If they choose $a_t = 0$, they instead pay a lower cost $(1 - \xi)cq_t$ with $\xi \in (0, 1)$ but the goods are only high quality with probability λ . Avoiding the action and cutting a share ξ of the cost may, for example, be interpreted as purchasing the goods from a cheaper manufacturer (or using a cheaper shipping technology) that only delivers with probability λ . Third, the good arrives and the firm observes the quality. We focus on grim trigger strategies in which both parties take their outside option if a low quality realization occurs, because these strategies provide maximal incentives.

Moral Hazard: While the action a_t is unobservable, the firm may be able to incentivize the supplier to choose $a_t = 1$ by promising of future rewards tied to repeated high quality realisations. In particular, a standard one-shot deviation argument shows that any equilibrium in grim trigger strategies with $a_t = 1$ must satisfy the following Dynamic Incentive Compatibility Constraint (DICC),

$$(1 - \lambda)\delta V_{t+1} \geq \xi cq_t \quad (\text{DICC})$$

where $V_{t+1} \equiv \sum_{n=1}^{\infty} \delta^n (\tau_{t+n} - cq_{t+n})$ is the (endogenous) discounted sum of future profits from the relationship. Intuitively, since the supplier has already received τ_t , they will only choose $a_t = 1$ if the increased probability $(1 - \lambda)$ of obtaining the future value of the relationship, δV_{t+1} , exceeds the additional cost ξcq_t . This constraint imposes a time-varying ceiling on the contractible quantity: to the extent that this constraint binds, the firm must limit their order size to ensure the supplier is not tempted to cheat.

Adverse Selection: There are two types of foreign suppliers: good and bad. A supplier's type is fixed over time, known to the supplier, and unobservable to firms. Good type suppliers are as described above. Bad types, in contrast, are unable to choose $a_t = 1$ (for example, because they are unskilled). We denote the firm's beliefs about the share of bad types after observing t realisations in which the goods were high quality as μ_t , with initial (correct) beliefs μ_0 . The firm updates this belief each period using Bayes' Rule. We assume that λ is sufficiently low that, if the firm knew that the supplier was a bad type, they would

prefer to exit the relationship and take their outside option.

Outside Options: The firm has outside option $\bar{U}$, given by the maximum of the expected value of searching again or of buying locally forever. Foreign suppliers have outside option 0, reflecting the idea that relationships are separable for the supplier.

Contracts and Enforcement Constraints: In period 0 of the relationship, the firm makes a take-it-or-leave-it offer of a long-term contract to the supplier, which consists of a sequence $\{q_t, \tau_t\}_{t=0}^{\infty}$. The contract is relational, meaning that it is not enforceable in court: both parties can agree on a long-term plan, but are unable to commit to it. This implies that both parties face an enforcement constraint that requires that in every period t it must be optimal for them to not walk away: $U_t \geq \bar{U}$ and $V_t \geq 0$ for all t . The enforcement constraint prevents the firm from (for example) promising very large bonus payments to the supplier in the distant future, because when the distant future arrives the firm will prefer to renege on these contracted payments and instead simply take their outside option.

Limited Liability: The contract must satisfy $\tau_t \geq cq_t$ for all t ; that is, the supplier must make weakly positive profits period-by-period. We impose this restriction to prevent large period-0 rent extractions that we rarely see in our empirical setting. Instead, in our discussions with suppliers, we often heard of selling at cost early in the relationship to establish a reputation, which this restriction permits (and which will occur in equilibrium). Note that this constraint implies that we can ignore the supplier's enforcement constraint.

Equilibrium: We focus on Perfect Public Equilibria in which the good type supplier chooses $a_t = 1$. We also restrict attention to pooling equilibria, which we show in Proposition 1 in Appendix B is without loss. Intuitively, the firm cannot commit to not immediately terminating the contract if it learns that the supplier is a bad type. Bad types thus earn 0 upon revealing their type, but can earn a positive payoff by mimicking the good type.⁶

Dynamic Program: It is convenient to define a scalar revenue function $r(q)$ over the input q from this supplier where all others inputs are chosen optimally conditional on q . The assumptions on the full revenue function ensure that $r(q)$ is strictly increasing, concave, and has $\lim_{q \rightarrow \infty} r'(q) = 0$. We can then write the firm's expected stage payoff as $y(q_t, \tau_t, \mu_t) \equiv (1 - \mu_t(1 - \lambda))r(q_t) + \mu_t(1 - \lambda)r(0) - \tau_t$, and the supplier's stage payoff as $\pi(q_t, \tau_t) \equiv \tau_t - cq_t$.⁷ We can then write the continuation values of the firm, U_t , and the

⁶This restriction is not especially important, as the separating equilibrium that might otherwise obtain is similar. This is because dynamic moral hazard is sufficient to generate an upward sloping quantity profile due to the well-known backloading intuition. We impose the restriction here because it seems closer in spirit to our empirical setting, in which many firms take their outside option (purchasing from a local wholesaler).

⁷Bad type suppliers behave mechanically, so we do not write out their payoff.

supplier, V_t , recursively as

$$\begin{aligned} U_t &\equiv y(q_t, \tau_t, \mu_t) + \delta [(1 - \mu_t(1 - \lambda))U_{t+1} + \mu_t(1 - \lambda)\bar{U}] , \\ V_t &\equiv \pi(q_t, \tau_t) + \delta V_{t+1}. \end{aligned}$$

Standard arguments in the dynamic moral hazard and endogenously incomplete markets literatures establish that it is equivalent—and substantially easier—to solve this problem by explicitly letting the firm choose these continuation values and encoding them as state variables.⁸ This implies that the firm solves the following recursive dynamic program:

$$W_t(U_t, V_t, \mu_t) = \max_{q_t, \tau_t, U_{t+1}, V_{t+1}} y(q_t, \tau_t, \mu_t) + \delta(1 - \mu_t(1 - \lambda))W_{t+1}(U_{t+1}, V_{t+1}, \mu_{t+1})$$

subject to the following constraints:

$$\begin{aligned} \delta(1 - \lambda)V_{t+1} &\geq \xi c q_t & (DICC) \\ U_{t+1} &\geq \bar{U} & (DEC) \\ \tau_t &\geq c q_t & (LL) \\ y(q_t, \tau_t, \mu_t) + \delta [(1 - \mu_t(1 - \lambda))U_{t+1} + \mu_t(1 - \lambda)\bar{U}] &\geq U_t & (PK_f) \\ \pi(q_t, \tau_t) + \delta V_{t+1} &\geq V_t, & (PK_s) \end{aligned}$$

with $\mu_{t+1} = \mu_t \lambda / (1 - \mu_t + \mu_t \lambda) < \mu_t$ if the good is high quality in t and $\mu_{t+1} = 1$ if the good is low quality in t , and with $U_0 = \bar{U}$, $V_0 = 0$, μ_0 given. We have already introduced the first three constraints. The final two constraints are known as promise-keeping constraints as they ensure that the continuation utilities promised in the previous period, U_t and V_t , are actually delivered through a combination of stage payoffs and future promises.

The solution to this program is generally not available in closed form. In Proposition 2 in Appendix B, we provide a detailed derivation of several properties of the solution, which we summarise briefly here. The optimal contract looks similar to a dynamic version of the typical “sell the firm to the agent” solution in static models with risk neutrality. In particular, there exists a finite T^* such that the supplier will earn zero stage profits for periods $t = 0, 1, 2, \dots, T^* - 1$ and then earn the entire surplus (net of the firm’s outside option) for all $t > T^*$. Intuitively, the firm makes the supplier the residual claimant for most of the

⁸Intuitively, this method, often called the “promised utility” approach, works because the continuation values fully summarize past binding incentive constraints. Thus, the principal in t need not worry about the full sequence of $\{q_s, \tau_s\}_{s>t}$ — they can just pick V_{t+1} directly alongside (q_t, τ_t) and let their future self choose the least-cost way to implement this promise. This approach was originally developed somewhat independently in different theoretical contexts by Spear and Srivastava (1987), Abreu, Pearce, and Stacchetti (1990), and Thomas and Worrall (1988).

relationship—which is the most efficient way to provide incentives—and extracts surplus in the early periods as these minimise incentive distortions. This result is stark, but not unreasonable: the supplier sells at cost at the beginning of the relationship while its reputation is being established, and reaps the benefits of its reputation later on. Moreover, backloading incentives is a very general prediction of models of reputation and dynamic moral hazard (Shapiro, 1983; Banerjee and Duflo, 2000; Martimort et al., 2017).

We now state three results relevant to intuition and the interpretation of the experiment.

Result 1 (Intensive Margin). *Trust frictions (adverse selection and moral hazard) distort quantity and value purchased from the foreign supplier downwards.*

Proof. See Proposition 2 in Appendix B. □

This result can be seen most intuitively by examining the following wedge equation relating marginal revenue and marginal cost that can be derived from the FOCs:

$$r'(q_t^*) = \underbrace{\frac{1}{1 - \mu_t(1 - \lambda)}}_{\text{Adverse Selection}} \underbrace{(1 + \xi\rho_t^*)}_{\text{Moral Hazard}} c, \quad (1)$$

where ρ_t^* is a term proportional to the Lagrange multiplier on the DICC. Both wedge terms are weakly greater than 1, and so distort q_t^* downwards. The adverse selection term captures the fact that the firm lowers quantity because with probability $\mu_t(1 - \lambda)$ the supplier is a bad type and delivers low quality. The moral hazard term captures the fact that the firm lowers quantity to ensure the supplier is not tempted to cheat and save a share ξ of the cost. Importantly, moral hazard only matters to the extent that the DICC binds, which is captured by the Lagrange multiplier term ρ_t^* . Treatments that alleviate these frictions should therefore increase quantity on the intensive margin. In addition, while we are not powered to operationalize this prediction, we also prove that this distortion decreases over time, both because of learning (μ_t decreases) and because it is optimal to backload incentives (ρ_t decreases).⁹

Result 2 (Extensive Margin). *The buyer's period-0 relationship value is decreasing in μ_0 and ξ . Moreover, the productivity cutoff for importing, $\bar{z}$, is increasing in μ_0 and ξ .*

Proof. See Proposition 3 and Lemma 6 in Appendix B. □

⁹Strictly speaking, ρ_t evolves non-monotonically in general. However, we prove that it can never increase by enough to offset the learning effect. Indeed, if μ_0 is sufficiently small, then ρ_t monotonically decreases.

This results highlights that adverse selection and moral hazard limit the gains from trade, which prevents some otherwise productive matches from developing relationships and causes some firms to opt out of searching for an international supplier altogether. Treatments that alleviate these frictions should therefore increase the share of firms that import directly.

Result 3 (Relaxing Trust Frictions is Complementary). *The buyer's period-0 relationship value satisfies $\partial^2 U_0(\mu_0, \xi) / \partial \mu_0 \partial \xi > 0$.*

Proof. See Proposition 4 in Appendix B. □

This result shows that interventions that lower the trust frictions are complements. To see this, suppose that the adverse selection problem is exogenously alleviated (μ_0 decreases). Equation (1) then implies that, all else equal, the buyer's desired q_t increases. But then the buyer is pushing up even more strongly against the quantity ceiling implied by the DICC (ρ_t^* increases), meaning that the return to alleviating moral hazard (lowering ξ) is higher. This is formalized in the positive cross-partial above.

3.4 Implications for the Experiment

The model makes precise the three frictions that we will study in the experiment: search, adverse selection, and moral hazard. Search frictions limit the share of firms that import from foreign suppliers, and Results 1 and 2 show that adverse selection and moral hazard not only reduce the share of firms that import but also lower the quantity and value ordered among firms that do import. To relax the search friction, we need a treatment that either lowers the cost of matching with foreign suppliers or improves the likelihood of finding a suitable match. To relax adverse selection, we need a treatment that improves beliefs (or improves the ability to learn) about a supplier or suppliers in general. To relax moral hazard, we need a treatment that either improves suppliers' incentives or changes firms' perceptions of suppliers' incentives. Finally, since our experiment will cross-randomize treatments for both adverse selection and moral hazard, Result 3 highlights that these treatments are complements in the model.

In addition to these predictions, the fact that the optimal contract is non-stationary also has a number of testable dynamic implications—e.g., that average order size within a relationship increases over time. However, we do not focus on testing these in the empirical section, as our primary goal in this paper is to use experimental variation to identify the

parameters governing the frictions rather than to test the model *per se*.¹⁰

4 Experimental Design

4.1 Treatment Conditions

The goal of the experiment is to generate variation that identifies the three frictions: search, adverse selection, and moral hazard (we refer to the latter two jointly as trust frictions).

Search 80% of firms receive the Search treatment. The model highlights that search costs can be identified by lowering the cost of matching with a supplier in Türkiye, so this treatment directly creates matches. We add treated firms to the supplier WhatsApp groups (described in Section 2.3) of 3 different suppliers. As is standard, only the supplier can post in the group, so this is a channel for firms to see what the supplier sells. The suppliers to match with are selected at random, subject to being a match to the firm’s sector. We do not give firms any information about the suppliers, except that they were recruited by a team in Türkiye in a similar manner to how the firm itself was recruited and that they are looking for customers in Senegal. We also make clear that we are not endorsing the suppliers or providing quality guarantees. We communicate to the control group that unfortunately we cannot add them to any supplier groups at this time, but that we might do so at the conclusion of the study.

Adverse Selection 50% of firms in the Search condition are treated with the Adverse Selection treatment. As the model highlights, identifying adverse selection requires either improving the ability to learn or providing information directly. This treatment does both. We add treated firms to a fourth WhatsApp group. This group does not contain any suppliers, but instead contains other firms in the study that were matched with the same suppliers. We explain that all members of this fourth group have been matched with the same three suppliers and that the purpose is to put such firms in touch with each other to share information about the suppliers (either directly in the group or privately). A member of the study team moderates discussion and periodically encourages firms to share information. As discussed in Section 2.3, these firm discussion groups are relatively common in this setting.

¹⁰We can nonetheless offer several pieces of suggestive evidence in favor of a non-stationary model. First, using our mobile money data (described in Section 5), we find that average quantity ordered from new relationships created by the experiment is indeed increasing over time (see Appendix Figure A2). Second, in a survey that we conducted with 141 suppliers, 89% require new customers to fully pay in advance, while only 45% require this for established customers. Third, in qualitative interviews, suppliers often told us that they are willing to sell at cost in early stages of a relationship (which occurs in equilibrium in our model). While non-stationarity also has dynamic implications for treatment effects, we are not powered to test these.

Since no firms have experience with the supplier at this point, we seed the groups with initial information. Treated firms receive a phone call 2-3 days after recruitment from a recommender.¹¹ The recommender is part of a team of firms—who are not subjects in the study—that we hired prior to the study to make mystery orders from all of the suppliers. The recommender describes their experience ordering from one of the suppliers that the firm was matched with and sends a photo of the item that they ordered. They explain that they are also in this fourth WhatsApp group, and post a similar message there. To ensure firms were not surprised to receive this call, we asked all firms at the end of the baseline survey if they would be willing to call a few other firms to discuss their experiences working with the study suppliers.

The primary function of this treatment is to shift firms' information sets about the suppliers' types while minimizing any impact on suppliers' incentives. To that end, suppliers are not informed of the groups' existence, and firms are explicitly told so. Although firms could, in principle, try to convince suppliers that these groups exist in an attempt to influence their behavior, it would be hard to do this credibly, and we found no evidence of such attempts in practice.

Moral Hazard 50% of firms in the Search treatment are treated with the Moral Hazard treatment, cross-randomised with the Adverse Selection treatment. As the model highlights, identifying moral hazard requires either shifting the suppliers' incentives or shifting firms' perceptions of suppliers' incentives. To do this, we read a statement to treated firms at the end of the baseline survey that emphasizes that we have given the study suppliers strong incentives. The full statement is printed in Appendix F.2, but, in brief, we explain that we will ask the firms to rate the study suppliers, and we will remove suppliers that receive bad feedback, meaning they will lose access to the large number of potential customers we have matched them with.¹² Importantly, we stress that the suppliers know this and therefore have strong incentives.

After delivering this message, the surveyor provides a business card to the firm. The business card has a phone number to call, and prominently highlights that this number should be used to rate the suppliers and/or to signal any problems. Untreated firms receive a similar card, but without any mention of ratings or suppliers—instead saying that the phone number is for questions about the study. Both cards can be seen in Appendix Figure A3.

¹¹ All firms not in this treatment condition instead receive a “placebo” phone call from a surveyor, asking them for their opinion about supplier WhatsApp groups in general.

¹² This punishment is credible as the supplier WhatsApp groups used in the Search treatment were created specifically for the study, meaning that we are administrators. We can therefore easily remove a problematic supplier and replace them with a supplier from the wait list.

The experiment does not randomise the incentives provided to suppliers—instead, it provides high-powered incentives to all suppliers and randomises whether we tell this to firms. This allows us to keep all interventions at the firm-level, which greatly simplifies interpretation, and is equivalent in the model because the firm would not propose or agree to a contract that they believe violates the Dynamic Incentive Compatibility Constraint. This also implies that we should not expect this treatment to affect supplier defaults (conditional on a relationship forming). As above, we explain that we are not endorsing the suppliers: we have simply provided them with incentives. To the control firms, we explain that our only involvement is the initial connection. Overall, the primary function of this treatment is to shift firms’ views about the suppliers’ incentives while providing minimal information about the suppliers’ types.¹³

Sub-Treatments We cross-randomised two additional sub-treatments within the pure control group. The first sub-treatment aims to test whether the binding constraint behind the limited use of formal B2B e-commerce platforms is that firms do not know how to use them. We thus provide a short training on Alibaba that covers how to install the app, how to search for products, how to contact suppliers, and how to make purchases and arrange delivery. The second sub-treatment is a placebo check for the fact that, in the Adverse Selection treatment condition, we have connected firms to each other. To ensure that results are not driven by connecting firms *per se* (as in [Cai and Szeidl \(2018\)](#)), we thus also create similar groups here, where none of the firms have been connected with any suppliers.¹⁴

4.2 Randomisation and Balance Check

Overall, there are five equally likely groups: Pure Control, Search Only, Search + Adverse Selection, Search + Moral Hazard, and Search + Adverse Selection + Moral Hazard. We randomly assigned firms to one of these five groups, stratifying on product (men’s clothing, women’s clothing, shoes & bags), whether the firm has a physical store, and whether the firm had prior direct importing experience. Any misfits, due to integer indivisibility or other issues, were unconditionally randomised across the five cells.

¹³In principle, since bad types will eventually draw a low quality realisation, firms could wait and attempt to infer types by observing whether the supplier is still around after several periods. We do not think this happens for two reasons. First, firms typically decide quickly whether to initiate a relationship with the study suppliers (indeed, this is embedded within the model as the firm makes a take-it-or-leave-it offer upon matching). Second, the script does not imply that the enforcement process is particularly fast.

¹⁴Consistent with our prior, we find no meaningful effects of the placebo groups (see Appendix Table A12). The nature of our intervention is quite different from [Cai and Szeidl \(2018\)](#), where firms met in person every month for a year. In contrast, in our study, we connected firms only via a WhatsApp group, which is well-suited for pooling information about a specific topic of this nature but is less likely to generate the kind of broader networking interactions that could affect outcomes.

Since this is an RCT, treatment is independent of pre-randomisation covariates by construction, absent errors in the randomisation protocol. To check that the randomisation operated as expected, we report a balance check in Appendix Table A1. The differences in means across groups are all small and insignificant, and a joint test has p -value 0.595.

5 Outcomes, Data, and Empirical Methodology

5.1 Data and Outcomes

Consumer Survey In March 2024, we conducted a 15-minute survey with 400 households in several areas of Dakar. Enumerators selected respondents by starting from a random point and then visiting every fifth household along the route. We use this to calculate two sets of summary statistics. First, we use it to measure the relationship between consumer willingness to pay and our measures of quality (described below). Second, we use it to calculate statistics on household clothing expenditure.

Baseline and One-Week Surveys Upon recruiting a firm (see Section 2.2) we conducted a 30 minute baseline survey between November 2023 and January 2024. The survey contained questions on their supplier relationships, social media usage, e-commerce usage, and 30-day profits and sales. In a short additional follow-up survey one week later, we asked for their views about their access to Turkish-made goods, and, if applicable, their beliefs about the trustworthiness of one of the suppliers we matched them with.¹⁵

Access to Foreign Goods To evaluate whether social media reduces search and trust frictions in a way that expands firms’ access to foreign goods, we designed a novel mystery shopping exercise. This allows us to separate between different theoretical mechanisms for potential gains: a horizontal dimension (access to a wider set of differentiated varieties), a vertical dimension (access to higher quality varieties), and a price dimension (cheaper access to otherwise available varieties). Around two weeks after recruitment, firms are contacted over WhatsApp by a mystery shopper, played by a trained surveyor. Firms are not aware that the customer is part of the survey team, but are expecting to be contacted, as we explain to them at the end of the baseline survey that we will put them in touch with customers. This exercise enables us to capture firms’ sourcing capabilities and pricing in a natural, incentive-compatible way, shortly after treatment exposure.

¹⁵As a measure of the effects of the search and trust treatments on these perceptions (albeit after only one week and self-reported), we report treatment effects on these outcomes in Appendix Table A2. All treatments significantly increase self-reported access to Turkish-made goods, and the trust treatments jointly increase self-reported trustworthiness (although only the Search + AS + MH group is individually significant).

The mystery shopper explains that they would like to purchase a certain high quality product for an event (see Appendix F.3 for the full protocol). Each product is defined by five horizontal criteria that are largely unrelated to quality, such as color, sleeve style, and presence of a graphic (see Appendix Figure A4 for two examples). The primary outcome for this horizontal component, pre-specified in our PAP, is an indicator for whether the firm had a good with at least three of the five criteria.¹⁶

If the firm has a good with at least three criteria, the mystery shopper proceeds to buy the good in a randomly selected 80% of cases (with the exception of certain conditions described in Appendix F.3). Then, once the good arrives in our office, two expert tailors assess its quality according to a 50-point scorecard that we developed (see Appendix F.4).¹⁷ To validate the quality measure, we also gave the surveyors conducting the consumer survey a subset of these goods to present and elicit willingness to pay (WTP). We show a binscatter of the relationship between the quality score and WTP in Panel (a) of Figure 5. There is a clear positive relationship, although it becomes flat in the left tail, reflecting the fact that beyond a certain point consumers simply view goods as “low quality”. In Panel (b), we classify goods as “high quality” or “low quality” (defined as whether a good is above or below the median quality score of its product type), and plot the CDF of consumer WTP separately by quality. The high quality CDF is shifted rightwards of the low quality CDF, with an average premium of 35%. The outcomes are these two measures: the high quality indicator and the raw 50-point quality score. The advantage of the binary outcome is that it is not vulnerable to a long left tail of quality scores that, as we saw in Figure 5, are not meaningful in terms of WTP.

We also attempt to infer whether the good was manufactured in Türkiye. As we showed in Figure 4, there is a large premium for Turkish-made goods since it is a strong signal of quality. Thus, while the other two vertical outcomes measure quality directly, in practice quality can be difficult for consumers to observe and so product origin plays an important role in consumer WTP.¹⁸ For most goods, we record this information from the label, and the

¹⁶Our definition of having a good was that the firm said that they could deliver it to our office within two weeks. First, this matches the natural cadence of retail trade for the firms without physical stores (and is also common for firms with a physical store). Second, our goal is to measure whether the treatments increase *access* to foreign goods. Third, as discussed later, a secondary goal of the mystery shopping was to provide firms with an opportunity to experiment with the new suppliers.

¹⁷We designed this scorecard together with these hired experts specifically for this study. Vitali (2024) takes a similar approach to measure the garment quality. Although the details are quite different, we benefited greatly from showing her scorecard to our experts as an example of what we had in mind. As a small share of our firms sell shoes, we also developed a scorecard for shoes and hired an expert shoemaker to assess quality.

¹⁸We pre-specified this outcome, but did not attach it to either the horizontal or vertical dimensions. Since the consumer preference for Turkish-made goods reflects a preference for quality, it seems more fitting to include it under the vertical dimension.

outcome is 1 if the label says “Made in Türkiye” and 0 if it says it was made elsewhere.¹⁹

Finally, the mystery shopping also had a secondary goal of providing treated firms with an opportunity to experiment with the study suppliers. Nothing in the procedure makes this explicit, and the mystery shopper does not know the treatment status of the firm, but if the firm was considering making an order, then the mystery shopper reduces the risk that they will be unable to find a buyer.²⁰

Followup Survey We conducted a 30-minute followup survey with similar questions to the baseline survey between February and April 2024, around 3 months after a firm is recruited to the study. We successfully surveyed 90% of the sample (1671 firms). The followup rate is very similar and not significantly different across the four treated groups, but is 5 percentage points higher and statistically significantly different in the pure control group.²¹ The main outcomes are questions about the number and location of the firms’ suppliers, their profits and sales, and their e-commerce use.

Mobile Money To complement the survey data, we also use real-time, transaction-level data from the largest mobile money provider in Senegal, Wave Mobile Money, made available for this study. This data contains the value, timestamp, sender, and receiver for the universe of transfers between the phone numbers of study firms and the phone numbers of study suppliers. This data complements survey-based measures and has several advantages: (1) we can see transaction profiles over time, (2) it continues 15 months after the followup survey, (3) it captures real transactions (i.e., it is not self-reported).

While we do not observe the full universe of transactions, we expect that a substantial share takes place through this mobile money provider, for two reasons. First, as described in Section 4.1, we hired a set of non-study firms to make mystery orders from the suppliers prior to the intervention. In 100% of cases, suppliers requested payment through this provider. Second, in the baseline survey, 86% of firms reported regularly using this provider to pay suppliers for remote transactions, with 35% exclusively so.²²

These patterns suggest that we likely capture the majority of small-to-medium sized supplier payments. In contrast, larger wholesale orders are more often conducted through

¹⁹For the small subset of goods where this was not possible, the expert tailors independently evaluated origin based on design and construction cues. If they both determine the good was made in Türkiye, we impute 1; otherwise, we impute 0. In Table A4, we show that the results are not sensitive to how we treat these goods.

²⁰We mystery shop at *all* firms, so this secondary goal does not confound our analysis.

²¹We present Lee (2009) bounds for the main survey outcomes in Appendix Table A11. The results are similar, in line with the differential attrition being relatively small and a higher followup rate in control than treatment (i.e., the lower bounds trim the left tail of the control distribution, which is typically bounded).

²²For comparison, 37% reported using the second-most-common provider, with 2% of them exclusively.

formal channels such as bank transfers or international remittance services (e.g., Western Union). This distinction is reflected in the survey data: wholesalers are significantly more likely than retailers to use such methods—12% versus 4% of them, respectively.

5.2 Empirical Methodology

Our primary empirical method, pre-specified in our PAP, is the following OLS specification

$$y_i = \alpha + \sum_{j=1}^4 \beta_j T_{ji} + \delta y_{0i} + \gamma_s + \rho' X_i + \varepsilon_i, \quad (2)$$

where y_i is the outcome for firm i and T_{ji} for $j = \{1, 2, 3, 4\}$ are indicators for treatment arms Search Only, Search + Adverse Selection, Search + Moral Hazard, and Search + Adverse Selection + Moral Hazard. y_{0i} is the outcome measured at baseline, if available. γ_s are stratum fixed effects. X_i are firm-level covariates selected by Double Lasso (Belloni et al. (2014)).²³ We also report a similar regression that pools the four treated groups.

Inference Our primary method of inference is randomisation inference, as recommended by Athey and Imbens (2017) and Young (2019). In particular, we compute two-sided p -values for the sharp null of zero treatment effect using 2,000 permutations of the t -statistic. We report conventional heteroskedasticity-robust standard errors in parentheses, but we do not use these for inference directly. As each regression estimates four coefficients, we also calculate Romano-Wolf multiple-testing adjusted p -values in square brackets (Romano and Wolf, 2005). Since there are three combinations of trust treatments, to test the null that none of these has any effect, we report the p -value of a joint test that all four coefficients are equal (computed by permuting the F -statistic).

Indexes To account for multiple hypothesis testing across outcomes, for any table that presents more than one outcome corresponding to the same family of outcomes, we also report the results on an index that aggregates the outcomes using the standardised inverse-variance weighted method of Anderson (2008). Since the disaggregated regressions may include different covariates y_{0i} and X_i , before indexing we first residualise each outcome using the covariates that were included in their respective regressions.

Quantile Regression Because prior evidence suggests that some of our outcomes, notably profit and sales, may have thick tails and have important distributional treatment

²³This means that, prior to each regression, we run lasso to predict y_i and each T_i and include the union of selected covariates. In practice, this selects few covariates and thus makes little difference. Nonetheless, in Appendix D, we reproduce the main tables using only treatment indicators. The results are very similar.

effects (Meager, 2022), we also included in our PAP that we may use quantile regression to examine distributional treatment effects. For these, we follow the same specification as above, except that we omit the stratum fixed effects (γ_s) and the vector of Double-Lasso-selected covariates (X_i) as quantile regressions are much more demanding and the covariate selection procedure in Belloni et al. (2014) is designed for linear models.

6 Results

In Section 6.1, we present results on access to foreign goods, as measured by our mystery shopping activity. In Section 6.2, we present results on supplier relationships. In Section 6.3, we present results on profit and sales. In most tables, we show the pooled regression in Panel A, and then disaggregate across the treated groups in Panel B, with standard errors in parentheses, Romano-Wolf multiple-testing adjusted p -values in square brackets, and the p -value of a joint test that all coefficients are equal at the bottom.

6.1 Access to Foreign Goods

As described in Section 5, we designed a mystery shopping exercise to obtain a revealed preference of access to foreign goods, and to separate potential gains across horizontal, vertical and price-based mechanisms. Table 1 reports the outcomes of this exercise.

Horizontal In Column 1, we report the main horizontal outcome, which is an indicator for whether the firm had a product with at least three horizontal criteria.²⁴ Pooling the treatments together, treated firms are 9.3 percentage points more likely to find a suitable good ($p = 0.002$). This is a 26.1% increase from the control mean of 35.7%. In Panel B, we see that the effect is broadly similar in all four treatment groups and we do not reject the joint null that they are all equal, which implies that the effect is not larger for the groups with the trust treatments. The results thus show that relaxing the Search friction by connecting firms with foreign suppliers via social media led to a sizable increase in the set of varieties that they can provide to real customers.

Vertical The outcome in Column 2 is an indicator for whether the product is “High Quality”, defined as whether the product’s quality score is greater than its product-group me-

²⁴Our PAP specified this indicator as the main outcome. For robustness, we use the raw number of criteria as an outcome in Appendix Table A3. The pattern of the results is the same. We also disaggregate this outcome into an extensive margin effect (agreeing to sell a product at all) versus an intensive margin effect (how many criteria the product matched), and find that the effect comes mostly through the intensive margin.

dian.²⁵ Pooling the treatments together, treated firms are 13.1 percentage points more likely to be high quality ($p = 0.053$), a 30.4% increase from the control mean of 43.1%. In Panel B, we see that the coefficient is positive and similar in all four treatment groups.

The outcome in Column 3 is the raw quality score out of 50. Here, the effect is both small and insignificant. As we noted in Section 5, this outcome is vulnerable to a long left tail having an outsize influence. This is indeed what happens: in Appendix Figure A5 we plot the CDF of quality score by treatment status, and we see that, while the treatment CDF is to the right of the control CDF from the 30th percentile onwards (as reflected in the binary outcome), it also contains a handful of very low scoring goods that skew the mean.

The outcome in Column 4 is an indicator for whether the product was made in Türkiye. Pooling the treatments together, treated firms are 16.2 percentage points more likely to supply a good made in Türkiye, a 34.0% increase from the control mean of 47.7% ($p = 0.022$).²⁶ In Panel B, we see that the effect is positive and similar for all four treatments.

Finally, to account for multiple hypothesis testing across outcomes, we aggregate these three outcomes into a vertical index using the standardised inverse-variance weighting method in Anderson (2008). The pooled coefficient is 0.412 standard deviations ($p = 0.004$), and similar across the four treatment groups. The results thus show that relaxing the search friction by connecting firms with suppliers abroad via social media led to a sizable increase in their access to higher quality goods.

Price The outcome in Column 6 is the price charged to the mystery shopper. The coefficient is positive, small, and statistically insignificant, suggesting no meaningful price effects in either direction. While this could reflect two offsetting effects if the observed increase in quality causes prices to rise by an amount that offsets a decrease through a price-based mechanism, it at least suggests that a price-based mechanism is not so large enough as to overcome this.

Summary: Relaxing Search Frictions Expands Access to Foreign Varieties Putting together these results, we find that all of the treated groups saw large and significant increases in access to foreign goods. Treated firms are able to sell a wider set of varieties

²⁵The sample size is lower for the vertical and price outcomes because the mystery shopper only proceeded to the negotiation phase of the procedure (to measure price) if the firm had a good matching at least 3 horizontal criteria, and then only purchased the good (to measure quality) under some additional conditions described in Appendix F.3. To investigate whether this introduces selection bias, in Appendix Table A5 we replicate Columns 2-6 of Table 1 but also include fixed effects for each of the 28 products, and find that the qualitative takeaways are the same. We also define an outcome unaffected by this issue that is 1 if the firm had a suitable and high quality good and 0 otherwise. We find positive and significant effects on this outcome.

²⁶As a robustness check, we report various alternative ways of defining whether a good was made in Türkiye in Appendix Table A4. The pattern of results is the same.

and higher quality varieties. The fact that we find no detectable effects on the price suggests that gains from relaxing the frictions come more from improving access to otherwise unavailable varieties than from lowering prices for current varieties.

Across all of these outcomes, the pattern is consistent: the results are driven by relaxing the search friction. We therefore conclude that finding a supplier abroad is costly, and that social media can alleviate this friction. This does not necessarily mean that the trust frictions do not exist: these are small orders with a customer already in place, so for many firms the risk may simply be quite low. However, for firms to realise meaningful gains from trade, they will have to eventually make larger orders and develop these new connections into relationships, which we look at next.

6.2 Building Supplier Relationships

6.2.1 Survey Data

In our follow-up survey, conducted after 3 months, we asked firms how many regular suppliers they had, and where those regular suppliers were based. We defined a regular supplier as any supplier from whom the firm had made at least two orders, and intended to continue the relationship. We analyse these outcomes in Table 2. In Column 1, the outcome is an indicator for whether the firm has a regular supplier in Türkiye. In Column 2, the outcome is the number of regular suppliers in Türkiye. In both cases, pooling all four treatment groups together shows that treatment caused firms to develop new relationships with suppliers in Türkiye. The pooled coefficients are 3.7 percentage points (an increase of 22.2% relative to control mean of 16.7%) with $p = 0.088$, and 0.083 suppliers (an increase of 37.4% relative to the control mean of 0.222) with $p = 0.023$.

Disaggregating by treatment arm, for both outcomes, the coefficient for Search + Adverse Selection + Moral Hazard is substantially larger: this group sees a 7.5 percentage point increase in the likelihood of having a supplier in Türkiye, and a 0.188 increase in the number of suppliers in Türkiye, both of which are highly significant (including after adjusting for multiple hypothesis testing). The effect is also larger for the Search + Adverse Selection group. Formally, we can reject the null that all four coefficients are equal. This suggests that relaxing trust frictions, and perhaps in particular relaxing them jointly, increased the likelihood that these new connections developed into relationships.

In the second part of Table 2, we provide evidence on whether these new relationships complement or substitute for existing relationships. In Column 3, the outcome is the total number of suppliers; in Column 4, the outcome is the number of suppliers in Senegal. The general direction looks closer to a world of substitutes: the coefficients on the total number

of suppliers are close to zero, and the coefficients on the number of suppliers in Senegal, while noisy, are of similar magnitude (but opposite sign) to the coefficients on the number of suppliers in Türkiye. Finally, Column 5 shows an indicator for whether the firm said that they have ended a relationship with a regular supplier in the past 3 months. The coefficients are generally positive, which is also suggestive of substitutes.

To assess how the fourth WhatsApp group (for the adverse selection treatment) influenced buyer perceptions and behavior, we included targeted questions in the follow-up survey. A majority (53%) reported that this group provided them with meaningful value. Specifically, 24% stated that the group increased their trust in the suppliers or helped them obtain useful information about them. Another 12% indicated that the group facilitated new connections, and an equal share (12%) reported that they had privately contacted other firms from the group to learn more about the study suppliers. In addition, 24% of firms reported knowing other firms who had placed orders with the study suppliers (compared to 13% in the absence of this treatment)—evidence pointing to the group’s role in disseminating information about peer purchasing behavior.

6.2.2 Mobile Money Data

As discussed in Section 5, we use data made available for this research from the largest mobile money provider in Senegal to directly observe transactions between study firms and study suppliers. Before turning to formal regression results, we first present graphical evidence from the raw data. In Figure 6, we plot cumulative order value over time from study suppliers aggregated across the four treatment groups.²⁷ The dashed line shows when we finished our mystery shopping activities. The figure shows two striking patterns. First, over the mystery shopping period, the total value ordered is very similar across the four treatment groups, which is in line with the mystery shopping results. As discussed in Section 5, interpreting this part of the graph is a little challenging because a secondary goal of the mystery shopping was to lower the cost for firms to experiment, although it is true that the total value ordered is much larger than would be implied by simply buying and re-selling to us. Second, almost immediately after the mystery shopping ends, the Search Only line flattens, suggesting that most of these relationships were not lasting. In contrast, in all three of the trust treatments, the total value ordered continues to increase well beyond when the mystery shopping ended, suggestive of continuing relationships.

We formally test these patterns in Table 3, where the omitted category is Search Only. In Column 1, we test the observation from Figure 6 that only the trust treatments appear to continue ordering after the mystery shopping ends. The outcome is the total value ordered

²⁷We omit Pure Control from the figure because they were not connected to any study suppliers.

after the mystery shopping ends. We did not pre-specify this outcome, but rather included it after observing the pattern in Figure 6. The coefficient pooling all three trust treatments implies an 209.8% increase ($p = 0.067$). When we disaggregate the treatments, we find that the coefficient is positive in all three trust treatment groups, although the p -value for the joint F -test is 0.162. As in the survey data, the coefficient is largest in the group with both trust treatments, although the standard errors are too large to distinguish between the trust groups. In Columns 2 and 3, we decompose total value into the number of orders and average value per order. The coefficients on the number of orders are positive and sizable (but insignificant), and those on average order value are positive, large, and significant.

We repeat the same three outcomes with the mystery shopping period included in Columns 4, 5, and 6. The broad pattern is similar, which is in line with the observation from Figure 6 that the differences only open up after the mystery shopping ends. In interpreting the magnitudes, note that these are unconditional averages and that most firms did not develop relationships, and that larger transactions tend to take place off-platform.

6.2.3 Summary: Trust Treatments Convert Matches into Supplier Relationships

Putting together the results from Sections 6.2.1 and 6.2.2, we find that trust treatments made it more likely for these new connections to develop into lasting relationships. In both datasets, the largest effects come from the Search + Adverse Selection + Moral Hazard group. This is consistent with the fact that the trust treatments are complements in the model, although we are not powered to formally test for a complementarity.

6.3 Profit and Sales

6.3.1 Average Treatment Effects

To see whether improved access to foreign goods and new relationships flow through to profits, we report the results on profits and sales in Table 4. For profit, we use the summary survey question from De Mel et al. (2009) for the last 30 days. For sales, we use a similar summary question.

Columns 1-2 show the results on raw profit and sales. We find large and significant effects. For profit, the pooled coefficient is 82.4 USD ($p = 0.025$), or a 43.8% increase from the control mean. For sales, the pooled coefficient is 245.2 USD ($p = 0.051$), or a 40.2% increase from the control mean. When we disaggregate the four treated groups in Panel B, we see that (for both outcomes), while the effect is positive in all four groups, it is much larger and only significant in the Search + Adverse Selection + Moral Hazard group (including after adjusting for multiple hypothesis testing). Formally, we can reject the null that the

four coefficients are equal. The same pattern holds when we index the two outcomes.

To limit the influence of outliers, in Columns 4-5 we report the results when we winsorize the outcomes at the top 1%. The coefficients decrease in magnitude by around half, but the same pattern remains: the coefficient on the Search + Adverse Selection + Moral Hazard group is large and significant, including after adjusting for multiple hypothesis testing.

6.3.2 Distributional Treatment Effects

It is now well established in the literature studying small firms in lower-income countries that profit and sales tend to be thick-tailed, and that these tails can have large effects on the coefficients in OLS regressions (Meager, 2022).²⁸ Thus, as discussed in Section 5.2 and specified in our PAP, we use quantile regression to examine distributional effects.

Quantile Treatment Effects In Figure 7 Panel (a), we show the quantile treatment effects for profit for percentiles 5-95. Across all four groups, the coefficients are small and generally insignificant for percentiles 5-65. However, starting from around the 75th percentile, the Search + Adverse Selection + Moral Hazard group coefficient becomes large and significant. The coefficients for Search Only and Search + Adverse Selection are also large at the 95th percentile, but are very noisy. The results for sales in Panel (b) are similar.²⁹

Threshold Regression As an alternative way to analyse distributional treatment effects, we also run OLS regressions on indicators that are 1 if the outcome is greater than t , for a range of t . This has the advantage of being unaffected by high variance in the tails as all that matters is whether the outcome is above the threshold t . The results, reported in Appendix Figure A6, are similar to the quantile regressions: large and positive effects near the top of the distribution, from above USD 200 profit, for the Search + Adverse Selection + Moral Hazard group, and some suggestive evidence of positive effects for the other groups at the very top.

²⁸To verify that treatment effects are not driven by measurement error in the tails, we called back all firms whose profit was more than 5 times that at baseline and exceeded a threshold at endline, and we asked them to confirm their previous response. Out of 13 such firms, 12 confirmed that their previous response was correct.

²⁹While quantile treatment effects are only interpretable as the effect for firms at a given quantile under rank preservation, we think that it is likely that these are treatment effects for firms that are larger at baseline. First, in Appendix Figure A7 we show treatment effects on the transition matrix between baseline and endline profit: the largest coefficients are in the northeast region, suggesting that treatment makes large firms more likely to remain large. Second, we show in Section 6.5 that the profit gains appear concentrated among wholesalers.

6.3.3 Summary: Substantial Profit Gains from Reducing Trust Frictions

In Table 4, we saw large average effects on profit and sales. As in Section 6.2, the effects are concentrated in the trust groups, and in particular in the group with both trust treatments. This suggests that trust frictions have quantitatively important effects on firm profits, and that alleviating them using social media can unlock large gains.

We validate that these effects are not driven by a handful of outliers that happen to be in the treatment group. First, gains appear from at least the top 5% of the distribution, well beyond a few extreme cases. Second, the threshold regressions use indicators as their outcomes and thus are immune to the risk of a few observations having outsize influence. Third, the p -values in Table 4 highlight that the patterns we observe are very unlikely under the sharp null, including after adjusting for multiple hypothesis testing. Fourth, a placebo using baseline profit (Appendix Figure A8) shows no such pattern.

A positive effect driven by the upper tail of the profit distribution is not unusual in the literature studying firms in lower- and middle-income countries.³⁰ In our setting, the effect concentrated among a relatively small number of firms is consistent with a small subset of firms developing meaningful relationships with the study suppliers, which is what we describe in Section 6.2.

6.4 Firms Prefer Social Media to Formal E-commerce Platforms

Our final set of outcomes relate to the Alibaba training sub-treatment. The goal of this treatment was to test whether the binding constraint explaining the very limited use of formal e-commerce platforms is that firms find these platforms too complex. Thus, we regress outcomes relating to Alibaba use against an indicator for whether the firm was in the Alibaba training group. Since we only randomised this training among firms that received none of the main treatments, we exclude firms that received any of the main treatments (although the results are similar if we include them).

We report the results in Table 5. The training has a first-stage: treated firms are 6.5 percentage points more likely to have heard of Alibaba, 11.3 percentage points more likely to have searched for goods on Alibaba, and 8.7 percentage points more likely to have compared prices on Alibaba with prices from their regular supplier. However, they are no more likely to have actually made a purchase from Alibaba. The coefficient is 1.4 percentage points and the standard errors are small enough to rule out modest to large effects. These

³⁰For example, Meager (2022) aggregates the results of six RCTs on microcredit, and concludes that the evidence suggests precise zero effects on profit throughout most of the distribution, and large but uncertain effects near the top. Another example is De Mel et al. (2013), which studies the effect of formalisation among small firms in Sri Lanka, and find profit results driven by the upper tail.

results provide strong evidence against the hypothesis that the binding constraint is that firms do not know how to use the platform. While our experiment was not designed to directly test social media against formal platforms, we can speculate that the fact that firms clearly prefer social media likely reflects something deeper about how social media—in this context—relaxes frictions in a way that formal B2B platforms currently do not.

6.5 Heterogeneous Treatment Effects: Higher Profit Gains for Wholesalers

We pre-specified a set of heterogeneity analyses to explore how treatment effects vary with firm characteristics, and we report detailed tables in Appendix C. Several patterns emerge.

First, we compare retailers and wholesalers. We define wholesalers as the 33% of the sample for whom some (or all) of their sales are wholesale to other firms. While we find no consistent heterogeneity across mystery shopping or relationship outcomes, the profit gains appear concentrated among wholesalers. This aligns with the distributional treatment effects in Section 6.3. One possible explanation is that wholesalers, who typically buy and sell in bulk, stand to gain more immediately from access to new suppliers. Alternatively, the effects may reflect reduced travel costs for sourcing, given that wholesalers are more likely to travel abroad for procurement. While we cannot definitively distinguish between these mechanisms, Appendix Table A9 provides suggestive evidence: wholesalers in the treatment group are less likely to have traveled internationally for business in the three months between the intervention and the follow-up survey. The estimated effects are large and mostly negative, with a few significant coefficients, though wide confidence intervals suggest they should be interpreted with caution.

Second, we examine whether treatment effects differ between online-only firms and those with a physical store. The horizontal mystery shopping effects (e.g., custom order responsiveness) are more pronounced among online-only firms, likely reflecting their greater flexibility in adapting to buyer requests. However, we find no systematic differences in vertical mystery shopping or relationship outcomes. Interestingly, the profit effects are instead concentrated among firms with physical stores, many (45%) of which are wholesalers. This suggests that while online-only firms may be more responsive to initial contact, firms with physical stores may be better positioned to convert new relationships into sustained profit gains. Since online-only firms are disproportionately women-operated (63% against 25% for physical stores), we observe a similar pattern when disaggregating by the gender of the firm owner.

Finally, we find no meaningful heterogeneity in treatment effects based on prior importing experience, initial exposure to Turkish-made goods, or existing membership in a Turkish supplier WhatsApp group. This suggests that the intervention may be useful for firms

regardless of their baseline access to international sourcing networks.

7 Model Estimation

In this section, we use the results of the experiment to estimate the model from Section 3. We have two main goals. First, we want to estimate the parameters governing the search and trust frictions, which are of general interest to users of these types of models. Second, since our treatments are only one particular way of alleviating the frictions, we want to evaluate the gains from trade available if the frictions were to be counterfactually alleviated in different combinations, to different magnitudes, or through different interventions.

Modifications to the Model In order to make the model estimable, we need to make a few modifications. First, we let the revenue function r be such that the firm faces and internalises a constant elasticity residual demand curve for goods, $zQ^{-1/\sigma} = P$, with $\sigma > 1$.³¹ The aggregate good Q is a linear combination of inputs from either a new foreign supplier that they are matched with, q_f , or their existing suppliers (who may also be foreign), q_e .³² For simplicity, we let firms' existing suppliers be summarized by constant price p_e . Timing is such that the firm chooses (q_f, q_e) , observes whether the order q_f is high quality and therefore whether $Q = q_f + q_e$ or $Q = q_e$, and then sells to their downstream buyers. The firm's stage game payoff is therefore the following,

$$(1 - \mu_t(1 - \lambda))z(\psi q_{ft} + q_{et})^{\frac{\sigma-1}{\sigma}} + \mu_t(1 - \lambda)zq_{et}^{\frac{\sigma-1}{\sigma}} - \tau_t - p_e q_{et}, \quad (3)$$

where $z > 0$ is firm-specific productivity and ψ is match-specific productivity. We can see from inspection that the firm will always choose $q_{et} > 0$, but may choose $q_{ft} = 0$ if either beliefs μ_t or the transfers $\{\tau_s\}_t^\infty$ required to incentivise the foreign supplier are sufficiently large.³³

³¹Since our experiment was not large enough to generate general equilibrium effects, we do not explicitly model competition between firms in the estimation or counterfactuals. Our counterfactuals remain valid so long as the shock does not affect very large numbers of firms. We do not expect that introducing general equilibrium effects would meaningfully affect the relative gains to alleviating the frictions (e.g., adverse selection vs moral hazard), but it would likely lower the absolute returns.

³²It is conceptually straightforward to allow these goods to be imperfect substitutes to introduce love of variety explicitly. However, making them perfect substitutes allows for analytical solutions for q_{ft} and q_{et} , which speeds up the computation as this problem must be solved within an inner loop.

³³To prevent a discontinuity in learning when going from $q_{ft} = 0$ to any positive value, while also ensuring the principal's choice set remains convex, we impose a technical assumption that for q_{ft} below some small $\underline{q}$, posteriors are given by $\mu_{t+1} = \mu_{t+1}^{\text{Bayes}} + (\mu_t - \mu_{t+1}^{\text{Bayes}}) \cdot (1 - q_{ft}/\underline{q})^p$, where μ_{t+1}^{Bayes} is the posterior implied by Bayes' Rule and $p \in (1, 2)$. Making learning a continuous function of q_{ft} ensures the problem satisfies appropriate regularity conditions. Intuitively, it captures the idea of a minimum order size (which is the case in our empirical setting).

Second, we need distributions for the heterogeneity. For the match-specific productivity, ψ , we normalize the foreign supplier's base cost, c , to 1 and then let ψ be distributed lognormal with parameters (ψ_μ, ψ_σ) , which we will estimate. We draw the firm-specific productivity, z , from the empirical distribution that rationalises the control group's profits (described in more detail below).

Third, we need to take a stand on the baseline equilibrium. The sequential search process implies that there exists a cutoff value $\bar{z}$ such that firms with $z > \bar{z}$ will search until they find a foreign supplier with sufficiently high match-specific productivity, where $\bar{z}$ is a function of parameters to be estimated. We interpret the baseline equilibrium as the long-run steady state of the model, meaning that all firms with $z > \bar{z}$ have found a satisfactory foreign supplier and are thus in the late phase of the relationship, which, as we show in Section 3, means they earn $(1 - \delta)\bar{U}(z)$ every period. For such firms, we then set p_e to be the constant price that delivers exactly $(1 - \delta)\bar{U}(z)$. For all other firms, p_e represents the cost of buying from a local supplier in Dakar, which we calibrate using the baseline survey. A firm's outside option, $\bar{U}$, is then defined by the value of purchasing at p_e forever,

$$\bar{U}(z) = \left(\max_{q_e} z q_e^{\frac{\sigma-1}{\sigma}} - p_e(z) q_e \right) / (1 - \delta).$$

Parameters Our goal is to estimate the parameters governing the three frictions: the search cost, s , the share of bad types, μ_0 , and the moral hazard effort cost, ξ . As the experiment has four treatment effects per outcome, we include two additional, related parameters to better match the combinations of treatments. First, we include the discount factor, δ , which is important for dynamic incentives and thus has been a target parameter in previous studies of dynamic moral hazard (e.g., Einav et al. (2015)). We interpret one period as one month. Second, we include the probability of low effort generating high quality goods, λ . This parameter affects the speed of learning and, indirectly, the strength of the interaction between adverse selection and moral hazard: when learning is slow, moral hazard becomes less consequential. Finally, we also need to estimate the parameters governing the lognormal distribution of match-specific productivity, (ψ_μ, ψ_σ) , which are related to the search friction as they govern the expected returns to searching. We thus estimate these seven parameters, and calibrate the remaining two, (σ, p_l) . We discuss this calibration in Appendix Table A10.

Moments Since the ultimate goal is to use the model to extrapolate from the treatments to consider counterfactuals where we vary the extent of the frictions, we estimate the model by simulating the impact of treatment in the model and then matching the reduced form

treatment effects. We select the four treatment effects on winsorized profit, the four treatment effects on likelihood of having a supplier in Türkiye, and the three treatment effects on mobile money order value post mystery shopping.³⁴

We simulate the three treatments in the model as follows. We implement search as the firm being matched at zero cost to a foreign supplier with match-specific productivity equal to the maximum of three draws from the distribution (to capture the idea that the firm is matched to three foreign suppliers in the experiment). We implement adverse selection as the firm receiving one high quality signal realisation without having to purchase anything, meaning that they update as a Bayesian and thus begin the relationship with belief that the supplier is a bad-type of $\mu_1 = \mu_0\lambda/(1 - \mu_0(1 - \lambda)) < \mu_0$. This intends to capture the recommendation call from the adverse selection treatment that explicitly told the firm about one positive order experience.³⁵ We implement moral hazard as treated firms solving for the optimal dynamic path under a joint-punishment strategy among study firms, as opposed to an individual grim-trigger strategy. That is, treated firms know that the suppliers stands to lose δNV_{t+1} instead of δV_{t+1} if low-quality is realised, where we calibrate $N = 5$ as the empirical ratio of the number of firms that made orders from study suppliers to the number of study suppliers.³⁶ Some algebra shows that the DICC for these firms can be rewritten as $(1 - \lambda)\delta V_{t+1} \geq \tilde{\xi}cq_t$, where $\tilde{\xi} \equiv \xi/N$, so this treatment is equivalent to lowering ξ by a factor of N .

Solving and Estimating the Model Dynamic models of moral hazard are non-stationary in general and solving them numerically is non-trivial. We use the method of [Marcet and Marimon \(2019\)](#), which involves rewriting the problem as an as-if planner’s problem with Pareto weights that evolve endogenously to summarize past utility promises.³⁷ This permits a recursive formulation of the original Lagrangean with the Pareto weights as additional co-states, which—under some regularity conditions—we can then solve by iterating on the value function.

³⁴For the mobile money moments, we match the treatment effects expressed in percent of control mean, as we do not know the exact share of transactions covered by the mobile money data.

³⁵We focus on the direct information provision, as opposed to the joint learning in the discussion WhatsApp group, as we believe this approach is more transparent. Modeling the WhatsApp group would introduce several additional assumptions (and parameters), such as those governing firms’ incentives to share information and firms’ beliefs about when other firms will make orders.

³⁶This implicitly assumes that firms treat other firms as identical to themselves (i.e., V_{t+1} is the same). This simplifies the computation significantly, as otherwise we would need to repeatedly solve for a fixed point in V_{t+1} (for all t), and can be reasonably interpreted as firms extrapolating from their own experience.

³⁷The Pareto weights are equal to the sum of the Lagrange multipliers from the incentives constraints in earlier periods. Intuitively, if the DICC binds in period 0, which implies that the agent must be delivered a certain amount of utils in the future, then the agent’s Pareto weight increases in proportion to the multiplier to ensure that the “planner” in the future delivers these utils. This allows the problem to be written recursively because the endogenous Pareto weights (which become state variables) fully summarizes previous constraints.

For estimation, we use Simulated Method of Moments (SMM) with a weighting matrix equal to the inverse of the variance-covariance matrix of the empirical moments, and with 50 randomized initial points for the numerical minimization algorithm. For each guess of the parameters, we do the following two-step procedure. First, we calculate the value of ψ_μ and the empirical distribution of z that rationalise the control group's share of firms with a supplier in Türkiye and profit distribution, respectively.³⁸ Second, we use these to draw from the (z, ψ) distribution and calculate the moments. We compute standard errors by bootstrapping the estimation procedure 200 times and calculating the standard deviation of the estimates. We provide a more detailed description of the procedure to solve and estimate the model in Appendix E.

Results and Model Fit We present the estimated parameters in Table 6. The value of $\hat{\mu}_0 = 0.57$ suggests an important adverse selection problem as over half of suppliers are bad types. The value of $\hat{\xi} = 0.91$ is close to a standard model of moral hazard (which would have $\xi = 1$, i.e., low effort production is costless), although how much moral hazard actually constrains the firm depends on how severely the DICC's bind.³⁹ The search cost of $s = 94$ USD is somewhat large, although it has a very high standard error. The estimation implies that the frictions must be relatively large to explain why firms do not search more, despite the sizable treatment effects observed experimentally.

We plot the targeted moments in Panel A of Appendix Figure A9. The model is able to match the treatment effects reasonably well. It cannot match the scale of the complementarity in the profit moments, but it is close in the survey relationship moments. As in the experimental data, the model generates a heavy right tail: in Panel B, we show that quantile treatment effects computed in the model, which are not directly targeted, look similar the experimental ones in Figure 7.

Counterfactuals We now use the estimated model to evaluate counterfactuals in which we reduce the trust frictions among all suppliers that a firm faces by setting key parameters to half their estimated values.⁴⁰ We show the results in Figure 8, Panel (a). We first reduce

³⁸Specifically, the model implies that profits for firm i are given by $\pi_i = z_i^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p_{ei}^{1-\sigma}$. Since the definition of baseline described above implies a (parameter-dependent) mapping from z to p_e , we can define a bijective function $\pi(z) = z^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p_e(z)^{1-\sigma}$. Then, we can construct a distribution of z by drawing π_i from the control group's empirical distribution and inverting this function to obtain z_i .

³⁹The moral hazard problem also appears in the monthly discount factor $\hat{\delta}$ being quite low at 0.78—this happens because it is hard for dynamic moral hazard to have meaningful quantitative impacts if there is a large amount of surplus in the relationship and the players are very patient. While this value is low if the parameter reflects pure impatience, it is not unreasonable if interpreted as a composite term that embeds an exogenous probability of the relationship breaking up, especially in a volatile informal trading environment.

⁴⁰We focus on the trust frictions as the search cost parameter is quite noisy, suggesting that search counterfactuals are likely to be particularly uncertain.

the adverse selection parameter, μ_0 , by half. This may be thought of as reflecting policies that allow firms to filter out half of the bad types, such as through an online platform or a matching service that does some pre-screening. Note that this is much stronger than our intervention as it is market-wide rather than among three specific suppliers. The effect is very large in percentage terms, an increase in lifetime discounted profits of 180%, although note that these are small firms and so the absolute magnitude of the gains (around USD 1900) is of the same order of magnitude as fixed costs associated with other solutions, such as traveling internationally to source supplies. A major mechanism underlying this effect is the fact that, when adverse selection is lower, firms endogenously increase their search behavior. This can be seen in Figure 8, Panel (b), where we plot firm lifetime profits, $U(z)$, over the distribution of firm productivity, z , at baseline and in the counterfactual. On the extensive margin, we can see that the cutoff determining which firms search, $\bar{z}$, shifts left, meaning that more firms engage in trade. On the intensive margin, even firms already importing increase their reservation utility and may re-search.

Next, we reduce the moral hazard parameter by half, reflecting policies that strengthen suppliers' incentives by making shirking more costly, such as services that offer digital traceability of the supply chain. The effect here, an increase in lifetime discounted profits of 27%, is smaller. This happens because, in the model, even though the discount factor is relatively low, moral hazard is not the binding constraint—firms are already reluctant to order from foreign suppliers because of adverse selection, so the DICC binds infrequently in practice. However, this is no longer the case in the last scenario, where we both remove half of the bad types and double supplier's incentives. As suggested by the experimental results, there is a strong complementarity: when firms are less worried about bad types, incentives become the binding constraint and thus improving them has a larger effect. Indeed, the average Lagrange multiplier on the DICC is around four times larger when adverse selection is lowered. This suggests—and helps explain why—practical solutions that address both adverse selection and moral hazard simultaneously (such as reviewer systems) can be especially powerful.

8 Conclusion

We study the extent to which search and trust frictions limit the ability of small firms to source inputs from foreign markets—and whether social media can help reduce these barriers. Finding and developing relationships with suppliers is a first-order issue, and thus understanding the nature and magnitude of these frictions is important. Moreover, the rapid recent growth of firms using social media to find and evaluate suppliers makes it especially timely to understand the role that these new digital tools play in alleviating

information frictions.

Our results have two main implications. First, search and trust frictions are important barriers for trade. Our mystery shopping exercise shows that lowering search costs by connecting firms to new suppliers abroad allows them to provide both a wider set of varieties and higher quality varieties to real customers. However, using both survey and mobile payments data, we find that these connections only develop into meaningful and lasting relationships when we also alleviate the trust frictions. This, in turn, leads to higher profits for some firms. Beyond the experiment, our structural exercise suggests that adverse selection may be the more important trust friction, with scope for large complementarities to alleviating both trust frictions jointly. While our experiment focuses on international trade, our results likely apply more generally when buyers and sellers are spatially distant or where alternative ways to mitigate the frictions are unavailable.

Second, the fact that we find these effects through interventions that deliberately mimic how firms use social media in the real world suggests that the rapid growth of trade through social media should be at least partially understood as a new way for firms to mitigate search and trust frictions. These technologies are changing the structure of supply chains: allowing retailers to import directly and wholesalers to reduce costly travel. Moreover, these changes are here to stay, as smartphone access and mobile connectivity continue to grow, and as social media companies themselves introduce e-commerce features to their products.

Yet, despite substantial demand for online commerce, firms in Africa appear to rarely use formal B2B platforms like Alibaba. Our e-commerce training had no effect on purchasing behavior, suggesting that the constraint is not just a lack of knowledge. The stronger appeal of social media may reflect deeper differences in how it reduces frictions in this setting in ways that formal platforms do not—for example, through greater familiarity, flexibility, or trust. We view understanding these differences as a promising direction for future research.

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Tables

Table 1: Access to Foreign Goods

	Horizontal	Vertical			Price	
	Has Product ≥ 3 Criteria	High Qual Dummy	Qual Score (/50)	Made in Turkey	Index	Price (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Pooled</i>						
Treatment	0.093*** (0.030)	0.131* (0.067)	-0.391 (0.562)	0.162** (0.067)	0.412*** (0.135)	0.710 (0.699)
<i>Panel B: Individual Treatments</i>						
Search Only	0.140*** (0.038) [0.002]	0.175** (0.083) [0.108]	0.259 (0.675) [0.898]	0.138* (0.082) [0.162]	0.426** (0.166) [0.033]	1.360 (0.896) [0.353]
Search + Adverse Selection	0.053 (0.037) [0.151]	0.159* (0.087) [0.155]	-0.027 (0.858) [0.971]	0.130 (0.086) [0.162]	0.398** (0.169) [0.035]	0.445 (0.833) [0.820]
Search + Moral Hazard	0.105*** (0.038) [0.018]	0.099 (0.082) [0.351]	-1.179 (0.789) [0.358]	0.183** (0.081) [0.060]	0.399** (0.162) [0.035]	1.064 (0.866) [0.473]
Search + AS + MH	0.072* (0.038) [0.111]	0.083 (0.084) [0.351]	-0.717 (0.814) [0.703]	0.211** (0.081) [0.033]	0.424** (0.171) [0.035]	-0.025 (0.818) [0.977]
Control Mean	0.357	0.431	43.064	0.477	0.000	19.990
% Increase (Pooled)	26.1%	30.4%	-0.9%	34.0%	N/A	3.6%
All Coefs Equal p -val	0.113	0.607	0.310	0.660	0.998	0.321
Adjusted R^2	0.09	0.04	0.34	0.09	0.02	0.40
N	1579	359	359	361	359	642

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is an indicator that is one if the firm has a good that matches at least 3 horizontal criteria, and is missing if the firm never replied to the mystery shopper or was otherwise unreachable. Column 2 is an indicator for whether the good's quality score is above the median product-group quality score. Column 3 is the raw quality score. Column 4 is an indicator for whether the good is made in Turkey, primarily inferred based on the label. See the text for full details of how this outcome is constructed. Column 5 is the [Anderson \(2008\)](#) index combining the vertical outcomes to account for multiple hypothesis testing across outcomes. Column 6 is the price in USD, which is only measured conditional on the firm finding a good matching at least three horizontal criteria.

Table 2: Supplier Relationships (Followup Survey)

	Regular Suppliers in Türkiye		Previous Suppliers		
	Any Sup in Türkiye (1)	Num Sup in Türkiye (2)	Num Sup Total (3)	Num Sup in Senegal (4)	Ended with Sup (5)
<i>Panel A: Pooled</i>					
Treatment	0.037* (0.021)	0.083** (0.035)	-0.102 (0.164)	-0.130 (0.170)	0.062*** (0.021)
<i>Panel B: Individual Treatments</i>					
Search Only	0.024 (0.027) [0.576]	0.081 (0.051) [0.263]	-0.101 (0.216) [0.885]	-0.111 (0.222) [0.924]	0.050* (0.028) [0.103]
Search + Adverse Selection	0.049* (0.029) [0.205]	0.063 (0.047) [0.321]	-0.171 (0.218) [0.836]	-0.245 (0.226) [0.636]	0.066** (0.028) [0.049]
Search + Moral Hazard	0.003 (0.026) [0.916]	0.003 (0.041) [0.936]	-0.118 (0.201) [0.885]	-0.064 (0.212) [0.924]	0.076** (0.029) [0.038]
Search + AS + MH	0.075** (0.028) [0.035]	0.188*** (0.057) [0.003]	-0.014 (0.222) [0.951]	-0.096 (0.225) [0.924]	0.053* (0.028) [0.103]
Control Mean	0.167	0.222	3.700	3.213	0.135
% Increase (Pooled)	22.2%	37.4%	-2.8%	-4.0%	45.9%
All Coefs Equal p -val	0.075	0.013	0.919	0.861	0.826
Adjusted R^2	0.14	0.12	0.32	0.26	0.05
N	1680	1680	1681	1681	1671

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm says that they have a regular supplier in Türkiye. Column 2 is the number of regular suppliers in Türkiye. Column 3 is the total number of regular suppliers. Column 4 is the number of regular suppliers in Senegal. Column 5 is 1 if the firm has ended a relationship with a regular supplier in the past 3 months. A regular supplier is defined as a supplier from whom the firm has made two or more orders with the intention to continue the relationship.

Table 3: Order Value (Mobile Money Data)

	Post Mystery Shopping			Entire Period		
	Total Order Value (USD)	Num Orders	Avg Order Value (USD)	Total Order Value (USD)	Num Orders	Avg Order Value (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Pooled</i>						
Trust Treatment	7.46*	0.05	2.24***	7.43	0.03	2.12**
	(4.74)	(0.07)	(0.74)	(5.12)	(0.09)	(1.00)
<i>Panel B: Individual Treatments</i>						
Search + Adverse Selection	3.92	-0.02	3.14**	4.99	0.00	3.04*
	(3.06)	(0.07)	(1.59)	(3.74)	(0.10)	(1.74)
	[0.247]	[0.881]	[0.058]	[0.235]	[0.996]	[0.143]
Search + Moral Hazard	6.71**	0.11	2.80**	6.97*	0.06	3.49**
	(3.69)	(0.09)	(1.21)	(4.47)	(0.11)	(1.67)
	[0.062]	[0.529]	[0.024]	[0.146]	[0.916]	[0.073]
Search + AS + MH	11.86	0.06	0.76	10.41	0.01	-0.23
	(12.66)	(0.12)	(0.61)	(13.21)	(0.13)	(0.88)
	[0.429]	[0.881]	[0.234]	[0.572]	[0.996]	[0.804]
Control Mean	3.56	0.14	0.89	8.93	0.33	3.65
% Increase (Pooled)	209.8%	34.5%	253.3%	83.2%	7.8%	58.0%
All Coefs Zero p -val	0.162	0.612	0.028	0.272	0.955	0.061
Adjusted R^2	-0.00	0.01	-0.00	0.01	0.04	0.00
N	1500	1500	1500	1500	1500	1500

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all trust treated groups, where Search Only is the omitted category. Panel B shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is the total value of orders from study suppliers after the mystery shopping period finished. Column 2 is the number of orders from study suppliers. Column 3 is the average order size from study suppliers after the mystery shopping period finished. Column 4 is the total value of orders from study suppliers over the entire period. Column 5 is the number of orders from study suppliers over the entire period. Column 6 is the average order size from study suppliers over the entire period. Mystery shopping took place during the first 3 months of the study, from 16 November 2023 to 22 February 2024. The entire period covered by the data comprises 18 months, from 16 November 2023 to 31 May 2025. All values are in USD.

Table 4: Profit and Sales

	Raw			Winsorized (1%)		
	Profit (USD)	Sales (USD)	Index	Profit (USD)	Sales (USD)	Index
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Pooled</i>						
Treatment	82.4** (31.6)	245.2* (108.4)	0.314** (0.107)	45.5** (21.2)	121.5 (78.6)	0.174** (0.076)
<i>Panel B: Individual Treatments</i>						
Search Only	31.7 (28.8) [0.561]	311* (177) [0.183]	.197* (.114) [0.187]	22.1 (25.4) [0.650]	178 (112) [0.270]	.21* (.108) [0.124]
Search + Adverse Selection	43.5 (37.7) [0.561]	80.8 (114) [0.729]	.193 (.129) [0.235]	30.8 (30.7) [0.650]	73.1 (100) [0.695]	.124 (.102) [0.387]
Search + Moral Hazard	15.8 (25.1) [0.562]	-24 (106) [0.829]	.036 (.085) [0.695]	8.21 (23.2) [0.741]	-25.3 (87) [0.786]	6.2e-03 (.078) [0.943]
Search + AS + MH	254*** (89.3) [0.004]	636** (267) [0.031]	.88*** (.315) [0.005]	128*** (39.2) [0.002]	269** (124) [0.084]	.371*** (.137) [0.018]
Control Mean	188.3	609.5	0.000	188.3	609.5	0.000
% Increase (Pooled)	43.8%	40.2%	N/A	24.2%	19.9%	N/A
All Coefs Equal p -val	0.051	0.041	0.041	0.024	0.054	0.027
Adjusted R^2	0.13	0.12	-0.01	0.25	0.29	-0.02
N	1351	1378	1298	1351	1378	1298

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is total profit from the past 30 days in USD. Column 2 is total sales from the past 30 days in USD. Column 3 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Column 4 is total profit from the past 30 days in USD, winsorizing the top 1%. Column 5 is total sales from the past 30 days in USD, winsorizing the top 1%. Column 6 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Profit is measured using the survey question from [De Mel, McKenzie, and Woodruff \(2009\)](#). Sales is measured using a similar survey question.

Table 5: Effect of Alibaba Training

	Heard of Alibaba (1)	Searched on Alibaba (2)	Compared Prices with Supplier (3)	Bought on Alibaba (4)
e-Commerce Treatment	0.065** (0.025)	0.113** (0.052)	0.087* (0.049)	0.014 (0.035)
Control Mean	0.908	0.423	0.319	0.135
% Increase	7.2%	26.8%	27.4%	10.7%
Adjusted R^2	0.07	0.09	0.14	0.22
N	340	340	340	340

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. Conventional robust standard errors are reported in parentheses. This table shows the effect of the Alibaba training treatment on Alibaba usage.

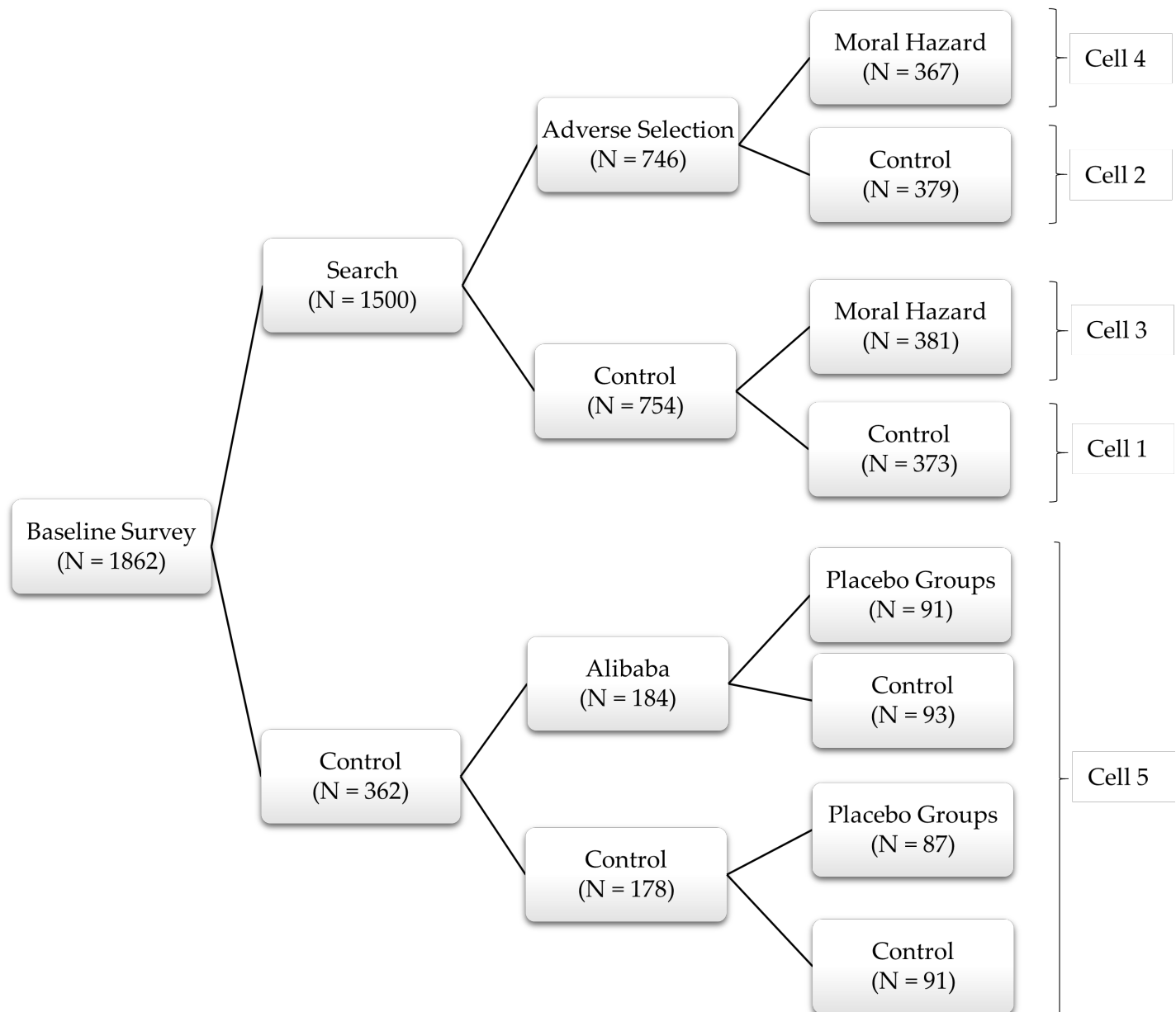
Table 6: Estimated Structural Parameters

Description	Parameter	Estimate
Share of bad types	μ_0	0.57 (0.09)
Moral hazard cost saving	ξ	0.91 (0.16)
Sequential search cost (USD)	s	94 (156)
Match productivity (mean)	ψ_μ	-6.26 (0.89)
Match productivity (std dev)	ψ_σ	1.66 (0.44)
Discount factor	δ	0.78 (0.11)
Bad type quality probability	λ	0.30 (0.05)

Notes: This table shows the estimated structural parameters. We estimate the parameters using Simulated Method of Moments, where we simulate the treatment in the model and match the model-implied treatment effects to the experimental treatment effects on profit, whether the firm has a supplier in Turkey, and orders from mobile money. We weight the moments by the inverse of the empirical variance-covariance matrix, and run the optimiser on 50 randomized starting points. We present bootstrapped standard errors in parentheses, which we calculate as the standard deviation of the results of the estimation procedure run on 200 sets of treatment effects obtained by bootstrapping the experimental data. The parameters for the Lognormal match-specific productivity distribution are for the underlying Normal. Please see Appendix E for more detail on the procedure to solve and estimate the model.

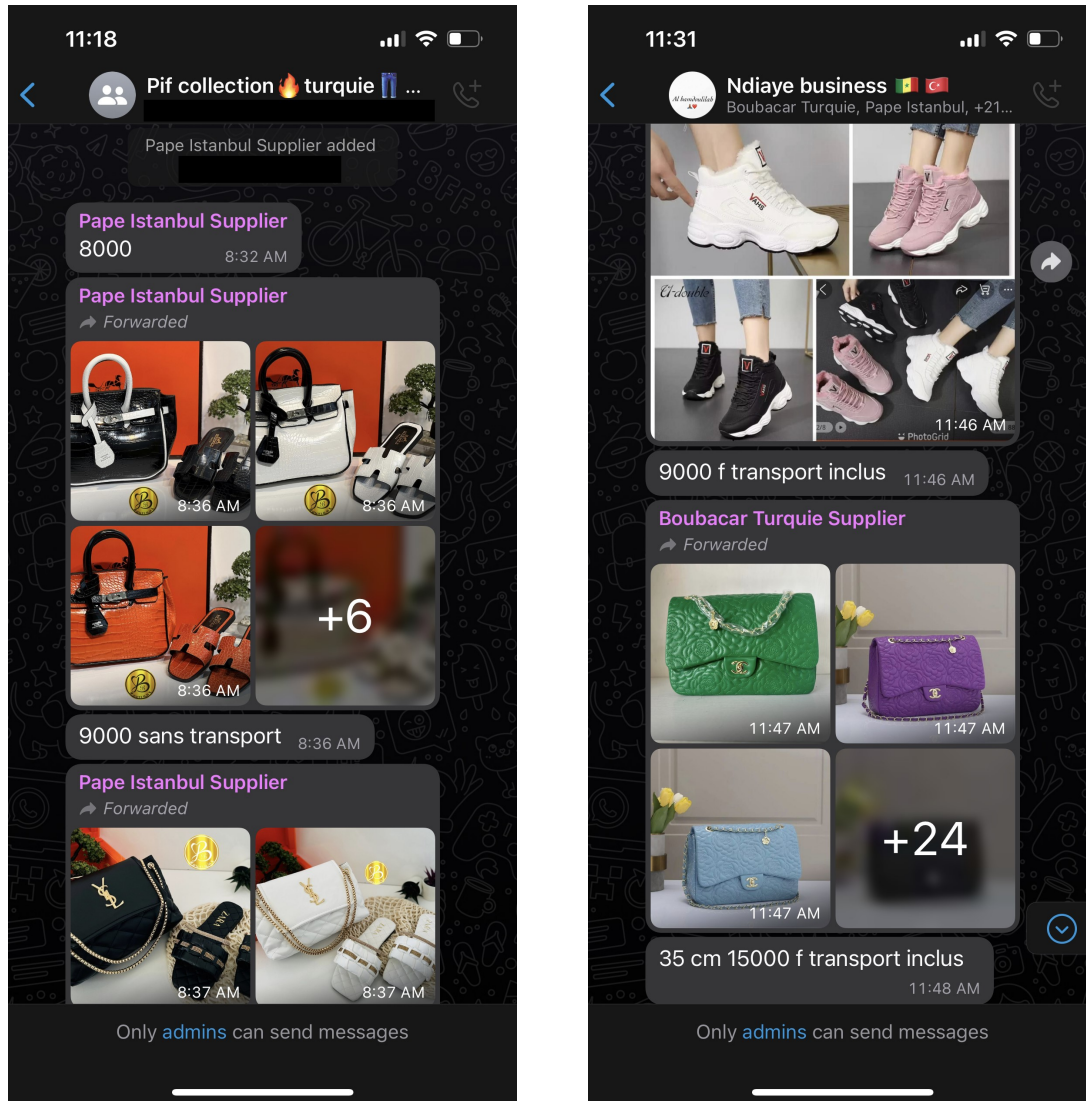
Figures

Figure 1: Experimental Design Tree



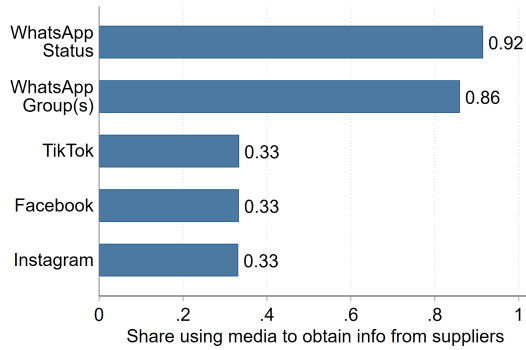
Note: This figure presents the design tree of the experiment, which identifies three frictions—search, adverse selection, and moral hazard—via randomized variation in information and messaging. Treatments are described in Section 4.1. Among recruited firms, 80% of them are randomly assigned to the Search treatment, in which they are matched via WhatsApp groups to three suppliers of Turkish-made goods. Within the Search group, 50% are cross-randomized into the Adverse Selection treatment, where they are placed in peer groups to share supplier experiences and receive seeded information from independent recommenders. Separately, 50% of firms in the Search group are cross-randomized into the Moral Hazard treatment, in which firms are told that suppliers are rated and subject to removal, increasing perceived incentives for supplier effort. Two additional sub-treatments—Alibaba training and placebo peer groups—are cross-randomized within the pure control group to test secondary hypotheses within the control group. The first provides a brief Alibaba training to assess whether limited platform knowledge constrains B2B e-commerce adoption. The second creates peer WhatsApp groups without supplier matches to test whether connecting firms alone drives outcomes, serving as a placebo for the Adverse Selection treatment. Neither sub-treatment yields meaningful effects.

Figure 2: Supplier WhatsApp Groups

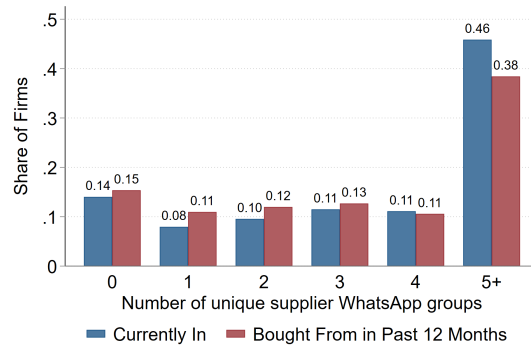


Note: These pictures show two typical supplier WhatsApp groups used in our study context, as described in Section 2.3. These groups function as virtual storefronts: the supplier regularly posts photos or videos showcasing available inventory, often with prices and product details. Clients—usually 50 to 100 per group—observe the posts but can rarely interact publicly; instead, they inquire or negotiate privately with the supplier via direct messages. Such groups could help reduce search costs by giving buyers access to a steady stream of updated product information and facilitating seamless communication. They might also reduce trust frictions to a lesser extent, as visible group membership and repeated engagement raise the reputational costs of cheating.

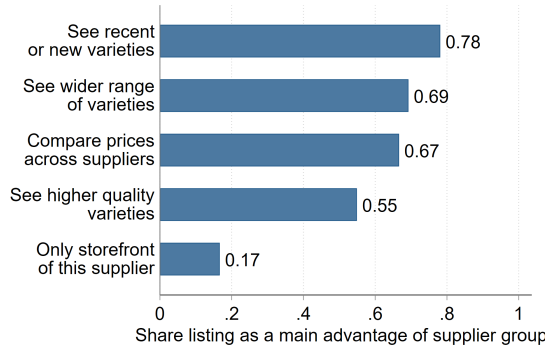
Figure 3: Firm Social Media Usage to obtain Information about Suppliers



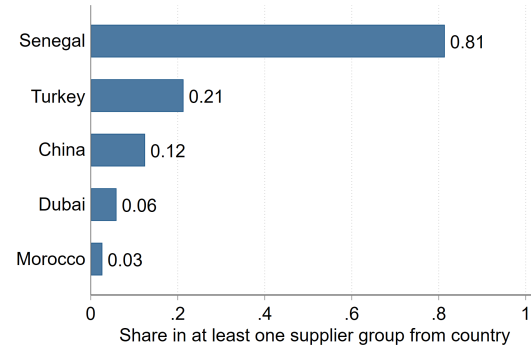
(a) Social Media Usage



(b) Supplier WhatsApp Groups



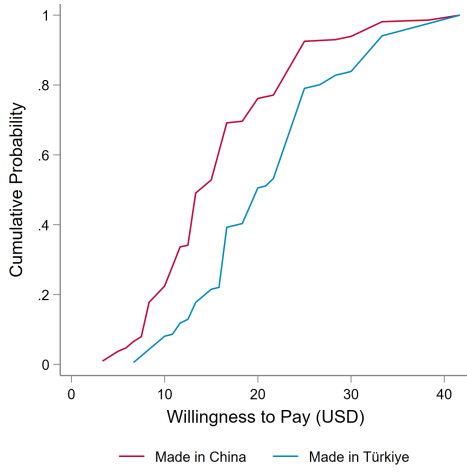
(c) Advantages of Supplier Groups



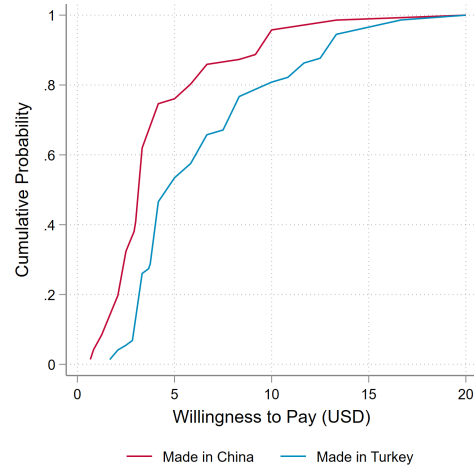
(d) Supplier Groups, by Supplier Country

Note: This figure shows a number of statistics about how firms in our sample use social media to obtain information about suppliers. All data is from our baseline survey with 1,862 firms. Panel (a) shows the results of a question asking firms to select all social media that they use to obtain information about suppliers for their business. Panel (b) shows the distribution of the number of supplier WhatsApp groups a firm is in at the time of the baseline survey, as well as the distribution of the number of such groups that the firm has directly made at least one purchase from in the past 12 months. Supplier WhatsApp groups are defined as WhatsApp groups in which the primary purpose is for suppliers to advertise their wares to downstream clients. Panel (c) shows the results of a question asking firms that use supplier WhatsApp groups to select all reasons why they find these groups useful. Panel (d) shows the share of firms who are in at least one supplier WhatsApp group where the supplier located and based in the country listed.

Figure 4: Willingness to Pay by Product Origin



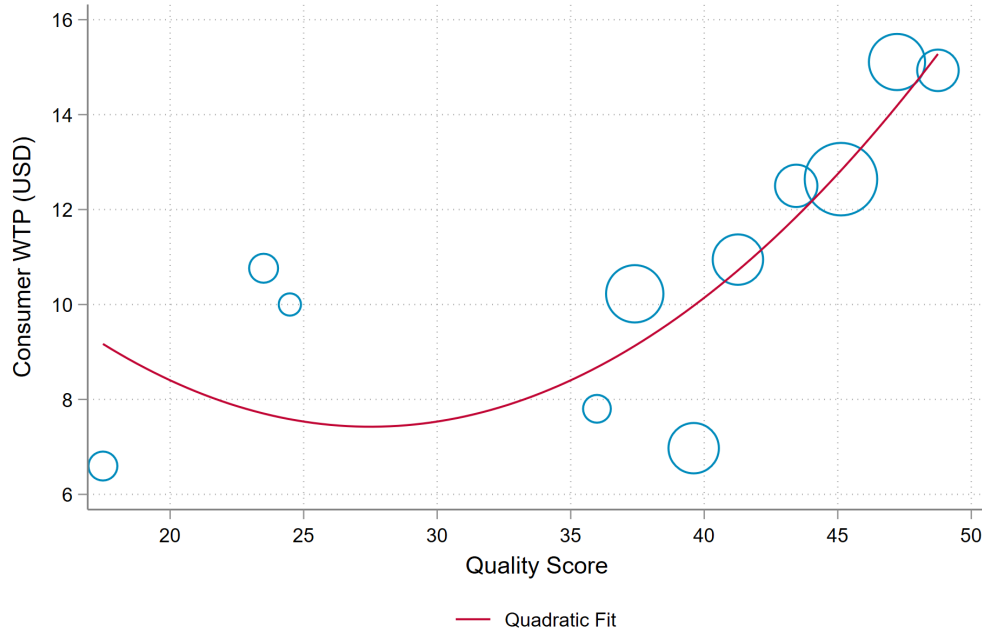
(a) Consumer Survey ($N = 400$)



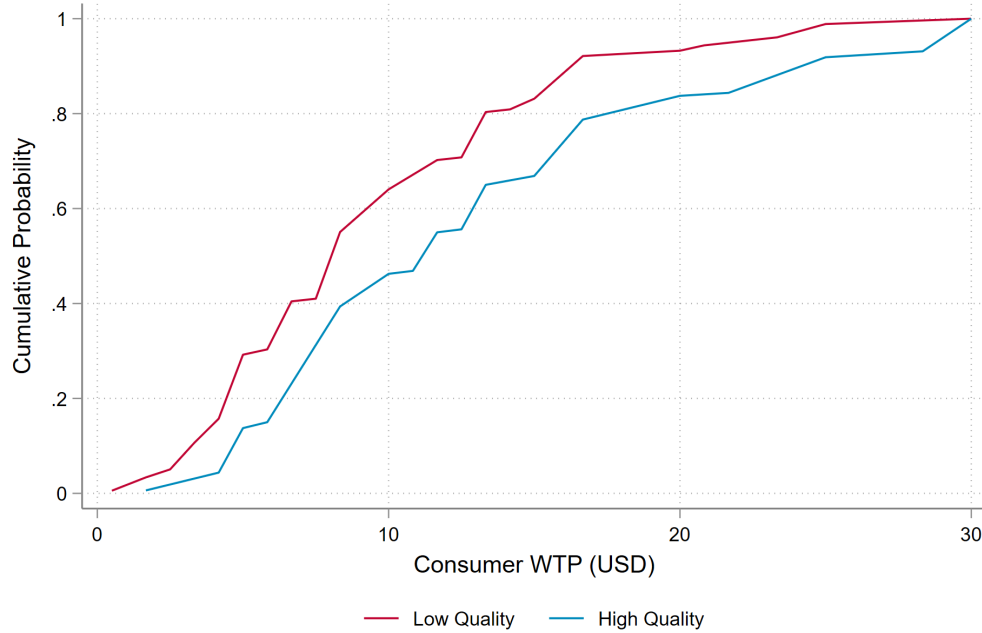
(b) Firm Survey ($N = 144$)

Note: This figure shows the results of two exercises in which surveyors showed various images of garments to respondents, randomising whether they stated that the good was made in Türkiye or made in China, and elicited willingness to pay for the garments. Panel (a) shows the CDF of WTP in the consumer survey, separately by whether the surveyor stated the good was made in Turkey or China. The distribution is truncated at 40 USD for ease of readability. Panel (b) shows the CDF of WTP for a small, separate survey of firms (for a different set of goods), with distribution truncated at 20 USD for ease of readability.

Figure 5: Consumer Willingness to Pay for Quality



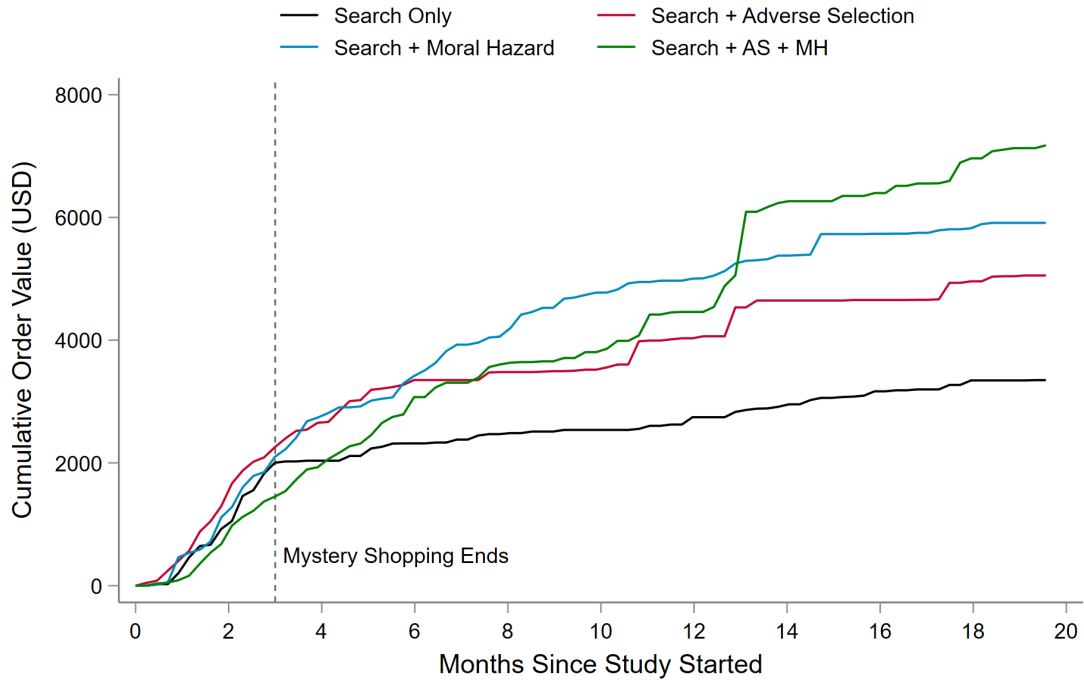
(a) Raw Score



(b) CDF of WTP by High or Low Quality

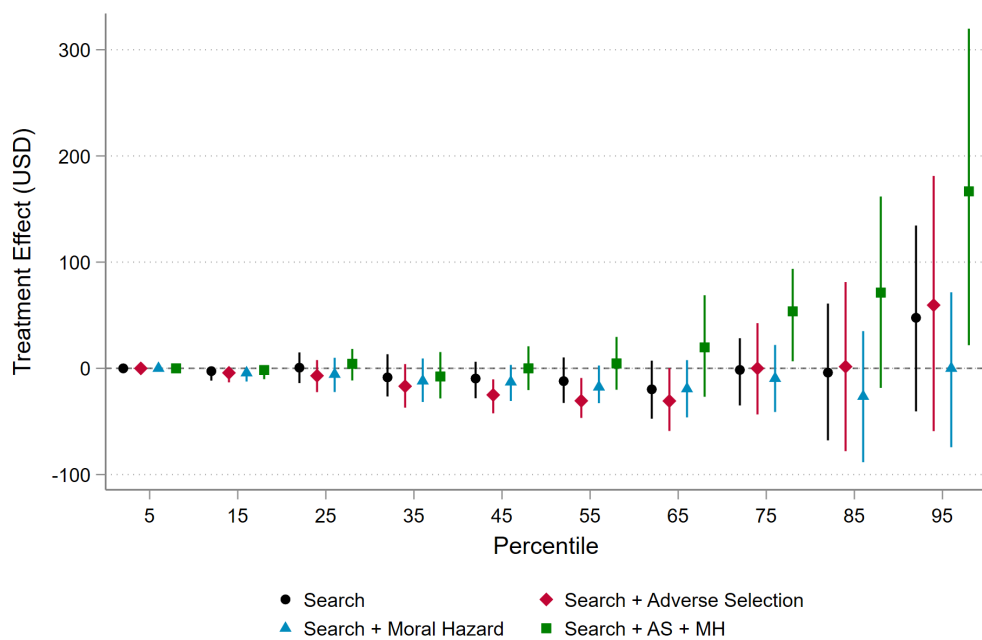
Note: Panel (a) shows a binscatter of consumer willingness to pay for garments (as measured by the consumer survey) against the quality score of the garments. The size of each bubble is proportional to the number of observations. Panel (b) shows the CDF of consumer willingness to pay separately based on whether the garment met our definition of high quality. We truncate willingness to pay at 30 USD to avoid unnecessarily stretching the x-axis. See the main text for full details on the consumer survey and variable construction.

Figure 6: Cumulative Order Value (Mobile Money Data)

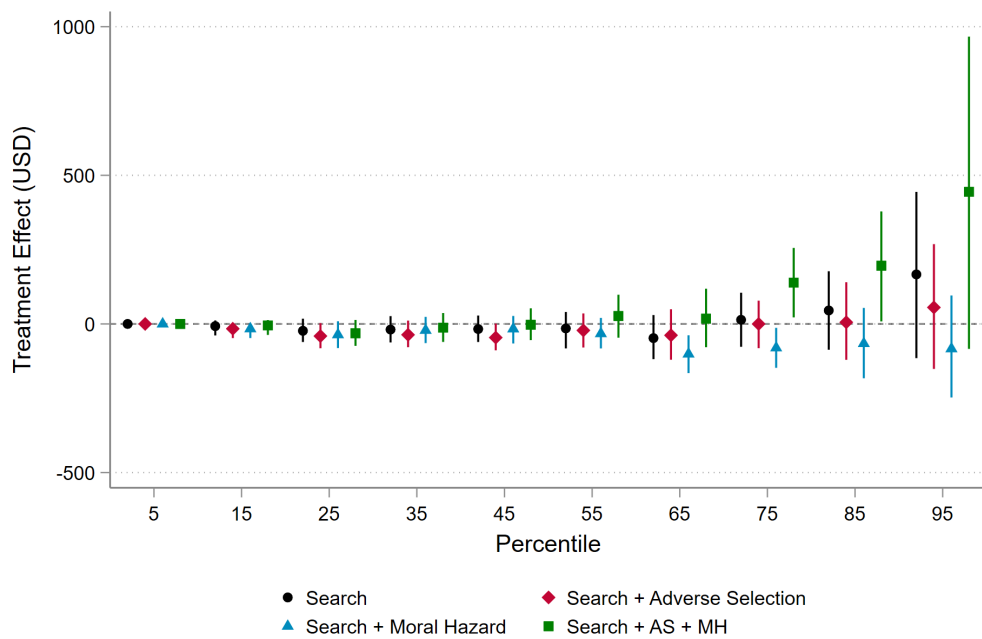


Note: This figure shows the total order value from study suppliers, according to the mobile money data, in each treatment group as a function of number of months since the study begun (16 November 2023). As the group sizes are slightly different due to integer indivisibility issues, we normalize group-level totals by N_j / N_{avg} , where N_j is the number of units in treatment group j and N_{avg} is the average number of units across all four treatment groups. Pure control is omitted as they were not connected to any study suppliers.

Figure 7: Quantile Treatment Effects



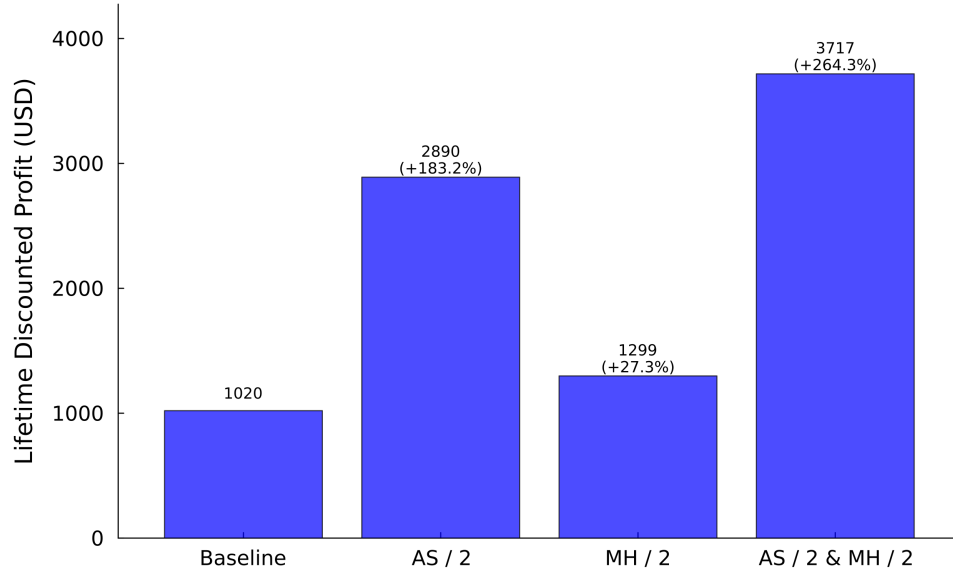
(a) Profit



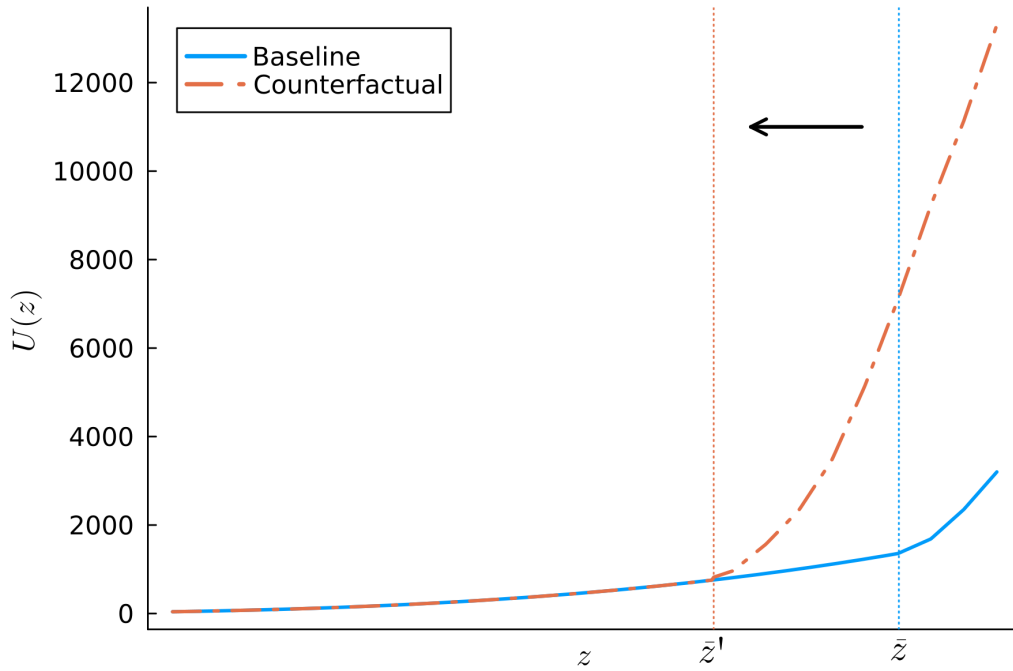
(b) Sales

Note: This figure shows the coefficients from quantile regressions of profit and sales on the four treatment groups. All quantile regressions include the outcome measured at baseline (if available), but otherwise do not include any covariates. We plot 95% confidence intervals constructed using randomisation inference, defined as the set of sharp nulls that do not reject at the 5% level.

Figure 8: Counterfactual Analysis



(a) Counterfactuals



(b) Importing productivity cutoff, $\bar{z}$, shifts left when adverse selection is lowered

Note: Panel (a) shows counterfactuals alleviating combinations of the frictions. The first column is the baseline scenario. The second column reduces the share of bad types, μ_0 , which governs the size of adverse selection, by half. The third column reduces ξ , the parameter governing how much a supplier can reduce the marginal cost by if they choose low effort, which scales moral hazard, by half. Finally, the last column reduces both μ_0 and ξ by half each. All values are expressed in the expected present discounted value of lifetime profits for the firm. Panel (b) shows firm discounted profits, $U(z)$, as a function of productivity, z , in both the Baseline and the AS / 2 scenarios. The vertical lines show the productivity cutoff, $\bar{z}$, in the two scenarios.

Appendix A - Additional Tables and Figures

Tables

Table A1: Balance Table

	Control	Search	Search AS	Search MH	Search AS MH	Joint <i>p</i> -value
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Female	0.49	0.51	0.49	0.53	0.49	0.7
Online Only	0.67	0.66	0.68	0.7	0.65	0.02
Business Age	4.85	4.74	5.2	4.72	5.21	0.48
Share Cust Turkey	0.43	0.45	0.45	0.44	0.45	0.95
Any Reg Supp Turkey	0.22	0.21	0.24	0.21	0.19	0.53
Travelled Business (5y)	0.09	0.08	0.11	0.09	0.11	0.41
Profit USD (30 Days)	221.35	221.08	262.49	195.36	235.63	0.45
Bought Alibaba Ever	0.16	0.13	0.15	0.18	0.13	0.23
<i>N</i>	362	373	379	381	367	

Note: The table shows the mean for each variable in each of the five treatment cells. The final column shows the *p*-value from regressing the variable on indicators for each treatment (where the control group is omitted) and conducting a test that all coefficients are zero. Finally, we run a multinomial logit of treatment group against all of the variables in the table, for which a joint test that all coefficients are zero has a *p*-value of 0.595.

Table A2: One Week Survey

	Indicator for Answers 1 or 2 (out of 5)	
	Find Products From Turkey (1)	Trust Supplier (2)
<i>Panel A: Pooled</i>		
Treatment	0.110*** (0.029)	0.021 (0.030)
<i>Panel B: Individual Treatments</i>		
Search Only	0.078** (0.038) [0.048]	
Search + Adverse Selection	0.140*** (0.037) [0.001]	0.009 (0.036) [0.795]
Search + Moral Hazard	0.087** (0.037) [0.048]	-0.025 (0.036) [0.686]
Search + AS + MH	0.136*** (0.038) [0.001]	0.078** (0.038) [0.087]
Control Mean	0.453	0.303
% Increase (Pooled)	24.3%	6.9%
All Coefs Equal p -val	0.237	N/A
All Coefs Zero p -val	N/A	0.041
Adjusted R^2	0.06	0.06
N	1636	1201

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is an indicator for whether they can find Turkish products. Column 2 is 1 if they reported a trust level 1 or 2, 0 if they reported a trust level of 3 or 4 or 5, and missing otherwise.

Table A3: Horizontal Outcomes (Detailed)

	Extensive vs Intensive Margin		Number of Criteria	
	Agree Search (1)	Find Product Conditional (2)	Num Criteria Unconditional (3)	Num Criteria Conditional (4)
<i>Panel A: Pooled</i>				
Treatment	0.029 (0.025)	0.101*** (0.035)	0.361*** (0.124)	0.357** (0.140)
<i>Panel B: Individual Treatments</i>				
Search Only	0.029 (0.031) [0.659]	0.144*** (0.043) [0.002]	0.664*** (0.161) [0.001]	0.705*** (0.178) [0.000]
Search + Adverse Selection	0.014 (0.031) [0.659]	0.067 (0.044) [0.126]	0.196 (0.160) [0.313]	0.182 (0.182) [0.324]
Search + Moral Hazard	0.045 (0.030) [0.349]	0.099** (0.043) [0.053]	0.379** (0.158) [0.040]	0.308* (0.175) [0.192]
Search + AS + MH	0.025 (0.032) [0.659]	0.092** (0.044) [0.064]	0.204 (0.158) [0.313]	0.232 (0.179) [0.324]
Control Mean	0.781	0.457	1.650	2.111
% Increase (Pooled)	3.7%	22.1%	21.9%	16.9%
All Coefs Equal p -val	0.741	0.340	0.016	0.020
Adjusted R^2	0.10	0.04	0.05	0.03
N	1579	1269	1579	1269

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm agrees to sell or search for the product, and is missing if the firm never replied to the mystery shopper or was otherwise unreachable. Column 2 is 1 if the firm has a suitable product, conditional on agreeing to sell or search for the product. Column 3 is the number of horizontal criteria of the product, and is 0 if the firm either did not agree to sell or search for a product or agreed but never sent any product. Column 4 is the number of horizontal criteria of the product, conditional on agreeing to sell or search for a product, and is 0 if the firm agreed but never sent any product.

Table A4: Vertical Outcomes (Detailed)

	From Turkey		
	Made in Turkey (Label) (1)	Made in Turkey (Tailor Judgement) (2)	Made in Turkey (Label + Tailors) (3)
<i>Panel A: Pooled</i>			
Treatment	0.210** (0.081)	0.099 (0.081)	0.159** (0.072)
<i>Panel B: Individual Treatments</i>			
Search Only	0.169* (0.096) [0.141]	0.134 (0.095) [0.437]	0.138 (0.086) [0.195]
Search + Adverse Selection	0.172* (0.100) [0.141]	0.010 (0.102) [0.927]	0.107 (0.089) [0.221]
Search + Moral Hazard	0.241** (0.094) [0.038]	0.132 (0.095) [0.437]	0.192** (0.084) [0.087]
Search + AS + MH	0.271*** (0.097) [0.028]	0.096 (0.097) [0.515]	0.189** (0.084) [0.087]
Control Mean	0.489	0.581	0.544
% Increase (Pooled)	42.9%	17.0%	29.2%
All Coefs Equal p -val	0.538	0.489	0.625
Adjusted R^2	0.10	-0.00	0.07
N	255	287	330

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the label says , 0 if the label says for X other than Turkey, and missing otherwise. Column 2 is 1 if both tailors independently determined that the product was made in Turkey, and is 0 if both tailors independently determined that the product was not made in Turkey. It is missing if the tailors disagreed. For shoes, as there was only one expert shoemaker, we take their opinion directly. Column 3 is an indicator that combines the label and tailor measures of whether the good was made in Turkey. It is equal to the label measure where available, and the tailor measure otherwise.

Table A5: Access to Foreign Goods (Product Fixed Effects)

	Horizontal	Vertical				Price
	Sold High Quality (1)	High Qual Dummy (2)	Qual Score (/50) (3)	Made in Turkey (4)	Index (5)	Price (USD) (6)
<i>Panel A: Pooled</i>						
Treatment	0.053** (0.022)	0.097 (0.070)	-0.391 (0.562)	0.119* (0.062)	0.308** (0.137)	0.594 (0.711)
<i>Panel B: Individual Treatments</i>						
Search Only	0.084*** (0.031) [0.025]	0.164* (0.085) [0.148]	0.259 (0.675) [0.899]	0.119 (0.078) [0.320]	0.398** (0.159) [0.058]	1.054 (0.872) [0.569]
Search + Adverse Selection	0.046 (0.030) [0.202]	0.181** (0.091) [0.148]	-0.027 (0.858) [0.979]	0.084 (0.080) [0.320]	0.358** (0.169) [0.095]	0.829 (0.819) [0.621]
Search + Moral Hazard	0.050* (0.030) [0.202]	0.057 (0.089) [0.750]	-1.179 (0.789) [0.379]	0.121 (0.076) [0.320]	0.266 (0.167) [0.204]	0.655 (0.884) [0.682]
Search + AS + MH	0.032 (0.029) [0.264]	0.002 (0.092) [0.976]	-0.717 (0.814) [0.698]	0.126 (0.080) [0.320]	0.210 (0.181) [0.255]	-0.214 (0.834) [0.789]
Control Mean	0.103	0.438	43.064	0.484	0.000	20.031
% Increase (Pooled)	51.5%	22.1%	-0.9%	24.6%	N/A	3.0%
All Coefs Equal p -val	0.454	0.141	0.309	0.953	0.664	0.350
Adjusted R^2	0.04	0.06	0.34	0.25	0.01	0.45
N	1244	356	359	358	356	641

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm sold a suitable high quality good, and 0 otherwise (including if the firm did not have a suitable good). Column 2 is an indicator for whether the good's quality score is above the median product-group quality score. Column 3 is the raw quality score. Column 4 is an indicator for whether the good is made in Turkey, primarily inferred based on the label. See the text for full details of how this outcome is constructed. Column 5 is the [Anderson \(2008\)](#) index combining the vertical outcomes to account for multiple hypothesis testing across outcomes. Column 6 is the price in USD, which is only measured conditional on the firm finding a good matching at least three horizontal criteria. All columns from 2 onwards include fixed effects for each of the 28 products.

Table A6: Supplier Relationships (Further Results on Substitution)

	Reg Supp in China		Media for Suppliers			Forward Media	
	Any Supp in China	Num Supp in China	Uses Facebook	Uses TikTok	Uses Instagram	Fwd Photo for Search	Fwd Photo for Price
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A: Pooled</i>							
Treatment	-0.025 (0.016)	-0.014 (0.035)	-0.084*** (0.027)	-0.025 (0.025)	-0.027 (0.026)	-0.054** (0.027)	-0.097*** (0.030)
<i>Panel B: Individual Treatments</i>							
Search Only	-0.030 (0.019) [0.335]	-0.023 (0.040) [0.929]	-0.090*** (0.033) [0.024]	-0.047 (0.032) [0.386]	-0.037 (0.032) [0.592]	-0.054 (0.036) [0.230]	-0.103*** (0.039) [0.019]
Search + Adverse Selection	-0.029 (0.019) [0.335]	-0.028 (0.045) [0.929]	-0.063* (0.034) [0.076]	-0.006 (0.032) [0.924]	-0.038 (0.033) [0.592]	-0.033 (0.035) [0.337]	-0.088** (0.038) [0.035]
Search + Moral Hazard	-0.018 (0.020) [0.424]	0.009 (0.044) [0.946]	-0.080** (0.033) [0.037]	-0.039 (0.032) [0.467]	-0.016 (0.033) [0.836]	-0.063* (0.035) [0.204]	-0.081** (0.038) [0.035]
Search + AS + MH	-0.023 (0.020) [0.424]	-0.013 (0.045) [0.946]	-0.103*** (0.033) [0.007]	-0.011 (0.033) [0.924]	-0.012 (0.033) [0.836]	-0.058* (0.035) [0.230]	-0.109*** (0.039) [0.018]
Control Mean	0.099	0.167	0.328	0.290	0.279	0.716	0.659
% Increase (Pooled)	-25.3%	-8.4%	-25.6%	-8.6%	-9.7%	-7.5%	-14.7%
All Coefs Equal p -val	0.910	0.840	0.630	0.515	0.774	0.837	0.889
Adjusted R^2	0.12	0.36	0.09	0.12	0.12	0.04	0.03
N	1680	1680	1671	1671	1671	1671	1565

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm says that they have a regular supplier in China. Column 2 is 1 if the firm says that they have a regular supplier in China. Column 3 is 1 if the firm says that they use Facebook to learn about suppliers. Column 4 is 1 if the firm says that they use TikTok to learn about suppliers. Column 5 is 1 if the firm says that they use Instagram to learn about suppliers. Column 6 is 1 if the firm says that they have forwarded a photo or video from a supplier group to a regular supplier to try to obtain a similar product in the past 3 months. Column 7 is 1 if the firm says that they have forwarded a photo or video from a supplier group to a regular supplier to try to obtain a better price in the past 3 months. A regular supplier is defined as a supplier from whom the firm has made two or more orders with an intention of continuing the relationship.

Table A7: Order Value (Mobile Money Data) – Poisson Regressions

	Post Mystery Shopping			Entire Period		
	Total Order Value (USD)	Num Orders	Avg Order Value (USD)	Total Order Value (USD)	Num Orders	Avg Order Value (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Pooled</i>						
Trust Treatment	1.110 (0.549)	0.291 (0.417)	1.275*** (0.356)	0.589 (0.355)	0.092 (0.253)	0.453* (0.215)
<i>Panel B: Individual Treatments</i>						
Search + Adverse Selection	0.695 (0.515) [0.442]	-0.147 (0.541) [0.889]	1.524*** (0.479) [0.016]	0.412 (0.314) [0.548]	0.039 (0.305) [0.988]	0.590* (0.290) [0.125]
Search + Moral Hazard	1.051* (0.493) [0.235]	0.575 (0.451) [0.564]	1.445*** (0.421) [0.010]	0.568 (0.325) [0.340]	0.173 (0.303) [0.904]	0.670** (0.274) [0.073]
Search + AS + MH	1.455 (0.900) [0.442]	0.312 (0.656) [0.889]	0.612 (0.426) [0.225]	0.762 (0.711) [0.548]	0.052 (0.400) [0.988]	-0.064 (0.245) [0.807]
Control Mean	3.56	0.14	0.89	8.93	0.33	3.65
% Increase (Pooled)	203.4%	33.8%	257.9%	80.2%	9.6%	57.3%
All Coefs Zero p -val	0.353	0.599	0.013	0.471	0.961	0.032
Adjusted R^2	0.03	0.03	0.04	0.01	0.04	0.06
N	1500	1500	1500	1500	1500	1500

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all trust treated groups, where Search Only is the omitted category. Panel B shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is the total value of orders from study suppliers after the mystery shopping period finished. Column 2 is the number of orders from study suppliers. Column 3 is the average order size from study suppliers after the mystery shopping period finished. Column 4 is the total value of orders from study suppliers over the entire period. Column 5 is the number of orders from study suppliers over the entire period. Column 6 is the average order size from study suppliers over the entire period. Mystery shopping took place during the first 3 months of the study, from 16 November 2023 to 22 February 2024. The entire period covered by the data comprises 18 months, from 16 November 2023 to 31 May 2025. All values are in USD.

Table A8: Profit and Sales (Poisson Regression)

	Profit		Sales	
	Profit 30 Days	Profit 30 Days	Sales 30 Days	Sales 30 Days
	(USD)	Winsorized 1% (USD)	(USD)	Winsorized 1% (USD)
	(1)	(2)	(3)	(4)
<i>Panel A: Pooled</i>				
Treatment	0.307*	0.188	0.217	0.118
	(0.132)	(0.112)	(0.142)	(0.126)
<i>Panel B: Individual Treatments</i>				
Search Only	0.088	0.074	0.315	0.198
	(0.147)	(0.133)	(0.210)	(0.158)
	[0.916]	[0.826]	[0.431]	[0.511]
Search + Adverse Selection	0.097	0.119	-0.041	-0.026
	(0.189)	(0.162)	(0.167)	(0.159)
	[0.916]	[0.826]	[0.830]	[0.876]
Search + Moral Hazard	-0.047	-0.009	-0.151	-0.147
	(0.138)	(0.143)	(0.174)	(0.143)
	[0.916]	[0.950]	[0.683]	[0.546]
Search + AS + MH	0.865***	0.480***	0.475**	0.260
	(0.214)	(0.142)	(0.212)	(0.162)
	[0.001]	[0.007]	[0.159]	[0.396]
Control Mean	188.3	188.3	609.5	609.5
% Increase (Pooled)	35.9%	20.7%	24.2%	12.5%
All Coefs Equal p -val	0.009	0.020	0.067	0.057
Adjusted R^2	0.34	0.38	0.46	0.48
N	1351	1351	1378	1378

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is total profit from the past 30 days in USD. Column 2 is total profit from the past 30 days in USD, winsorizing the top 1%. Column 3 is total sales from the past 30 days in USD. Column 4 is total sales from the past 30 days in USD, winsorizing the top 1%. Profit is measured using the survey question from [De Mel, McKenzie, and Woodruff \(2009\)](#). Sales is measured using a similar survey question.

Table A9: Travel

	Wholesalers			Retailers		
	Any Travel	Travel China	Travel Turkey	Any Travel	Travel China	Travel Turkey
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Pooled</i>						
Treatment	-0.033 (0.036)	-0.035 (0.030)	-0.020 (0.019)	-0.004 (0.014)	-0.021* (0.011)	0.003 (0.005)
<i>Panel B: Individual Treatments</i>						
Search Only	-0.001 (0.049) [0.980]	-0.009 (0.039) [0.936]	-0.020 (0.026) [0.856]	-0.005 (0.017) [0.984]	-0.022 (0.013) [0.272]	-0.004 (0.004) [0.692]
Search + Adverse Selection	0.007 (0.041) [0.980]	-0.011 (0.034) [0.936]	-0.012 (0.022) [0.856]	-0.001 (0.018) [0.997]	-0.025** (0.012) [0.126]	0.004 (0.008) [0.813]
Search + Moral Hazard	-0.051 (0.038) [0.425]	-0.050* (0.030) [0.272]	-0.014 (0.020) [0.856]	-0.008 (0.017) [0.977]	-0.015 (0.014) [0.298]	0.010 (0.008) [0.586]
Search + AS + MH	-0.052 (0.037) [0.425]	-0.043 (0.032) [0.399]	-0.016 (0.020) [0.856]	-0.001 (0.020) [0.997]	-0.022 (0.014) [0.272]	0.001 (0.006) [0.865]
Control Mean	0.130	0.090	0.040	0.041	0.033	0.004
% Increase (Pooled)	-25.4%	-38.9%	-50.0%	-9.8%	-63.6%	75.0%
All Coefs Equal p -val	0.168	0.248	0.987	0.986	0.866	0.151
Adjusted R^2	0.18	0.28	0.09	0.04	0.07	0.01
N	546	546	546	1125	1125	1125

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm travelled internationally for business in the past 3 months. Column 2 is 1 if the firm travelled for business to China in the past 3 months. Column 3 is 1 if the firm travelled for business to Turkey in the past 3 months. Column 4 is 1 if the firm travelled internationally for business in the past 3 months. Column 5 is 1 if the firm travelled for business to China in the past 3 months. Column 6 is 1 if the firm travelled for business to Turkey in the past 3 months. Travel is 1 if either the firm owner or someone closely involved with the firm travelled internationally for firm-specific business in the past 3 months.

Table A10: Calibrated Parameters

Parameter	Value	Origin
σ	3.02	Average markup from baseline survey.
p_l	12.29	Average input price from baseline survey among firms without a foreign supplier.

Note: This table presents the values of the parameters that are calibrated in the model.

Table A11: Lee Bounds on Main Survey Outcomes

Outcome	Treatment	Coef	SE	Lee LB	LB SE	Lee UB	UB SE
Any Sup Turkey	Pooled	0.039*	0.021	0.030	0.029	0.089***	0.032
Any Sup Turkey	S Only	0.024	0.027	0.013	0.035	0.062*	0.037
Any Sup Turkey	S+AS	0.051*	0.029	0.044	0.036	0.093**	0.040
Any Sup Turkey	S+MH	0.006	0.027	-0.005	0.034	0.061	0.038
Any Sup Turkey	S+AS+MH	0.075***	0.029	0.068*	0.035	0.143***	0.042
Num Sup Turkey	Pooled	0.085**	0.035	0.073	0.046	0.171***	0.047
Num Sup Turkey	S Only	0.080	0.051	0.064	0.062	0.164***	0.059
Num Sup Turkey	S+AS	0.069	0.046	0.059	0.056	0.138**	0.062
Num Sup Turkey	S+MH	0.002	0.040	-0.012	0.051	0.098**	0.049
Num Sup Turkey	S+AS+MH	0.193***	0.058	0.185***	0.066	0.278***	0.078
Profit (W. 1%)	Pooled	42.346**	21.414	21.431	22.939	69.399*	40.765
Profit (W. 1%)	S Only	19.276	26.129	-16.096	112.056	10.220	30.194
Profit (W. 1%)	S+AS	30.947	31.180	14.094	33.439	57.659	54.219
Profit (W. 1%)	S+MH	-4.981	24.780	-31.444	27.323	26.762	49.715
Profit (W. 1%)	S+AS+MH	131.630***	44.353	103.253**	46.973	189.377***	46.504
Sales (W. 1%)	Pooled	133.128	83.904	60.980	88.451	300.616***	75.961
Sales (W. 1%)	S Only	185.777	123.823	113.197	127.067	328.926**	143.754
Sales (W. 1%)	S+AS	70.475	109.672	16.607	116.684	250.794**	111.430
Sales (W. 1%)	S+MH	-59.735	92.478	-141.587	99.540	108.860	132.227
Sales (W. 1%)	S+AS+MH	349.621**	148.693	268.666*	158.297	558.849***	146.320

Note: This table shows Lee (2009) bounds for the main outcome variables from the survey. Columns 1 and 2 present the main coefficient and standard error. Column 3 presents the Lee lower bound and Column 4 presents its standard error. Column 5 presents the Lee upper bound and Column 6 presents its standard error. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. As Lee bounds are based on bivariate regressions and do not easily permit covariates, we only include the baseline value of the outcome: we use it directly to tighten the bounds if it is binary, and otherwise we convert it to binary by creating an indicator for above or below the median.

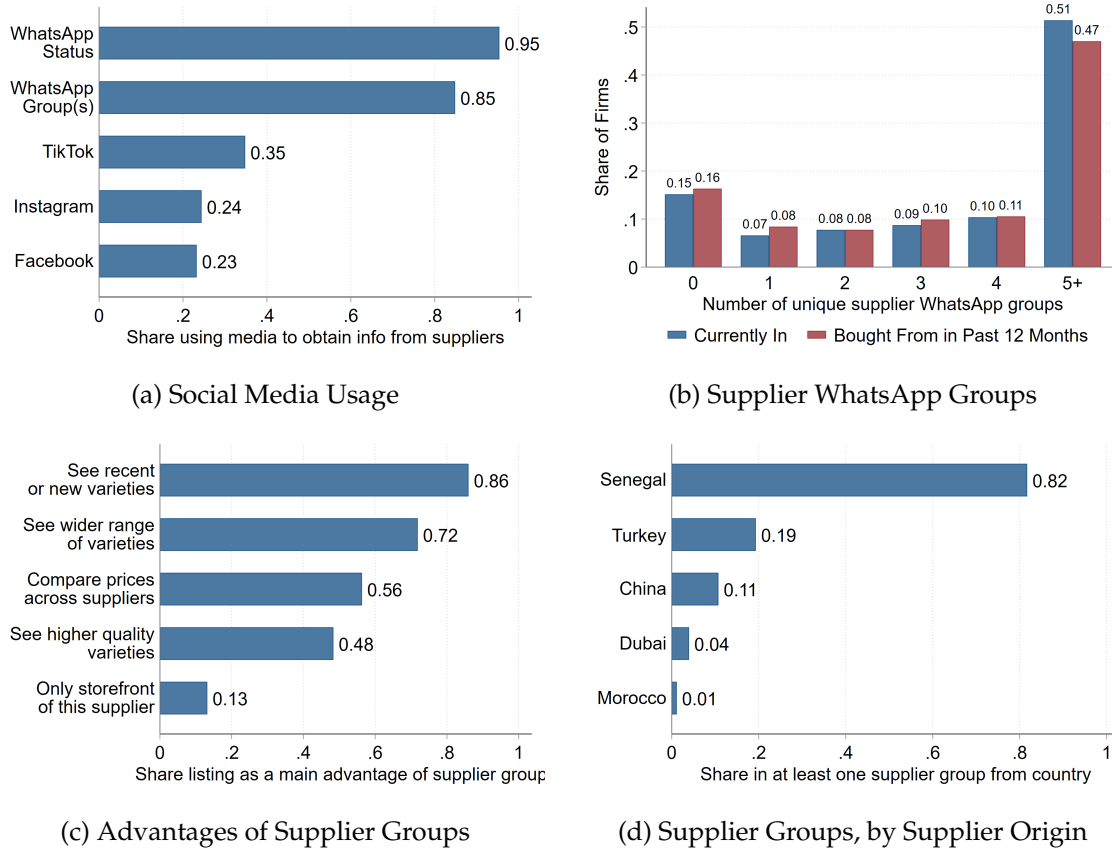
Table A12: Effect of Placebo Firm Groups

	Horizontal Dummy (1)	Vertical Dummy (2)	Num Supp In Turkey (3)	Profit Sales Index (1%) (4)
Placebo Group Treatment	-0.034 (0.054)	-0.013 (0.125)	0.013 (0.063)	0.012 (0.123)
Control Mean	0.374	0.438	0.216	-0.006
% Increase	-9.2%	-3.0%	6.0%	N/A
Adjusted R^2	-0.00	-0.02	-0.00	-0.00
N	311	65	342	266

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. Conventional robust standard errors are reported in parentheses. This table shows the effect of the placebo firm discussion groups on the main outcomes (see Section 4.1 for details). The omitted category is the half of the pure control group that did not receive as this treatment. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 2 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

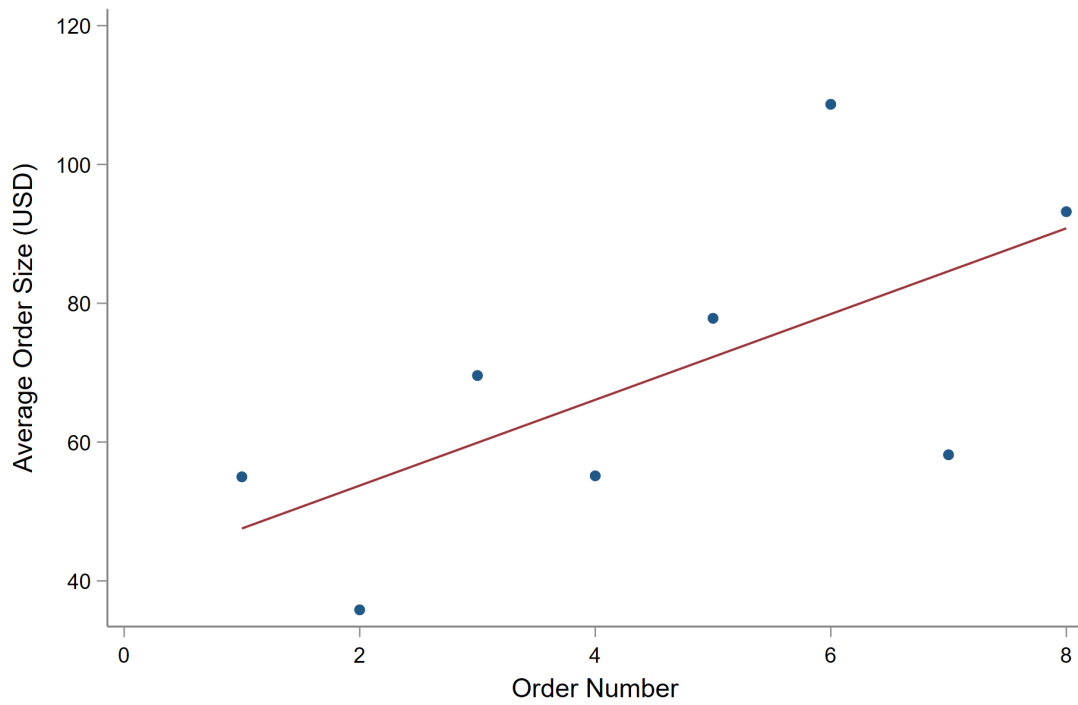
Figures

Figure A1: Social Media Usage (Physical Store Only)



Note: This figure shows a number of statistics about how firms in our sample use social media to obtain information about suppliers. It is the same as Figure 3, but instead calculates statistics only for the 607 firms that have physical stores. Panel (a) shows the results of a question asking firms to select all social media that they use to obtain information about suppliers for their business. Panel (b) shows the distribution of the number of supplier WhatsApp groups a firm is in at the time of the baseline survey, as well as the distribution of the number of such groups that the firm has directly made at least one purchase from in the past 12 months. Supplier WhatsApp groups are defined as WhatsApp groups in which the primary purpose is for suppliers to advertise their wares to downstream clients. Panel (c) shows the results of a question asking firms that use supplier WhatsApp groups to select all reasons why they find these groups useful. Panel (d) shows the share of firms who are in at least one supplier WhatsApp group where the supplier located and based in the country listed.

Figure A2: Average Quantity over Time



Note: This figure shows the mean order value from study suppliers, according to the mobile money data, for the first order that a firm made from a given supplier, then the second order, and so on. As the data only covers payments, and multiple payments may correspond to the same order (e.g., if the firm pays half in advance and half on arrival), we aggregate transfers to the firm-supplier-month level and define the order value as this aggregate. To avoid potential selection issues driven by more optimistic firms being more likely to make both larger and repeat orders, we restrict the analysis to firms who made at least six orders (i.e., made transfers in six separate months) from a given supplier.

Figure A3: Business Cards for Moral Hazard Treatment



Note: This figure shows the business cards used in the Moral Hazard treatment. In the treatment version, the card prominently displays a phone number for firms to rate suppliers or report problems, reinforcing the message that supplier performance will be monitored and can have consequences. In the control version, the card contains the same phone number but refers only to general questions about the study, with no mention of ratings or suppliers.

Figure A4: Mystery Shopping Goods (Examples)



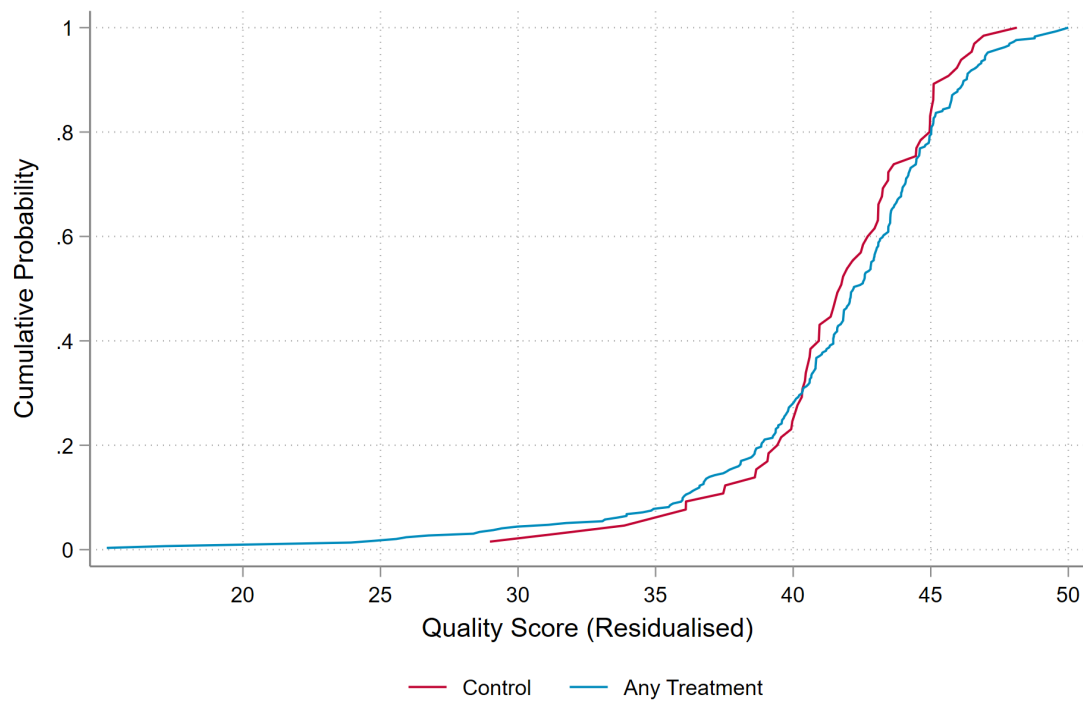
1. Red
2. One colour
3. No large logo/print
4. 3 black buttons
5. Size L



1. Black
2. Long sleeves
3. Knee length
4. Turtleneck
5. Ribbed Cotton

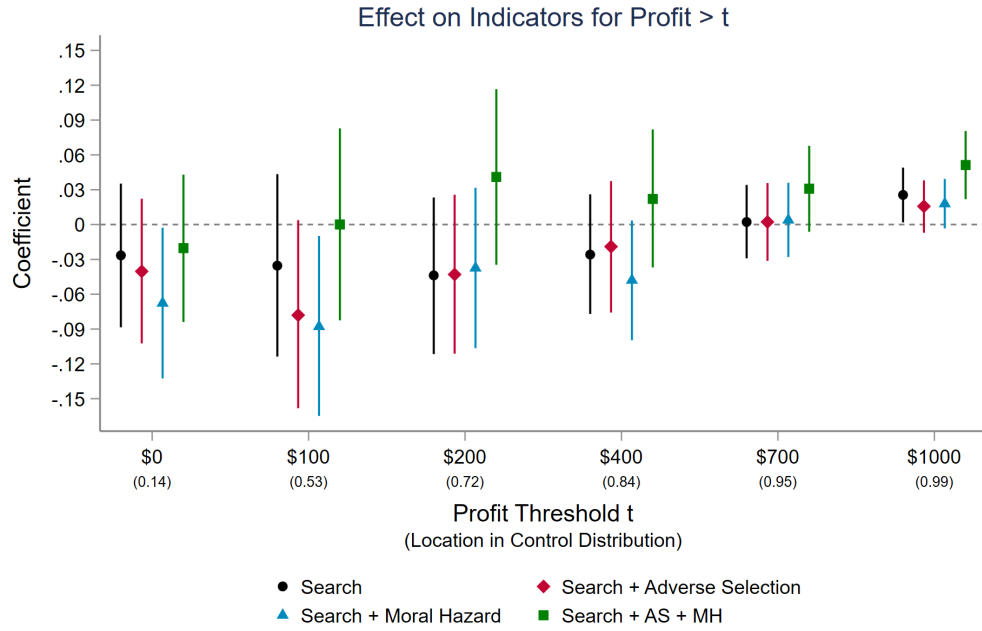
Note: Examples of goods requested in the mystery shopping exercise. In total, there were 28 different goods.

Figure A5: Quality Score Distribution

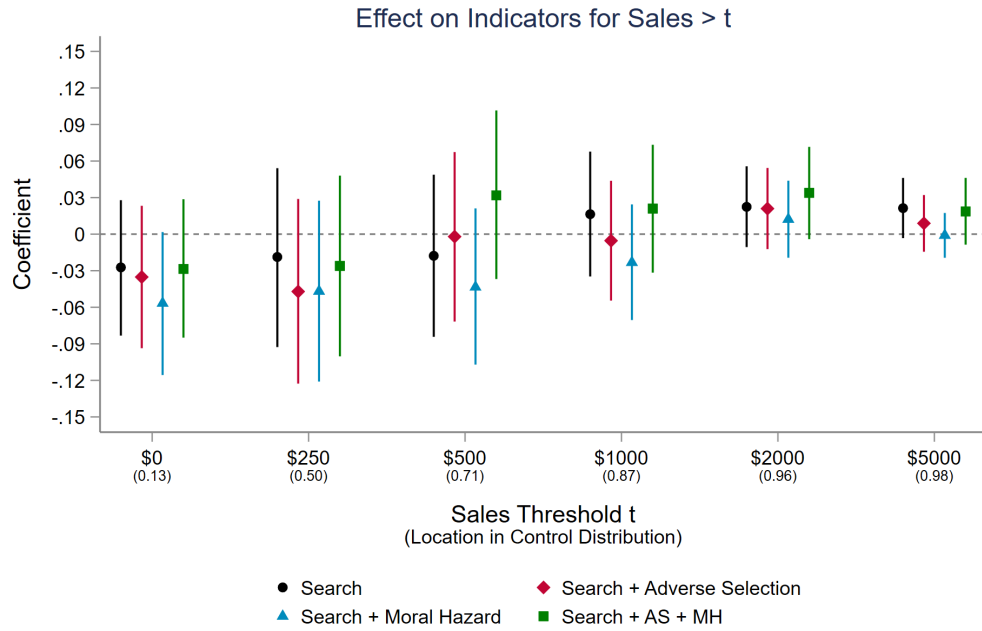


Note: This figure shows CDF of the quality score separately by treatment status, with all four treatment groups (Search Only, Search + Adverse Selection, Search + Moral Hazard, Search + AS + MH) pooled for visual ease. To be consistent with the regression in the table, we first residualise quality using stratum fixed effects and any covariates selected by PDS Lasso in the regression.

Figure A6: Threshold Regressions for Profit and Sales



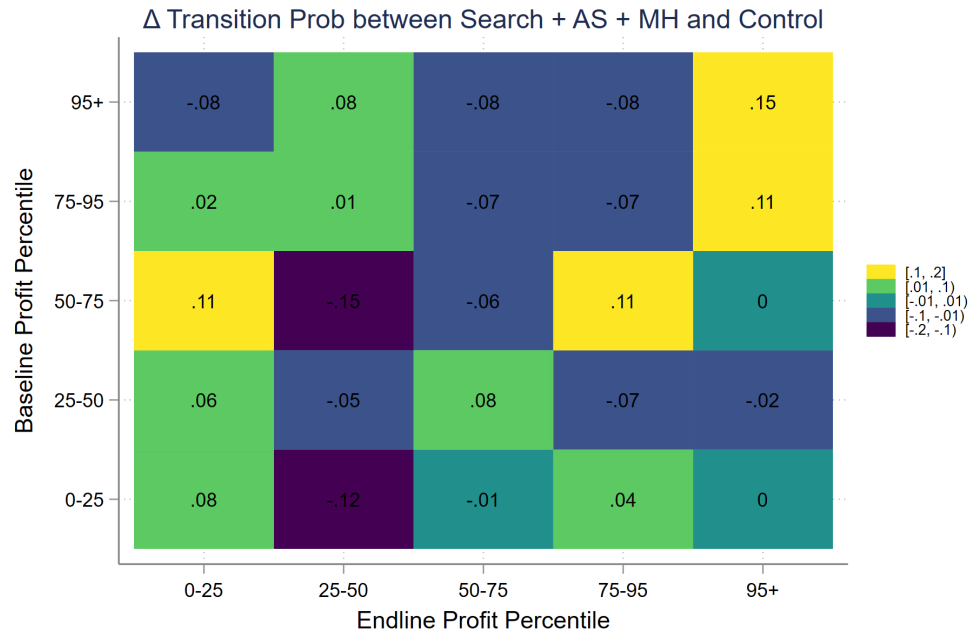
(a) Profit



(b) Sales

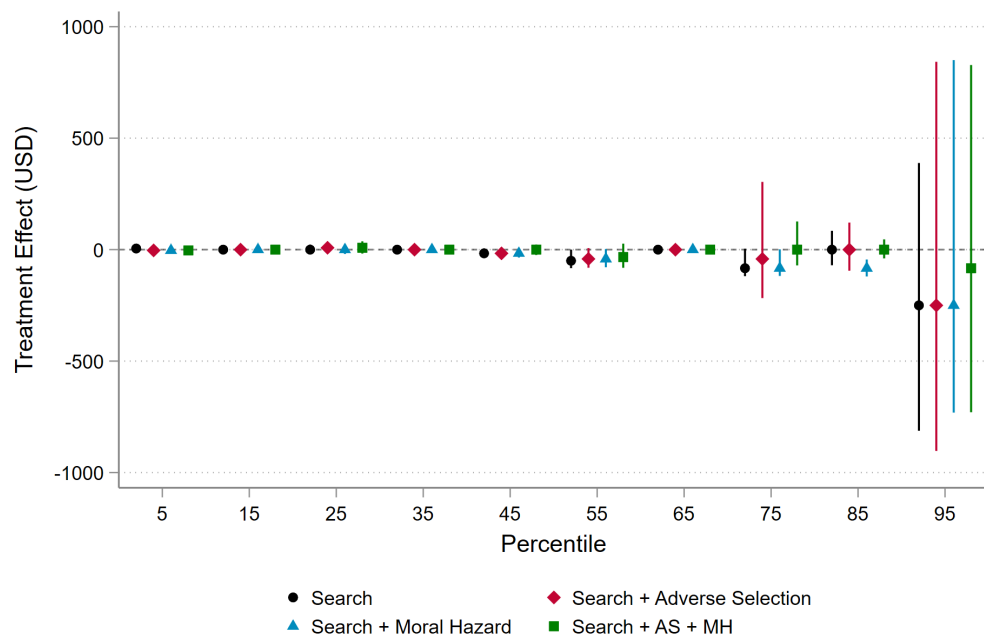
Note: This figure shows the coefficients from regressions of indicators for whether profit and sales are above some threshold t , for a range of t . All regressions include covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as the outcome measured at baseline (if available). The numbers in parentheses show the percentiles at which t is located in the distribution of the pure control group. We plot 95% confidence intervals constructed using randomisation inference, defined as the set of sharp nulls that do not reject at the 5% level, using the procedure in Young (2024).

Figure A7: Treatment Effects on Transition Matrix



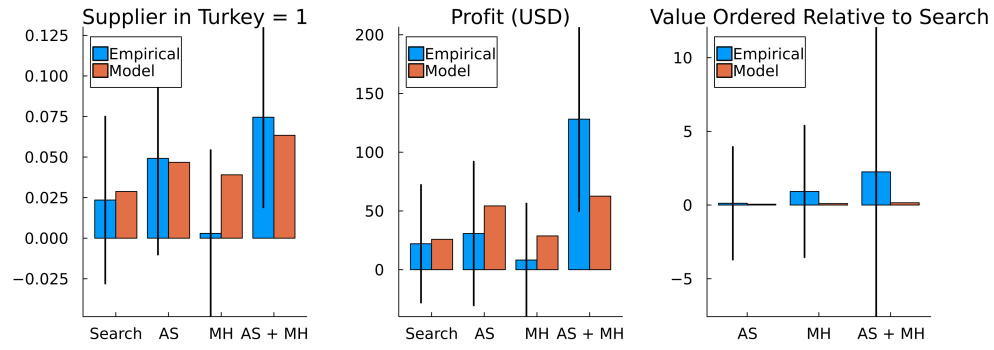
Note: This figure shows the treatment effects for the Search + Adverse Selection + Moral Hazard treatment on indicators for whether a firm transitioned between different percentile profit buckets between baseline and endline.

Figure A8: Placebo Check: Quantile Treatment Effects on Baseline Profit

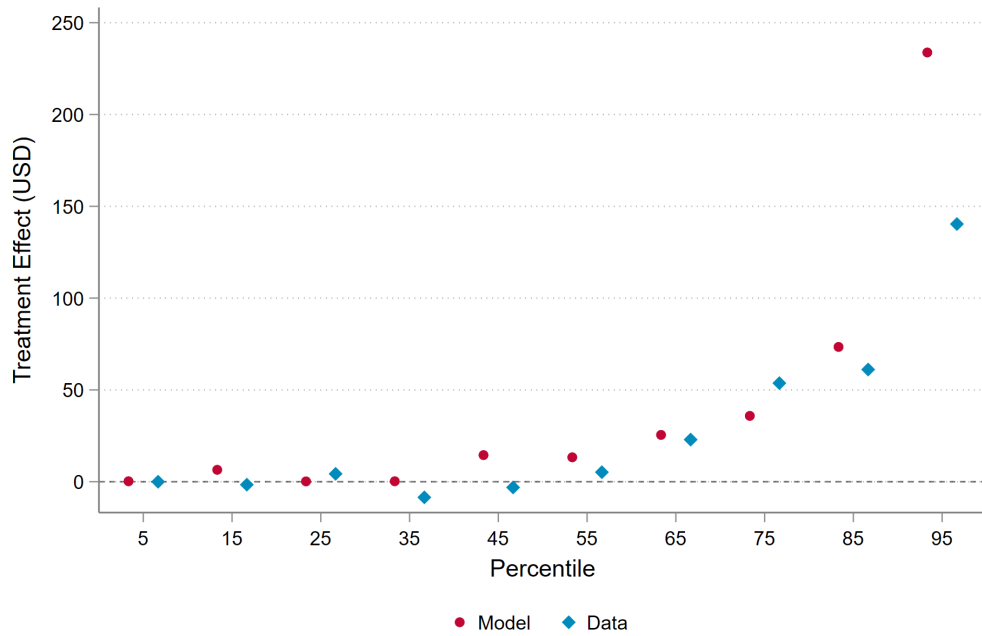


Note: This figure shows the coefficients from quantile regressions of baseline profit on the four treatment groups, intended as a placebo test. As the only covariate included in the main quantile regressions is the outcome measured at baseline, which is itself the outcome here, we do not include any covariates. We plot 95% confidence intervals constructed using randomisation inference, defined as the set of sharp nulls that do not reject at the 5% level.

Figure A9: Model Fit



(a) Fit on Targeted Moments



(b) Fit on Quantile Treatment Effects

Note: Panel A shows the 11 targeted moments in the structural estimation, with confidence intervals on the empirical moments (computed by bootstrapping the experimental treatment effects) shown to highlight the relative precisions. We weight moments by the inverse of the variance-covariance matrix of the treatment effects. Panel B compares quantile treatment effects between the model and the data, which were not explicitly targeted moments, for the group with both trust treatment. The Data moments are the same as in Figure 7.

Appendix B - Mathematical Appendix

In this Appendix, we prove further detail on the model.

Further Details on the Optimal Contract

Assumption 1. (*No trade with bad types*)

$$\max_q \lambda r(q) - (1 - \xi)cq < \max_q r(q) - p_l q$$

This assumption states that if a firm knew that the supplier was a bad type, they would prefer to order from the local supplier.

Proposition 1. *It is not optimal to offer a menu of contracts that separates good and bad suppliers.*

Proof. Suppose the contrary, and consider the state of the world where the supplier is a bad type. Since the menu fully separates the types, the firm's posterior is then that the supplier is a bad type with probability 1. Because the supplier has limited liability, the maximum that the firm can earn under any such contract is the full surplus, i.e., $(\max_q \lambda r(q) - (1 - \xi)cq)/(1 - \delta)$. Assumption 1 implies that the firm can always do better than this, because at the very least they can buy from a local supplier in every period. Since the contract is relational, the firm would thus renege before sending the first transfer. Thus, the expected payoff to the bad type from accepting the revealing contract is 0. So long as the contract recommended to the good type involves positive quantity, the bad type can always earn a positive expected payoff by accepting the good type's contract, because limited liability ensures that $\tau_t \geq cq_t > (1 - \xi)cq_t$ for all t . Therefore, the bad type would not accept the contract that reveals their type, which is a contradiction. $\square$

The original program is as follows:

$$\begin{aligned} L = & \min_{\{\rho_t\}, \{\eta_t\}, \{\gamma_t\}} \max_{\{q_t\}, \{\tau_t\}} \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) ((1 - \mu_t(1 - \lambda)r(q_t) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}) \\ & + \sum_{t=0}^{\infty} \delta^t \rho_t \left[\sum_{\tau=t+1}^{\infty} \delta^{\tau-t} (R_{\tau} - cq_{\tau}) - \xi cq_t \right] \\ & + \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \eta_t \left[\sum_{\tau=t}^{\infty} \delta^{\tau-t} (1 - \mu_{\tau}(1 - \lambda^{\tau-t})) ((1 - \mu_{\tau}(1 - \lambda)r(q_t) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}) - \bar{U}) \right] \\ & + \sum_{t=0}^{\infty} \delta^t \gamma_t [\tau_t - cq_t] \end{aligned}$$

The modified program is as follows.

$$\begin{aligned}
W_t(U_t, V_t, \mu_t) = & y(q_t, \tau_t, \mu_t) + \rho_t (\delta(1 - \lambda)V_{t+1} - \xi c q_{t+1}) \\
& + \eta_t (U_{t+1} - \bar{U}) \\
& + \gamma_t (\tau_t - c q_t) \\
& + \nu_t^b (y(q_t, \tau_t, \mu_t) + \delta(1 - \mu_t(1 - \lambda))U_t + \delta\mu_t(1 - \lambda)\bar{U} - U_t) \\
& + \nu_t^s (\tau_t - c q_t + \delta V_{t+1} - V_t) \\
& + \delta(1 - \mu_t(1 - \lambda))W_{t+1}(U_{t+1}, V_{t+1}, \mu_{t+1})
\end{aligned}$$

The FOCs are as follows

$$(1 - \mu_t(1 - \lambda))r'(q_t)(1 + \nu_t^b) = \rho_t \xi c + (\gamma_t + \nu_t^s) c \quad (q_t)$$

$$1 + \nu_t^b = \gamma_t + \nu_t^s \quad (\tau_t)$$

$$\rho_t(1 - \lambda) + \nu_t^s = -(1 - \mu_t(1 - \lambda)) \frac{\partial W_{t+1}}{\partial V_{t+1}} \quad (V_{t+1})$$

$$\eta_{t+1} + \delta(1 - \mu_t(1 - \lambda))\nu_t^b = -\delta(1 - \mu_t(1 - \lambda)) \frac{\partial W_{t+1}}{\partial U_{t+1}} \quad (U_{t+1})$$

Substituting the FOC for τ_t into the FOC for q_t and re-arranging gives

$$r'(q_t) = \frac{1}{1 - \mu_t(1 - \lambda)} \left(1 + \xi \frac{\rho_t}{1 + \nu_t^b} \right) c.$$

The Envelope Condition implies that $\frac{\partial W_{t+1}}{\partial V_{t+1}} = -\nu_{t+1}^s$ and $\frac{\partial W_{t+1}}{\partial U_{t+1}} = -\nu_{t+1}^b$. Combining FOCs 2-4 then gives the following equation relating γ_t and γ_{t+1} that we will make extensive use of in the following proofs.

$$\rho_t(1 - \lambda) + 1 - \gamma_t + \mu_t(1 - \lambda)\nu_t^b = (1 - \gamma_{t+1})(1 - \mu_t(1 - \lambda)) + \frac{\eta_{t+1}}{\delta}. \quad (4)$$

Remark 1. $\nu_{t+1}^b > \nu_t^b \iff \eta_{t+1} > 0$

Proof. This follows immediately from the FOC for U_{t+1} after substituting in the Envelope Condition. $\square$

The optimal contract is generally not available in closed form, but we state and prove some properties in the following proposition.

Proposition 2. *There exists finite T^* such that:*

1. *The agent earns zero stage profits for all $t < T^*$ (i.e., LL binds for all $t < T^*$).*

2. The principal earns $(1 - \delta)\bar{U}$, that is, zero stage profits net of their outside option, for all $t > T^*$ (i.e., DEC binds for all $t > T^*$).
3. q_t is strictly increasing for $t < T^*$.
4. ICC binds for at least some $t < T^*$.

Proof. We prove this through a series of Lemmas, which we state and prove below. For parts 1 and 2, see Lemma 3. For part 3, see Lemma 1. For part 4, see Lemma 5. $\square$

Before formally stating the Lemmas, we first provide an intuitive sketch of the approach. First, we show that q_t must be strictly increasing whenever LL binds. Intuitively, a (weakly) decreasing q_t despite beliefs improving would imply that ICC strongly becomes “more binding” over time. But when LL is binding, the supplier is earning zero stage profits, so the ICC must be getting less binding over time.

Second, we show that if LL in t binds, then DEC in t cannot bind. Intuitively, both parties cannot be earning their outside option at the same time, as belief improving and q_t growing would imply that in other periods one of them must be making a loss.

Third, we show that the problem can be divided into two phases: LL will bind for all early periods and be slack for all late periods. “Backloading” results of this kind are standard in the dynamic moral hazard literature. Backloading happens for two reasons. The first reason is that incentives must be given to the agent at some point, and backloading incentives is efficient because it improves both early and late ICCs (whereas frontloading or even-loading still improves early ICCs but improves late ICCs less). The second reason is that adverse selection means that the “good type” agent is more patient than the principal, as the good type knows their own type. This means that it is always cheaper for the principal to backload payments. As a corollary, the DEC must bind for all late periods, as the principal wants to backload as much as possible, and will continue to do so until the DEC binds.

Finally, we show that ICC must bind for some t in the early section. Intuitively, if it didn’t, then the principal would just extend the early period—where they earn all the surplus—for longer. The only reason to ever end this early phase is precisely because an ICC eventually binds (the principal has to pay the agent eventually, and in absence of the ICC would always prefer not to).

Lemma 1. *If LL binds in $t + 1$, then $q_{t+1} > q_t$.*

Proof. Suppose that LL binds in $t + 1$ and $q_{t+1} \leq q_t$. The FOC for q_t is

$$r'(q_t) = \frac{1}{1 - \mu_t(1 - \lambda)} \left(1 + \xi \frac{\rho_t}{1 + \nu_t^b} \right) c.$$

We already know that $\nu_{t+1}^b \geq \nu_t^b$, and that $\mu_{t+1} \leq \mu_t$. Then, the hypothesis $q_{t+1} \leq q_t$ implies

that $\rho_{t+1} > \rho_t$. Then, we can write

$$\delta(1-\lambda)V_{t+1} < (1-\lambda)V_{t+1} = (1-\lambda)(R_{t+1} - cq_{t+1} + \delta V_{t+2}) = \delta(1-\lambda)V_{t+2} = \xi cq_{t+1} \leq \xi cq_t.$$

The first equality is a definition, the second equality follows from the fact that LL binds in $t+1$, the third equality follows from the fact that $\rho_{t+1} > \rho_t$ implies that $\rho_{t+1} > 0$, which implies that ICC binds in $t+1$. The final inequality follows from the hypothesis that $q_{t+1} \leq q_t$. Thus, we have shown that

$$\delta(1-\lambda)V_{t+1} < \xi cq_t.$$

But this implies that ICC in t is violated, which is a contradiction. $\square$

Lemma 2. *If LL in $t+1$ binds, then DEC in $t+1$ is slack.*

Proof. Suppose that DEC in $t+1$ binds. Then, we can write

$$\begin{aligned} U_t &= (1 - \mu_t(1 - \lambda))r(q_t) - \tau_t + \delta(1 - \mu_t(1 - \lambda))U_{t+1} + \delta\mu_t(1 - \lambda)\bar{U} \\ &= (1 - \mu_t(1 - \lambda))r(q_t) - \tau_t + \delta\bar{U} \\ &\leq (1 - \mu_t(1 - \lambda))r(q_t) - cq_t + \delta\bar{U} \\ &< (1 - \mu_{t+1}(1 - \lambda))r(q_t) - cq_t + \delta\bar{U} \\ &\leq (1 - \mu_{t+1}(1 - \lambda))r(q_{t+1}) - cq_{t+1} + \delta\bar{U} \\ &\leq (1 - \mu_{t+1}(1 - \lambda))r(q_{t+1}) - cq_{t+1} + \delta(1 - \mu_{t+1}(1 - \lambda))U_{t+2} + \delta\mu_{t+1}(1 - \lambda)\bar{U} \\ &= (1 - \mu_{t+1}(1 - \lambda))r(q_{t+1}) - R_{t+1} + \delta(1 - \mu_{t+1}(1 - \lambda))U_{t+2} + \delta\mu_{t+1}(1 - \lambda)\bar{U} \\ &= U_{t+1} \\ &= \bar{U}. \end{aligned}$$

The first line is the definition of U_t . The second line follows from DEC binding in $t+1$. The third line follows from LL in t . The fourth line follows from $\mu_{t+1} < \mu_t$. The fifth line follows from Lemma 1. The sixth line follows from DEC in period $t+2$, i.e., $U_{t+2} \geq \bar{U}$. The seventh line follows from the hypothesis that LL binds in $t+1$, i.e., $\tau_t = cq_t$. The eighth line is the definition of U_{t+1} . The ninth line follows from DEC binding in $t+1$.

The above thus establishes that $U_t < \bar{U}$. But this is a violation of DEC in period t , which is a contradiction. $\square$

Corollary 1. *If LL is slack in t , then it is also slack in $t+1$.*

Proof. LL slack in t means $\gamma_t = 0$. Suppose it binds in $t+1$, which means $\gamma_{t+1} > 0$. Lemma

2 then implies that $\eta_{t+1} = 0$. But, Equation (4) gives

$$\gamma_{t+1}(1 - \mu_t(1 - \lambda)) = \frac{\eta_{t+1}}{\delta} - \rho_t(1 - \lambda) - \mu_t(1 - \lambda)(1 + \nu_t^b).$$

It must then be that $\eta_{t+1} > 0$, which is a contradiction. $\square$

Corollary 2. *If LL is slack in t , then DEC binds in $t + 1$.*

Proof. Corollary 1 implies that $\gamma_t = \gamma_{t+1} = 0$. Then, the FOC implies

$$\frac{\eta_{t+1}}{\delta} = \rho_t(1 - \lambda) + \mu_t(1 - \lambda)(1 + \nu_t^b) \geq \mu_t(1 - \lambda)(1 + \nu_t^b).$$

The final term is strictly positive, which implies that $\eta_{t+1} > 0$. $\square$

Lemma 3. *If any trade occurs, then there exists finite $T^* \geq 1$ such that (i) LL binds for all $t < T^*$ and is slack for all $t \geq T^*$, and (ii) DEC is slack for all $t < T^*$ and binds for all $t > T^*$.*

Proof. For (i): Corollary 1 shows that $\gamma_t = 0 \implies \gamma_{t+1} = 0$. Thus, if there exists T^* such that $\gamma_t = 0$, then $\gamma_s = 0$ for all $s \geq T^*$. We already know that $\gamma_0 = 1$, so this is not the case for $t = 0$. Suppose that $\gamma_t > 0$ for all t . Then, the supplier earns zero profit, which implies that all ICCs will fail unless $q_t = 0$ for all t , which violates the supposition that trade occurs at some point. Thus, there must be at least one $t \geq 1$ such that LL is slack in t . If there are multiple, define T^* as the earliest such t .

For (ii): Since LL is slack for all $t \geq T^*$, Corollary 2 implies that DEC binds for all $t > T^*$. $\square$

Corollary 3. $\nu_t^b = 0$ for all $t \leq T^*$, and $\nu_{t+1}^b > 0$ for all $t > T^*$.

Proof. This follows from Remark 1 and Lemma 3. $\square$

Lemma 4. $\gamma_{t+1} \leq \gamma_t$, with inequality strict if $\gamma_t \in (0, 1)$.

Proof. If $\gamma_{t+1} = 0$, then this holds trivially. We thus need to establish the claim for $\gamma_{t+1} > 0$. Suppose then that $\gamma_{t+1} > \gamma_t$ with $\gamma_{t+1} > 0$. Then, Equation (4) implies

$$0 > \rho_t(1 - \lambda) - \frac{\eta_{t+1}}{\delta} + \mu_t(1 - \lambda)(1 + \nu_t^b - \gamma_{t+1}).$$

Since $\gamma_{t+1} > 0$, Lemma 2 implies that $\eta_{t+1} = 0$. We are thus left with

$$\begin{aligned} 0 &> \rho_t(1 - \lambda) + \mu_t(1 - \lambda)(1 + \nu_t^b - \gamma_{t+1}) \\ &= \rho_t(1 - \lambda) + \mu_t(1 - \lambda)(1 + \nu_{t+1}^b - \gamma_{t+1}) \\ &= \rho_t(1 - \lambda) + \mu_t(1 - \lambda)\nu_{t+1}^s \end{aligned}$$

where the first equality follows from the fact that the FOC for U_{t+1} implies that $\eta_{t+1} = 0 \implies \nu_t^b = \nu_{t+1}^b$, and the second equality follows from the FOC for R_{t+1} . But the RHS is weakly positive, so this is a contradiction, which establishes that $\gamma_{t+1} \leq \gamma_t$.

Then, to establish the claim about strict inequality, suppose instead that $\gamma_{t+1} = \gamma_t$. We instead have

$$0 = \rho_t(1 - \lambda) + \mu_t(1 - \lambda)\nu_{t+1}^s,$$

which is only possible if $\rho_t = \nu_{t+1}^s = 0$. But this implies that $\gamma_{t+1} = 1 + \nu_t^b \geq 1$. If $\gamma_t \in (0, 1)$, this implies $\gamma_{t+1} > \gamma_t$, which is a contradiction. $\square$

Lemma 5. *ICC binds for some $t < T^*$.*

Proof. We prove this by iterating forward Equation (4). Since $\eta_t = \nu_t^b = 0$ for all $t < T^*$, the equation can be written

$$\gamma_t = \mu_t(1 - \lambda) + \rho_t(1 - \lambda) + \gamma_{t+1}(1 - \mu_t(1 - \lambda)).$$

Starting with $\gamma_0 = 1$ and iterating this until $T^* - 1$, for which $\gamma_{t+1} = 0$, we get

$$1 + (1 - \mu_0(1 - \lambda^{T^*}))\frac{\eta_{T^*}}{\delta} = \sum_{t=0}^{T^*-1} (1 - \mu_0(1 - \lambda^t))[\mu_t(1 - \lambda) + \rho_t(1 - \lambda)].$$

Note that $LHS \geq 1$. The first term on the RHS simplifies to

$$\sum_{t=0}^{T^*-1} (1 - \mu_0(1 - \lambda^t))\mu_t(1 - \lambda) = (1 - \lambda)\mu_0 \sum_{t=0}^{T^*-1} \lambda^t = \mu_0(1 - \lambda^{T^*}) < 1.$$

Thus, it cannot be that $\rho_t = 0$ for all $t \leq T^* - 1$, as otherwise $LHS > RHS$. So there must be some $t \leq T^*$ for which the ICC binds. $\square$

Proposition 3. *The period-0 value of the relationship is decreasing in μ_0 and ξ .*

Proof. We first show the case for ξ . Using the Envelope Theorem to differentiate the original infinite horizon Lagrangian with respect to ξ gives

$$\frac{dL}{d\xi} = - \sum_{t=0}^{\infty} \delta^t \rho_t c q_t.$$

Since Proposition 2 established that the DICC binds for at least some t , we know that $\rho_t > 0$ for at least some t . So this is strictly negative except for the trivial case where $q_t = 0$ for all t , in which case it is zero.

Second, for μ_0 , it is easier to work with the recursive formulation. Differentiating the

period-0 program gives

$$\frac{dW_0}{d\mu_0} = \underbrace{-(1-\lambda)r(q_0)(1+\nu_0^b) - \delta(1-\lambda)\nu_0^b(U_1 - \bar{U}) - \delta(1-\lambda)W_1}_{\equiv \phi_0} + \delta(1-\mu_0(1-\lambda))\frac{d\mu_1}{d\mu_0}\frac{\partial W_1}{\partial \mu_1}.$$

The first three terms are all negative. The only term that we cannot immediately sign is $\partial W_1/\partial \mu_1$. However, since the problem is recursive, we can simply repeatedly lead ϕ_t to obtain

$$\frac{dW_0}{d\mu_0} = \sum_{t=0}^{\infty} \delta^t \prod_{s=0}^t (1 - \mu_s(1 - \lambda)) \frac{d\mu_{s+1}}{d\mu_s} \phi_t.$$

This is strictly negative, except in the trivial case where the firm never purchases anything with the foreign supplier. $\square$

Proposition 4. *The buyer's period 0 value satisfies $\partial^2 U_0(\mu_0, \xi)/\partial \mu_0 \partial \xi > 0$.*

Proof. This simply comes from the fact that the stage payoff satisfies increasing differences. It is easiest to work directly with the SPFE as we can then use standard contraction mapping arguments. As described in detail in Appendix ??, the problem can be rewritten as

$$\begin{aligned} W(\zeta_t^b, \zeta_t^s, \mu_t) = & \min_{\eta_t, \rho_t, \gamma_t} \max_{q_t, \tau_t} (\zeta_t^b + \eta_t)h_0^b(q_t, \tau_t, \mu_t) + \eta_t h_1^b(q_t, \tau_t, \mu_t) \\ & + (\zeta_t^s + \gamma_t)h_0^s(q_t, \tau_t) + \rho_t h_1^s(q_t, \tau_t) + \delta W(\zeta_{t+1}^b, \zeta_{t+1}^s, \mu_{t+1}), \end{aligned}$$

with

$$\begin{aligned} h_0^b(q_t, \tau_t, \mu_t) &\equiv [(1 - \mu_t(1 - \lambda))r(q_t) + \mu_t(1 - \lambda)r(0) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}] \\ h_1^b(q_t, \tau_t, \mu_t) &\equiv -\bar{U} \\ h_0^s(q_t, \tau_t) &\equiv \tau_t - cq_t \\ h_1^s(q_t, \tau_t) &\equiv -\xi cq_t \end{aligned}$$

subject to

$$\begin{aligned} \zeta_{t+1}^b &= (\zeta_t^b + \eta_t)(1 - \mu_t(1 - \lambda)) \\ \zeta_{t+1}^s &= \zeta_t^s + \rho_t, \end{aligned}$$

and the evolution of μ_t .

The stage payoff function satisfies weak increasing differences in $(-\mu, -\xi)$ because the cross-partial of these terms is zero (in particular, the ICC constraint does not contain μ). However, for strictness, note that the payoff function has increasing differences in $(-\mu, q)$, so q^* is increasing in $-\mu$, and the marginal value of lowering ξ is also proportional to q . Then, since ξ does not affect the transition function for μ and the feasible sets are parameter

invariant, and since the stage payoff satisfies the regularity conditions for standard mini-max theorems, standard contraction-mapping arguments establish that the value function inherits increasing differences in $(-\mu, -\xi)$. Since the value function in period 0 is equal to the buyer's utility as $\zeta_0^b = 1$ and $\zeta_0^s = 0$, and the function is continuous, we thus have $\partial U_0(\mu_0, \xi)^2 / \partial \mu_0 \partial \xi > 0$.

□

Further Details on the Search Process

In this sub-section, we prove (1) that the sequential search process behaves according to the standard results despite the fact that the outside option, $\bar{U}$, enters the utility function directly and (2) that if there is a firm-specific heterogeneity term z , then there exists a cutoff $\bar{z}$ such that only firms with $z > \bar{z}$ engage in search.

The search process works as follows. To match with a random foreign supplier, a firm can pay a one-time search cost $s > 0$. Upon matching, both parties immediately observe a realization of a match-specific productivity term, $\psi \sim G$, that is fixed over time. We denote the period-0 value of a relationship with a foreign supplier with match-specific productivity ψ as $U_0(\psi, \bar{U})$, where we include $\bar{U}$ to make explicit that U_0 depends on the outside option directly. It does so through two channels. First, the adverse selection problem means that the firm will eventually earn $\bar{U}$ on path if the supplier is a bad type (because a low quality realization will eventually occur), which will cause U_0 to be increasing in $\bar{U}$. Second, the limited commitment problem means that the firm cannot make promises that would result in earning less than $\bar{U}$ (i.e., the DEC), which limits the enforceable surplus—this causes U_0 to be decreasing in $\bar{U}$. Since this is direct dependence on the outside option is non-standard in sequential search models, we prove the standard results here.

Proposition 5. *The optimal policy in the search process is a cutoff rule, where the cutoff value equals the expected utility from the search process.*

Proof. First, note that the search problem is recursive. That is, the entire history of searching is summarized by the highest value of ψ drawn to date, which, we denote with ψ_b . Second, U_0 is monotonic in ψ , strictly so in the region where the firm develops a relationship. As is standard, these together imply that the optimal policy is to search until the firm receives a draw above an endogenous cutoff $\bar{\psi}$.

For a firm whose best current draw is ψ_t , the expected benefit from obtaining one more draw is

$$\int_{\psi}^{\infty} (U_0(\psi, \bar{U}) - U_0(\psi_b, \bar{U})) dG(\psi),$$

where $\bar{U}$ is the expected utility of terminating the current relationship and re-searching. The expected cost is s . By definition, the cutoff value of $\bar{\psi}$ equates expected benefits with expected costs,

$$\int_{\bar{\psi}}^{\infty} (U_0(\psi, \bar{U}) - U_0(\bar{\psi}, \bar{U})) dG(\psi) = s.$$

As the stopping time is geometrically distributed, the expected number of draws until the firm obtains a realisation above $\bar{\psi}$ is $1/(1 - G(\bar{\psi}))$. This means that the expected utility from engaging in search, $\bar{U}$, is

$$\bar{U} = \frac{\int_{\bar{\psi}}^{\infty} U_0(\psi, \bar{U}) dG(\psi)}{1 - G(\bar{\psi})} - \frac{s}{1 - G(\bar{\psi})}.$$

Re-arranging, we can obtain

$$\begin{aligned} s &= \int_{\bar{\psi}}^{\infty} U_0(\psi, \bar{U}) dG(\psi) - \bar{U} \int_{\bar{\psi}}^{\infty} dG(\psi) \\ s &= \int_{\bar{\psi}}^{\infty} (U_0(\psi, \bar{U}) - \bar{U}) dG(\psi), \end{aligned}$$

which, when compared with the definition of the cutoff, implies that $\bar{U} = U_0(\bar{\psi}, \bar{U})$. Thus, despite the fact that $\bar{U}$ enters the utility function directly, it does not change the fundamental structure of the search process. $\square$

Proposition 6. *The cutoff $\bar{\psi}$ is decreasing in s .*

Proof. It is easier to eliminate the $\bar{\psi}$ in the integral limit by rewriting the cutoff rule definition as follows

$$\int_{-\infty}^{\infty} \max\{U_0(\psi, \bar{U}) - \bar{U}, 0\} dG(\psi) = s,$$

where $\bar{U} = U_0(\bar{\psi}, \bar{U})$. Note that

$$\frac{ds}{d\bar{U}} = \int_{-\infty}^{\infty} \left(\frac{\partial U_0(\psi, \bar{U})}{\partial \bar{U}} - 1 \right) 1\{U_0(\psi, \bar{U}) - \bar{U} > 0\} dG(\psi).$$

In standard sequential search models, the first term in parentheses is zero, so this is clearly negative. Here, we will show that this new term is always strictly smaller than 1, which is sufficient to establish the claim.

We know that the U_0 is given by the following dynamic program (and we know that a solution exists for the particular draw of ψ under consideration as we only need to consider

the cases where $U_0(\psi, \bar{U}) - \bar{U} > 0$,

$$\begin{aligned}
L = & \min_{\{\rho_t\}, \{\eta_t\}, \{\gamma_t\}} \max_{\{q_t\}, \{\tau_t\}} \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) ((1 - \mu_t(1 - \lambda)r(\psi q_t) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}) \\
& + \sum_{t=0}^{\infty} \delta^t \rho_t \left[\sum_{\tau=t+1}^{\infty} \delta^{\tau-t} (R_{\tau} - cq_{\tau}) - \xi cq_t \right] \\
& + \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \eta_t \left[\sum_{\tau=t}^{\infty} \delta^{\tau-t} (1 - \mu_{\tau}(1 - \lambda^{\tau-t})) ((1 - \mu_{\tau}(1 - \lambda)r(\psi q_t) - \tau_t + \delta\mu_{\tau}(1 - \lambda)\bar{U}) - \bar{U} \right] \\
& + \sum_{t=0}^{\infty} \delta^t \gamma_t [\tau_t - cq_t].
\end{aligned}$$

The Envelope Theorem implies that $\frac{dL}{d\bar{U}}$ is given by

$$\sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \delta\mu_t(1 - \lambda) + \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \eta_t \left[\sum_{\tau=t}^{\infty} \delta^{\tau-t} (1 - \mu_{\tau}(1 - \lambda^{\tau-t})) \delta\mu_{\tau}(1 - \lambda) - 1 \right].$$

Since Bayesian updating implies that we can write $\mu_t = \mu_0 \lambda^t / (1 - \mu_0(1 - \lambda^t))$, we can rewrite the first term as $\mu_0 \delta (1 - \lambda) \sum_{t=0}^{\infty} \delta^t \lambda^t$, or $\frac{\mu_0 \delta (1 - \lambda)}{1 - \delta \lambda}$, which is strictly less than 1. Intuitively, this is a (discounted) sum of probabilities and thus cannot exceed 1—indeed, it is easy to see that if we set δ to 1 then this simplifies to just μ_0 . The first term is therefore strictly less than 1. The second term is negative, because the first term in square brackets is strictly less than 1 by the exact same argument. Since this argument did not rely on the value of ψ , we have that $dL/d\bar{U} < 1$ for all ψ for which $U_0(\psi, \bar{U}) - \bar{U} > 0$, which implies that

$$\frac{ds}{d\bar{U}} = \int_{-\infty}^{\infty} \left(\frac{\partial U_0(\psi, \bar{U})}{\partial \bar{U}} - 1 \right) 1\{U_0(\psi, \bar{U}) - \bar{U} > 0\} dG(\psi) < 0.$$

□

Note that a firm-specific multiplicative utility shifter that takes the form is equivalent to firms having heterogeneous search costs s/z as we simply divide through the cutoff definition equation by z . This proposition then implies that the cutoff utility is increasing in z , and thus that if there is a “global” outside option (from not searching at all) U_l , then there will exist some $\bar{z}$ such that the cutoff $\bar{U}(\bar{z})$ exactly equals U_l . In particular, the cutoff is defined such that,

$$\int_{-\infty}^{\infty} \max\{U_0(\psi, U_l) - U_l, 0\} dG(\psi) = s/\bar{z}.$$

Lemma 6. *The cutoff $\bar{z}$ is increasing in μ_0 and ξ .*

Proof. Since Proposition 3 establishes that U_0 is decreasing in μ_0 and ξ , the LHS of the

equation defining $\bar{z}$ is decreasing in μ_0 and ξ , which means that $\bar{z}$ must be increasing in μ_0 and ξ to ensure that the RHS decreases. $\square$

Appendix C - Heterogeneity Analysis

Table C1: Heterogeneity by Retailer vs Wholesaler

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.194*** (0.047)	0.158 (0.097)	0.021 (0.034)	0.004 (0.104)
Search + AS	0.049 (0.047)	0.221** (0.105)	0.053 (0.036)	0.013 (0.121)
Search + MH	0.131*** (0.047)	0.119 (0.100)	-0.006 (0.033)	0.004 (0.091)
Search + AS + MH	0.036 (0.049)	0.058 (0.110)	0.054 (0.037)	0.024 (0.103)
S Only * Wholesaler	-0.158* (0.083)	-0.071 (0.194)	-0.011 (0.067)	0.469 (0.324)
S + AS * Wholesaler	0.048 (0.083)	-0.179 (0.193)	0.001 (0.067)	0.255 (0.323)
S + MH * Wholesaler	-0.024 (0.084)	-0.139 (0.190)	0.016 (0.066)	-0.144 (0.231)
S + AS + MH * Wholesaler	0.114 (0.083)	0.005 (0.187)	0.023 (0.068)	1.209*** (0.437)
All Inter Zero p -val	0.020	0.803	0.991	0.040
Adjusted R^2	0.01	-0.01	0.00	0.05
N	1579	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm sells at least partially wholesale, defined as having a positive share of sales that are wholesale to other firms. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 5000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C2: Heterogeneity by Physical Store vs Online Only

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.208*** (0.048)	0.113 (0.097)	0.040 (0.038)	-0.058 (0.076)
Search + AS	0.115** (0.047)	0.139 (0.104)	0.065* (0.039)	-0.017 (0.098)
Search + MH	0.182*** (0.047)	0.074 (0.097)	0.005 (0.036)	-0.053 (0.060)
Search + AS + MH	0.130*** (0.049)	0.033 (0.104)	0.081** (0.040)	-0.026 (0.066)
S Only * Physical Store	-0.205** (0.082)	0.090 (0.200)	-0.066 (0.056)	0.653* (0.337)
S + AS * Physical Store	-0.168** (0.081)	0.048 (0.194)	-0.035 (0.060)	0.519 (0.376)
S + MH * Physical Store	-0.222*** (0.082)	-0.079 (0.209)	-0.036 (0.056)	0.165 (0.285)
S + AS + MH * Physical Store	-0.152* (0.082)	0.098 (0.185)	-0.031 (0.061)	1.582*** (0.497)
All Inter Zero p -val	0.059	0.915	0.835	0.027
Adjusted R^2	0.04	-0.01	0.02	0.12
N	1579	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm had a physical store at baseline. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 5000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C3: Heterogeneity by Direct Import Experience

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.149*** (0.048)	0.114 (0.104)	0.050* (0.030)	0.074 (0.121)
Search + AS	0.067 (0.048)	0.120 (0.106)	0.070** (0.031)	0.008 (0.102)
Search + MH	0.106** (0.048)	0.045 (0.106)	0.024 (0.028)	-0.135* (0.075)
Search + AS + MH	0.081* (0.049)	0.023 (0.108)	0.072** (0.031)	0.416** (0.192)
S Only * Experience	-0.021 (0.083)	0.076 (0.177)	-0.104 (0.068)	0.180 (0.278)
S + AS * Experience	-0.013 (0.082)	0.110 (0.190)	-0.053 (0.069)	0.246 (0.329)
S + MH * Experience	0.044 (0.082)	0.091 (0.178)	-0.080 (0.067)	0.285 (0.230)
S + AS + MH * Experience	-0.014 (0.083)	0.114 (0.179)	-0.014 (0.072)	0.319 (0.430)
All Inter Zero p -val	0.937	0.974	0.532	0.761
Adjusted R^2	0.01	-0.01	0.05	0.02
N	1579	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm has direct online importing experience, defined as having bought goods from another country online directly at least once in the past 12 months. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 5000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C4: Heterogeneity by Baseline Turkish Share

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.063 (0.063)	0.180 (0.145)	0.024 (0.043)	0.038 (0.173)
Search + AS	-0.006 (0.061)	0.121 (0.155)	0.052 (0.044)	-0.033 (0.176)
Search + MH	0.034 (0.061)	0.200 (0.150)	-0.020 (0.038)	-0.168 (0.125)
Search + AS + MH	0.034 (0.063)	0.354** (0.141)	0.041 (0.045)	0.255 (0.261)
S Only * High Share	0.137* (0.081)	-0.053 (0.178)	-0.012 (0.059)	0.161 (0.236)
S + AS * High Share	0.110 (0.080)	0.046 (0.187)	0.015 (0.061)	0.204 (0.248)
S + MH * High Share	0.128 (0.080)	-0.177 (0.182)	0.023 (0.056)	0.214 (0.178)
S + AS + MH * High Share	0.067 (0.082)	-0.427** (0.176)	0.037 (0.062)	0.420 (0.359)
All Inter Zero p -val	0.420	0.071	0.937	0.700
Adjusted R^2	0.02	0.01	0.01	0.01
N	1542	353	1636	1261

Note: This table shows the main results with treatment interacted with an indicator for whether the share of the firm's customers in the past 30 days that buy Turkish-made products is above the median. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 5000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C5: Heterogeneity by Sex of Owner

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.028 (0.056)	0.111 (0.128)	-0.028 (0.037)	0.439** (0.216)
Search + AS	-0.047 (0.054)	0.148 (0.127)	-0.010 (0.038)	0.240 (0.206)
Search + MH	0.044 (0.056)	-0.033 (0.120)	-0.056 (0.036)	0.005 (0.152)
Search + AS + MH	-0.036 (0.054)	0.040 (0.122)	0.039 (0.041)	0.904*** (0.321)
S Only * Female	0.226*** (0.078)	0.052 (0.173)	0.090 (0.058)	-0.574** (0.232)
S + AS * Female	0.222*** (0.077)	0.017 (0.177)	0.134** (0.060)	-0.313 (0.242)
S + MH * Female	0.148* (0.078)	0.200 (0.170)	0.097* (0.056)	-0.047 (0.182)
S + AS + MH * Female	0.232*** (0.078)	0.046 (0.173)	0.060 (0.062)	-0.822** (0.343)
All Inter Zero p -val	0.011	0.771	0.215	0.030
Adjusted R^2	0.02	-0.01	0.02	0.05
N	1579	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm owner is female. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 5000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C6: Heterogeneity by Baseline Turkish Supplier Group

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.135*** (0.045)	0.184* (0.097)	0.053** (0.027)	0.177 (0.127)
Search + AS	0.037 (0.044)	0.245** (0.099)	0.054** (0.028)	0.196 (0.148)
Search + MH	0.104** (0.044)	0.133 (0.096)	0.028 (0.026)	-0.044 (0.093)
Search + AS + MH	0.061 (0.045)	0.125 (0.100)	0.095*** (0.030)	0.438** (0.182)
S Only * In Turkish Group	0.019 (0.095)	-0.173 (0.191)	-0.144 (0.089)	-0.221 (0.314)
S + AS * In Turkish Group	0.072 (0.096)	-0.295 (0.210)	-0.033 (0.088)	-0.547** (0.239)
S + MH * In Turkish Group	0.092 (0.096)	-0.221 (0.200)	-0.116 (0.088)	0.042 (0.267)
S + AS + MH * In Turkish Group	0.072 (0.096)	-0.237 (0.192)	-0.126 (0.090)	0.421 (0.564)
All Inter Zero p -val	0.855	0.662	0.397	0.099
Adjusted R^2	0.01	-0.00	0.10	0.01
N	1562	357	1662	1286

Note: This table shows the main results with treatment interacted with an indicator for whether the firm was in a supplier WhatsApp group with a supplier based in Türkiye at baseline. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 5000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Appendix D - Main Tables without Covariates

We pre-specified that we would use the specification in Equation (2). Nonetheless, in this section, we replicate all of the main tables in the analysis using the following simpler regression specification that does not include any covariates:

$$y_i = \alpha + \sum_{j=1}^4 \beta_j T_{ji} + \varepsilon_i.$$

Table D1: Access to Foreign Goods (No Covariates)

	Horizontal	Vertical				Price
	Has Product ≥ 3 Criteria	High Qual Dummy	Qual Score (/50)	Made in Turkey	Index	Price (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Pooled</i>						
Treatment	0.101*** (0.031)	0.107 (0.068)	-0.636 (0.591)	0.145** (0.068)	0.379*** (0.138)	1.656** (0.840)
<i>Panel B: Individual Treatments</i>						
Search Only	0.141*** (0.039) [0.002]	0.139 (0.084) [0.241]	-0.177 (0.683) [0.954]	0.112 (0.081) [0.266]	0.389** (0.169) [0.063]	2.444** (0.991) [0.042]
Search + Adverse Selection	0.062 (0.039) [0.108]	0.154* (0.087) [0.234]	-0.026 (0.845) [0.976]	0.108 (0.085) [0.266]	0.358** (0.173) [0.063]	1.409 (1.049) [0.288]
Search + Moral Hazard	0.121*** (0.039) [0.004]	0.076 (0.084) [0.574]	-1.284 (0.843) [0.373]	0.154** (0.080) [0.126]	0.360** (0.165) [0.063]	1.746* (0.976) [0.152]
Search + AS + MH	0.076** (0.039) [0.085]	0.062 (0.085) [0.574]	-0.934 (0.846) [0.582]	0.203** (0.081) [0.038]	0.405** (0.176) [0.063]	0.932 (0.994) [0.348]
Control Mean	0.357	0.431	43.064	0.477	0.000	19.990
% Increase (Pooled)	28.3%	24.8%	-1.5%	30.4%	N/A	8.3%
All Coefs Equal p -val	0.148	0.621	0.475	0.530	0.987	0.367
Adjusted R^2	0.01	0.00	0.32	0.09	0.00	0.23
N	1579	359	359	361	359	642

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms.

Column 1 is an indicator that is one if the firm has a good that matches at least 3 horizontal criteria, and is missing if the firm never replied to the mystery shopper or was otherwise unreachable. Column 2 is an indicator for whether the good's quality score is above the median product-group quality score. Column 3 is the raw quality score. Column 4 is an indicator for whether the good is made in Turkey, primarily inferred based on the label. See the text for full details of how this outcome is constructed. Column 5 is the [Anderson \(2008\)](#) index combining the vertical outcomes to account for multiple hypothesis testing across outcomes. Column 6 is the price in USD, which is only measured conditional on the firm finding a good matching at least three horizontal criteria.

Table D2: Supplier Relationships (Followup Survey) (No Covariates)

	Regular Suppliers in Türkiye			Previous Suppliers		
	Any Sup in Türkiye (1)	Num Sup in Türkiye (2)	Index (3)	Num Sup Total (4)	Num Sup in Senegal (5)	Ended with Sup (6)
<i>Panel A: Pooled</i>						
Treatment	0.035 (0.023)	0.079** (0.038)	0.121* (0.062)	-0.032 (0.193)	-0.092 (0.191)	0.070*** (0.022)
<i>Panel B: Individual Treatments</i>						
Search Only	0.018 (0.029) [0.761]	0.069 (0.055) [0.336]	0.085 (0.080) [0.477]	0.092 (0.269) [0.867]	0.088 (0.266) [0.918]	0.062** (0.029) [0.052]
Search + Adverse Selection	0.056* (0.030) [0.150]	0.077* (0.048) [0.245]	0.146* (0.084) [0.214]	-0.154 (0.253) [0.867]	-0.236 (0.249) [0.722]	0.067** (0.029) [0.050]
Search + Moral Hazard	-0.001 (0.029) [0.971]	-0.009 (0.043) [0.829]	0.009 (0.078) [0.910]	-0.197 (0.248) [0.832]	-0.168 (0.244) [0.825]	0.091*** (0.029) [0.006]
Search + AS + MH	0.069** (0.031) [0.085]	0.183*** (0.062) [0.007]	0.249*** (0.086) [0.016]	0.139 (0.258) [0.867]	-0.049 (0.251) [0.918]	0.057** (0.029) [0.052]
Control Mean	0.167	0.222	0.000	3.700	3.213	0.135
% Increase (Pooled)	21.0%	35.6%	N/A	-0.9%	-2.9%	51.9%
All Coefs Equal p -val	0.072	0.012	0.050	0.487	0.642	0.707
Adjusted R^2	0.00	0.01	0.01	-0.00	-0.00	0.00
N	1680	1680	1680	1681	1681	1671

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms.

Column 1 is 1 if the firm says that they have a regular supplier in Türkiye. Column 2 is the number of regular suppliers in Türkiye. Column 3 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Column 4 is the total number of regular suppliers. Column 5 is the number of regular suppliers in Senegal. Column 6 is 1 if the firm has ended a relationship with a regular supplier in the past 3 months. A regular supplier is defined as a supplier from whom the firm has made two or more orders with the intention to continue the relationship.

Table D3: Order Value (Mobile Money Data) (No Covariates)

	Post Mystery Shopping			Entire Period		
	Total Order Value (USD)	Num Orders	Avg Order Value (USD)	Total Order Value (USD)	Num Orders	Avg Order Value (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Pooled</i>						
Trust Treatment	7.24*	0.05	2.28***	7.16	0.02	2.11**
	(4.48)	(0.07)	(0.73)	(4.93)	(0.09)	(0.99)
<i>Panel B: Individual Treatments</i>						
Search + Adverse Selection	3.57	-0.02	3.18**	4.55	-0.00	3.01*
	(2.82)	(0.07)	(1.59)	(3.64)	(0.10)	(1.72)
	[0.281]	[0.900]	[0.053]	[0.330]	[0.997]	[0.143]
Search + Moral Hazard	6.61**	0.12	2.87***	6.83*	0.06	3.46**
	(3.46)	(0.09)	(1.20)	(4.30)	(0.11)	(1.66)
	[0.066]	[0.459]	[0.017]	[0.152]	[0.915]	[0.075]
Search + AS + MH	11.68	0.05	0.75	10.20	0.01	-0.21
	(12.51)	(0.12)	(0.58)	(13.11)	(0.14)	(0.87)
	[0.450]	[0.900]	[0.215]	[0.586]	[0.997]	[0.820]
Control Mean	3.56	0.14	0.89	8.93	0.33	3.65
% Increase (Pooled)	203.5%	35.2%	257.7%	80.2%	6.6%	57.9%
All Coefs Zero p -val	0.162	0.576	0.025	0.297	0.952	0.064
Adjusted R^2	-0.00	-0.00	0.00	-0.00	-0.00	0.00
N	1500	1500	1500	1500	1500	1500

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all trust treated groups, where Search Only is the omitted category. Panel B shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments.

Column 1 is the total value of orders from study suppliers after the mystery shopping period finished. Column 2 is the number of orders from study suppliers. Column 3 is the average order size from study suppliers after the mystery shopping period finished. Column 4 is the total value of orders from study suppliers over the entire period. Column 5 is the number of orders from study suppliers over the entire period. Column 6 is the average order size from study suppliers over the entire period. Mystery shopping took place during the first 3 months of the study, from 16 November 2023 to 22 February 2024. The entire period covered by the data comprises 18 months, from 16 November 2023 to 31 May 2025. All values are in USD.

Table D4: Profit and Sales (No Covariates)

	Raw			Winsorized (1%)		
	Profit (USD)	Sales (USD)	Index	Profit (USD)	Sales (USD)	Index
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Pooled</i>						
Treatment	57.3*	204.0*	0.311**	24.1	83.0	0.172**
	(28.8)	(108.1)	(0.101)	(20.9)	(83.6)	(0.075)
<i>Panel B: Individual Treatments</i>						
Search Only	.566	254	.194*	-5.26	107	.208*
	(29.1)	(187)	(.114)	(26.2)	(125)	(.108)
	[0.984]	[0.377]	[0.203]	[0.847]	[0.582]	[0.132]
Search + Adverse Selection	30.6	56	.194	18.4	38.1	.124
	(36.4)	(119)	(.122)	(31.5)	(111)	(.099)
	[0.644]	[0.660]	[0.204]	[0.775]	[0.725]	[0.352]
Search + Moral Hazard	-27.4	-103	.037	-27.4	-121	6.7e-03
	(24.8)	(103)	(.082)	(24.8)	(93.6)	(.078)
	[0.565]	[0.516]	[0.702]	[0.543]	[0.394]	[0.935]
Search + AS + MH	244***	640**	.873***	120***	325**	.367***
	(93.2)	(282)	(.31)	(45.2)	(152)	(.135)
	[0.011]	[0.050]	[0.004]	[0.019]	[0.089]	[0.027]
Control Mean	188.3	609.5	0.000	188.3	609.5	0.000
% Increase (Pooled)	30.4%	33.5%	N/A	12.8%	13.6%	N/A
All Coefs Equal p -val	0.032	0.027	0.036	0.022	0.018	0.024
Adjusted R^2	0.01	0.01	0.01	0.01	0.01	0.01
N	1351	1378	1298	1351	1378	1298

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms.

Column 1 is total profit from the past 30 days in USD. Column 2 is total sales from the past 30 days in USD. Column 3 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Column 4 is total profit from the past 30 days in USD, winsorizing the top 1%. Column 5 is total sales from the past 30 days in USD, winsorizing the top 1%. Column 6 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Profit is measured using the survey question from [De Mel, McKenzie, and Woodruff \(2009\)](#). Sales is measured using a similar survey question.

Appendix E - Further Details on Solving and Estimating the Model

In this Appendix, we provide more detail on our algorithm to numerically solve and estimate the model.

Solving the Model

As is often the case in dynamic optimisation, the original infinite horizon program is very difficult to work with directly. It is much more tractable to find a way to work with a recursive formulation. The challenge is that, unlike standard dynamic problems encountered in macro, we have two constraints (the DEC and DICC) that are forward looking, and, in particular, forward-looking to an infinite horizon.

The literatures on dynamic moral hazard and limited commitment typically deal with this in one of two ways. One way is to define continuation values as state variables, which completely summarise the future and thus allow the constraints to be written recursively. This “promised utility” approach was originally developed somewhat independently in different theoretical contexts by [Spear and Srivastava \(1987\)](#), [Abreu, Pearce, and Stacchetti \(1990\)](#), and [Thomas and Worrall \(1988\)](#), and in fact we take this approach in Appendix B when we derive qualitative properties of the optimal contract.

The other approach, pioneered by [Marcet and Marimon \(2019\)](#), follows the idea that the original Lagrangean can be rewritten recursively as a pseudo planner’s problem, where the Pareto weights are state variables that evolve endogenously to completely summarise historical binding constraints. Intuitively, if the DICC is binding in period 0, which implies that the agent must be delivered a certain amount of utils at some point in the future, the Pareto weight on the agent increases over time to ensure that the planner delivers precisely the required amount of utils. This allows the problem to be written recursively because the principal can trade off the benefit of making a constraint “more binding” today against the cost of increasing next period’s Pareto weight on the agent. The recursive formulation delivers a Saddle Point Functional Equation, which is analogous to the familiar Bellman Equation but for saddle point problems, which satisfies a number of familiar properties that permit the use of dynamic programming techniques.

The key advantage of the [Marcet and Marimon \(2019\)](#) approach over the promised utility approach is that the feasible set of Pareto weights is known. This is important, because one needs to know the feasible set in order to numerically solve the model. In contrast, in the promised utility approach, we would need to know the feasible set of continuation values, which are endogenous objects that likely depend in complicated ways upon the model parameters. This is not a problem for proving properties of the model, which is why we use this approach in Appendix B, but is a problem for numerically solving it. It is also not an insurmountable obstacle: [Abreu, Pearce, and Stacchetti \(1990\)](#) provide an algorithm that can be used within an inner loop to compute the feasible set, and many papers in the literature fruitfully pursue this approach. Nonetheless, the Pareto weight approach completely sidesteps this issue, which is why we use it here.

We first rewrite our problem as a recursive Lagrangean and thus derive the Saddle Point

Functional Equation. To avoid cumbersome algebra and focus on the core intuition of the method, the derivations that follow assume that the firm can only purchase from one supplier at a time. We write the full version, allowing for purchases from both a new and existing supplier, at the end. Define $y(q_t, \tau_t, \mu_t) \equiv (1 - \mu_t(1 - \lambda))r(q_t) + \mu_t(1 - \lambda)r(0) - \tau_t$. Then, the original dynamic program is as follows

$$\begin{aligned} \max_{\{q_t\}, \{\tau_t\}} & y(q_0, \tau_0, \mu_0) + \delta \mu_0(1 - \lambda) \bar{U} \\ & + \delta(1 - \mu_0(1 - \lambda)) [y(q_1, \tau_1, \mu_1) + \delta \mu_1(1 - \lambda) \bar{U}] \\ & + \delta(1 - \mu_1(1 - \lambda)) [y(q_2, \tau_2, \mu_2) + \delta \mu_2(1 - \lambda) \bar{U}] \\ & + \delta(1 - \mu_2(1 - \lambda)) [\dots] \end{aligned}$$

subject to

$$\begin{aligned} \sum_{n=1}^{\infty} \delta^n (\tau_{t+n} - cq_{t+n}) & \geq \xi cq_t \quad \forall t \\ U_t & \geq \bar{U} \quad \forall t \\ \tau_t & \geq cq_t \quad \forall t, \end{aligned}$$

and with μ_t evolving according to Bayes' Rule. Rewriting the objective function as an infinite sum and including the constraints with Lagrange multipliers, the program can be expressed as follows

$$\begin{aligned} L = & \min_{\{\rho_t\}, \{\eta_t\}, \{\gamma_t\}} \max_{\{q_t\}, \{\tau_t\}} \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) ((1 - \mu_t(1 - \lambda)r(q_t) + \mu_t(1 - \lambda)r(0) - \tau_t + \delta \mu_t(1 - \lambda) \bar{U}) \\ & + \sum_{t=0}^{\infty} \delta^t \rho_t \left[\sum_{s=t+1}^{\infty} \delta^{s-t} (\tau_s - cq_s) - \xi cq_t \right] \\ & + \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \eta_t \left[\sum_{s=t}^{\infty} \delta^{s-t} (1 - \mu_s(1 - \lambda^{s-t})) ((1 - \mu_s(1 - \lambda)r(q_s) + \mu_s(1 - \lambda)r(0) - \tau_s + \delta \mu_s(1 - \lambda) \bar{U}) \right. \\ & \left. + \sum_{t=0}^{\infty} \delta^t \gamma_t [\tau_t - cq_t] \right] \end{aligned}$$

Then, with some algebra, we can collect the Lagrange terms directly inside the first infinite sum to express the Lagrangian as a function of "Pareto weights", Lagrange multipliers, state variables, and time-invariant functions.

$$\begin{aligned} L = & \min_{\{\rho_t\}, \{\eta_t\}, \{\gamma_t\}} \max_{\{q_t\}, \{\tau_t\}} \sum_{t=0}^{\infty} \delta^t [(\zeta_t^b + \eta_t) h_0^b(q_t, \tau_t, \mu_t) + \eta_t h_1^b(q_t, \tau_t, \mu_t) \\ & + (\zeta_t^s + \gamma_t) h_0^s(q_t, \tau_t) + \rho_t h_1^s(q_t, \tau_t)] \end{aligned}$$

where $\zeta_{t+1}^b \equiv (\zeta_t^b + \eta_t)(1 - \mu_t(1 - \lambda))$, $\zeta_{t+1}^s \equiv \zeta_t^s + \rho_t$, and

$$\begin{aligned} h_0^b(q_t, \tau_t, \mu_t) &\equiv [(1 - \mu_t(1 - \lambda))r(q_t) + \mu_t(1 - \lambda)r(0) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}] \\ h_1^b(q_t, \tau_t, \mu_t) &\equiv -\bar{U} \\ h_0^s(q_t, \tau_t) &\equiv \tau_t - cq_t \\ h_1^s(q_t, \tau_t) &\equiv -\xi cq_t \end{aligned}$$

ζ_t^b and ζ_t^s behave like Pareto weights and are equal to the sum of all prior Lagrange multipliers from the forward-looking constraints for the principal (the buyer) and the agent (the seller), respectively. This is the sense in which they fully summarise the shadow cost of constraints from earlier periods of the problem. For example, if DICC is “very binding” in early periods, meaning ρ_t is large for early t , then this is reflected in a large value of ζ_t^s for later t , which causes the “planner” to endogenously choose a high value of τ_t and thus give the agent utility. Note that there is a small difference relative to how the weights evolve in [Marcet and Marimon \(2019\)](#) as the buyer’s weight here includes the term $(1 - \mu_t(1 - \lambda))$ in the evolution, which appears because the Adverse Selection problem means that we only reach the next period in the relationship with probability $(1 - \mu_t(1 - \lambda))$.

Distinguishing between the foreign good, q_t , and the existing supplier q_{et} , we can then write this recursively as a Saddle Point Functional Equation (SPFE) as follows,

$$\begin{aligned} W(\zeta_t^b, \zeta_t^s, \mu_t) &= \min_{\eta_t, \rho_t, \gamma_t} \max_{q_t, q_{et}, \tau_t} (\zeta_t^b + \eta_t)h_0^b(q_t, q_{et}, \tau_t, \mu_t) + \eta_t h_1^b(q_t, \tau_t, \mu_t) \\ &\quad + (\zeta_t^s + \gamma_t)h_0^s(q_t, \tau_t) + \rho_t h_1^s(q_t, \tau_t) + \delta W(\zeta_{t+1}^b, \zeta_{t+1}^s, \mu_{t+1}), \end{aligned}$$

with

$$\begin{aligned} h_0^b(q_t, \tau_t, \mu_t) &\equiv [(1 - \mu_t(1 - \lambda))r(q_t, q_{et}) + \mu_t(1 - \lambda)r(0, q_{et}) - \tau_t - p_{et}q_{et} + \delta\mu_t(1 - \lambda)\bar{U}] \\ h_1^b(q_t, \tau_t, \mu_t) &\equiv -\bar{U} \\ h_0^s(q_t, \tau_t) &\equiv \tau_t - cq_t \\ h_1^s(q_t, \tau_t) &\equiv -\xi cq_t \end{aligned}$$

subject to

$$\begin{aligned} \zeta_{t+1}^b &= (\zeta_t^b + \eta_t)(1 - \mu_t(1 - \lambda)) \\ \zeta_{t+1}^s &= \zeta_t^s + \rho_t, \end{aligned}$$

and the evolution of μ_t . For the evolution of μ_t , to prevent a discontinuity in learning when going from $q_t = 0$ to any positive value, while also ensuring the principal’s choice set remains convex, we impose a technical assumption that for q_t below some small $\bar{q}$, posteriors after observing a high quality realisation are given by $\mu_{t+1} = \mu_{t+1}^{\text{Bayes}} + (\mu_t - \mu_{t+1}^{\text{Bayes}})$.

$(1 - q_t/q)^p$, where μ_{t+1}^{Bayes} is the posterior implied by Bayes' Rule and $p \in (1, 2)$. We set $p = 1.4$ as numbers around this value tended to perform well in numerical experiments. Making learning a continuous function of q_t ensures the problem satisfies appropriate regularity conditions as it makes the buyer's choice set convex. Intuitively, it captures the idea of a minimum order size (which is the case in our empirical setting), and we set q to 6 as packets of 6 are a very common minimum order size in our empirical setting. This learning function makes little difference in practice as it only applies in rare cases where the optimal q is below $\underline{q}$, but it is important to ensure the problem satisfies the appropriate regularity conditions.

The Saddle Point Functional Equation is analogous to the Bellman Equation for saddle point problems, and [Marcet and Marimon \(2019\)](#) prove that—under some regularity conditions—it has the usual desirable properties associated with dynamic programming problems. In particular, this means that we can obtain solutions to the original problem by using dynamic programming techniques to solve for the value function and policy functions associated with the SPFE.

We solve the model by iterating on the value function, W . Since W is homogeneous of degree one in the Pareto weights, we can write $W(\zeta_t^b, \zeta_t^s, \mu_t) = \zeta_t^b W(1, \frac{\zeta_t^s}{\zeta_t^b}, \mu_t)$. This means that we can eliminate one state variable. We define a discrete grid over the two state variables and linearly interpolate over the grid.⁴¹ Within each iteration, we solve the min max as follows. First, the FOC with respect to τ_t gives implies $\zeta_t^b + \eta_t = \zeta_t^s + \gamma_t$. Since we know from [Appendix B](#) that we will never have $\eta_t > 0$ and $\gamma_t > 0$ simultaneously, it must then be that $\eta_t = \max(\zeta_t^s - \zeta_t^b, 0)$ and $\gamma_t = \max(\zeta_t^b - \zeta_t^s, 0)$. We can then obtain τ_t as either cq_t if $\gamma_t > 0$ or $(1 - \mu_t(1 - \lambda))r(q_t, q_e) + \mu_t(1 - \lambda)r(0, q_e) + \delta\mu_t(1 - \lambda)\bar{U} - p_e q_e + \delta(1 - \mu_t(1 - \lambda))\frac{\partial W}{\partial \zeta_{t+1}^b} - \bar{U}$ if $\eta_t > 0$. (We discuss the edge-case where $\eta_t = \gamma_t = 0$ below). To obtain q_t, q_{et} , for a given ρ_t we can solve for q_t and q_{et} analytically using the FOCs if $q_t > \underline{q}$; otherwise we have to solve for them numerically as a term $\frac{\partial W}{\partial \mu_{t+1}}$ appears in this region as the choice of q affects μ_{t+1} through the smoothed learning function. The FOC with respect to ρ_t includes $\frac{\partial W}{\partial \zeta_{t+1}^s}$, which must be evaluated numerically and thus we cannot solve for ρ_t analytically. We thus solve for it numerically in an outer loop, with the solver for q_t, q_{et} in the inner loop.

Once we have obtained the value function, we iterate forward from the initial conditions at $t = 0$ to get the solution path. Since the agent's utility is linear in τ_t , the value function has a kink in the neighbourhood of $\zeta_t^b = \zeta_t^s$, which [Marimon and Werner \(2021\)](#) show results in non-unique policy functions and thus inconsistent utility promises. This kink is also the point at which the surplus shifts from the buyer to the seller, and the period where it occurs is exactly $T^* - 1$ (see [Appendix B](#)), where τ_t is interior. We resolve this by doing the following. First, since $\pi_t = 0$ for all $t < T^* - 1$, we know that $V_0 = \delta^t V_t$ for any $t < T^* - 1$. Starting with $t = 1$, we can calculate $V_0 - \delta^t V_t$ for each t , and find $T^* - 1$ by checking when this object switches sign. Second, once we know $T^* - 1$, we iterate forward normally until $T^* - 2$, at which point we set ρ_{T^*-2} to be such that $\zeta_{T^*-1}^b = \zeta_{T^*-1}^s$ and we calculate τ_{T^*-1} as the value that ensures that the Envelope Selection Condition in [Marimon and Werner](#)

⁴¹We experimented with both linear interpolation and cubic splines, and generally found that linear interpolation performed better.

(2021) holds.

Estimation

The above sub-section describes how we solve the model for a given guess of the parameters. In order to estimate the parameters, we need to find the parameters that best match the empirical moments, and thus solve the model for many combinations of parameters. We do this in two steps. First, we solve the model for a grid of $(\mu_0, \xi, c, \delta, \lambda)$, where c is the cost of foreign supplier (varying c and ψ are isomorphic and one can simply map between them by setting $\psi = 1/c$), and linearly interpolate over the grid. This gives *functions* for all of the relevant theoretical objects, such as $y_t(\mu_0, \xi, c, \delta, \lambda, \bar{U})$, which means that we do not need to further solve the model as we can simply evaluate these functions at a given $(\mu_0, \xi, c, \delta, \lambda)$. The model converges for 99.6% of the parameter combinations, and we impute the values for the combinations that do not converge using a simple nearest-neighbour means algorithm.

Second, we use the above functions for estimation. We define 11 theoretical moments. The first four are the treatment effects on winsorized profit, i.e., the coefficients in Column 4 of Table 4. The next there are the post-mystery shopping total value ordered treatment effects in Column 1 of Table 3. For these three moments, because the mobile money data does not capture all transactions (and is likely top-censored), we do not match the coefficients directly but rather the relative treatment effects—i.e., the coefficients divided by the control mean reported in the table. The next four moments are the four treatment effects on the survey outcome of whether a firm has a regular supplier in Turkey, i.e., the coefficients in Column 1 in Table 2. This gives 11 treatment effect moments which provide exogenous variation to identify the parameters governing the frictions. We calculate the variance-covariance matrix of these 11 empirical moments by drawing 1,000 bootstrap sub-samples of the experimental data and use the inverse of this as the weighting matrix.

As the moments only identify differences, we ensure that the model matches the control means through the choice of the z distribution and the ψ_μ parameter. For z , since our assumption on the baseline equilibrium is that each firm can be represented as having one supplier, the model implies that profit for a firm i with one input is given by $\pi_i = z_i^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p_i^{1-\sigma}$, where p_i is their input price. The input price is a function of z : firms with $z < \bar{z}$ have p_l ; other firms have an endogenous price determined by their reservation utility from the search game (which is a function of z). We thus have a many-to-one function that maps z onto p , which implies that there is a one-to-one mapping from z to π given by $\pi(z) = z^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p(z)^{1-\sigma}$. We thus draw at random from the empirical π_i distribution of the control group and invert this mapping to obtain an empirical distribution of z that rationalizes the baseline control distribution (and thus control mean). As firms often report profit in round numbers, there are several mass points in this distribution, so to improve numerical stability in the optimization routine, whenever we draw z we jitter it by adding $u \sim U[-1, 1]$. For ψ_μ , note that the share of firms importing in the control group is determined by the share of firms with $z > \bar{z}$. Thus, for a given guess of the parameters, we can solve for ψ_μ that sets this share to match the control mean. Intuitively, for any guess of the parameters, we can always shift the location of the distribution of match-specific

productivities of the foreign suppliers until exactly a share d falls above the cutoff.

The two dimensions of heterogeneity are the firm-specific productivities, z , and the match-specific productivities, ψ . For distribution of match-specific productivity, we directly estimate the parameters of the LogNormal, (ψ_μ, ψ_σ) . We winsorize the upper tail such that $c = 1/\psi$ never falls below 2. Note that the absolute level is not meaningful—the model is homogenous of degree $1 - \sigma$ in c and p_l , so all that matters is the ratio c/p_l , and since we calibrate $p_l = 12.29$, a relative price of less than one sixth is already very low. We do this both because heavy tails make estimation sensitive to implementation details of the SMM and because the value function becomes very large as c becomes small (causing numerical issues with convergence). We actually compute the moments by Monte Carlo integration using $\Omega = 10,000$ draws of ψ .⁴²

As this is a non-stationary dynamic model, the questions of what is baseline and how changes in the environment affect existing relationships are non-trivial. We assume that the baseline equilibrium is the long-run steady state of the model. That means that all firms with $z \geq \bar{z}$, where $\bar{z}$ is the endogenous cutoff above which a firm will engage in search for a foreign supplier, have already searched and are in the “terminal” phase of the relationship. Recall from Section 3 that the optimal contract features a unique T^* such that the firm earns exactly their outside option (which is the maximum of re-searching or buying locally forever) for all $t > T^*$, which we refer to as the terminal phase. Thus, all firms with $z \leq \bar{z}$ earn $U_l = \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} z^\sigma p_l^{1-\sigma} / (1 - \delta)$, while all firms with $z > \bar{z}$ earn $\bar{U}(z) > U_l$, where $\bar{U}(z)$ is the expected value of participating in the search game, defined implicitly by $\int_\psi \max\{U(\psi, z) - \bar{U}(z), 0\} dF(\psi) = s$. For simplicity, we simply assume that these firms earning $\bar{U}(z)$ also have access to a technology that allows them to frictionlessly purchase at a constant price $p_e(z)$ defined such that $\bar{U}(z) = \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} z^\sigma p_e(z)^{1-\sigma} / (1 - \delta)$, i.e., the constant price that gives the same utility as they are currently earning. This simplifies the model as they can then combine orders from this technology with orders from the new supplier without us having to take a stand on how or when they might renegotiate their existing dynamic contract. Thus, the baseline is defined by firms with $z < \bar{z}$ having existing price $p_e = p_l$, while firms with $z > \bar{z}$ have existing price $p_e(z) < p_l$. Using the fact that the model is homogeneous of degree $1 - \sigma$ in p_e and c , we can then evaluate the interpolated objects from the grid of solved model runs.

For the theoretical moments, we calculate the theoretical profit in each treatment group, with the treatments implemented as described in the main text. For profit, we calculate the mean profit in the first three periods, $\frac{1}{3} \sum_{t=0}^2 y_t$, as the survey asked about profits over the past 30 days and typically took place within 2-3 months after recruitment. For the binary outcome relationships, we set this to 1 if they already have an existing foreign supplier (i.e., firms with $z > \bar{z}$) and 0 if they have $z < \bar{z}$ and they choose $q_{f0}(c_{\min}) = 0$, where $c_{\min}$ is the minimum of the three free cost draws from the search treatment (or, equivalently, the maximum of the three productivity draws, $\psi = 1/c$). For firms with $q_{f0} > 0$ and $z < \bar{z}$, we set the moment equal to the probability that the relationship reaches the third period (as a function of μ_0). Finally, for mobile money, we calculate $\sum_{t=3}^{18} \tau_t$, reflecting the total value

⁴²We experimented with different numbers of draws, and found that the value of the objective function becomes fairly stable across seeds starting at around Ω of 5,000.

ordered after mystery shopping finished (and, as discussed above, we divide the treatment means by the control—i.e., search only—mean). The final moments are shown in Figure [A9](#).

Appendix F – Survey and Protocol Details

F.1 Firm Eligibility Criteria

To ensure the relevance of the study, we administered a short screening survey with the following eligibility criteria. Firm owners were included if they: (i) managed their own orders with suppliers; (ii) averaged at least five ready-to-wear sales per week over the past 30 days; (iii) had access to WhatsApp; and (iv) had used WhatsApp to sell to clients in the past 30 days, including through posts or status updates. Respondents who did not meet these criteria were thanked for their interest but deemed ineligible to participate.

Among those who agreed to participate, 85% of online sellers and 89% of physical-store sellers met the eligibility criteria. This indicates that our final sample represents a substantial share of these two key populations of interest, while being appropriately filtered to focus on active garment retailers engaged in a minimum level of digital commerce.

F.2 Moral Hazard Treatment Details

The Moral Hazard treatment, as the model highlights, requires either shifting the suppliers' incentives or shifting firms' perceptions of suppliers' incentives. To do this, we read the information to treated firms at the end of the baseline survey to emphasize that study suppliers have strong incentives:

I have one last piece of information to give you. As you know, you have been added to WhatsApp groups of Senegalese suppliers in Turkey.

We work with many suppliers in our study. We want to assure you that they are motivated.

We would like to collect feedback on these suppliers so that we can recommend the best ones in the future. To do this, we will ask the merchants in the study [such as yourself] to rate your experience with the suppliers we have presented to you on a scale of 1 to 5 on product arrival and quality. These reviews help identify the best suppliers, which is beneficial to them and allows us to continue recommending them to others. They are therefore motivated.

If a supplier gets bad ratings, we will investigate and remove them from the study if they did not do a good job. They will therefore lose access to around 150 merchants if they do not do a good job.

I will give you a phone number that you can use to give your rating or report a problem.

Lastly, I want to emphasise that the suppliers are aware that they are being rated and that, if they receive bad ratings, they will be removed from the study. We can thus assure you that they are motivated.

F.3 Mystery Shopping Protocol

To evaluate firms' access to high-quality and differentiated foreign goods, we implemented a structured mystery shopping protocol. Surveyors contacted firms over WhatsApp posing as genuine customers interested in purchasing garments for an upcoming event. The goal was to simulate realistic sourcing and negotiation behavior while standardizing the interaction across sellers.

Initial contact. Two weeks after the baseline survey, trained surveyors messaged each firm over WhatsApp, referencing a product image and asking whether the firm had something similar. Firms had previously been told that they might be contacted by potential customers who were seeking high-quality goods.

Horizontal match criteria. The referenced product images were designed to include five horizontal characteristics that are orthogonal to quality (e.g., color, presence of graphic, sleeve style, neckline). If the firm proposed a good that met at least three of the five criteria, this was coded as a horizontal match. This is the primary outcome for the horizontal dimension.

Negotiation protocol. If a horizontal match was found, then the mystery shopper proceeded to negotiating the price and delivery details. All price interactions followed a standardized script:

- Surveyors first asked the seller for a quote (including delivery).
- They then proposed a counteroffer equal to 75% of the quoted price (rounded down to the nearest 500 FCFA), capped at 20,000 FCFA.
- The seller's final offer was recorded, along with whether they accepted payment on delivery or required an advance (and how much).

Purchase and fallback procedure. If a match was found, the seller agreed to a price at or below the 20,000 FCFA cap, and the seller accepted either 50% or full payment on delivery, the mystery shopper proceeded to buy the good in a randomly selected 80% of cases. In the 20% of cases where the random assignment indicated to not buy the good, they explained that their plans had changed, and offered a nominal payment of about 2 USD (1,000 FCFA) as a gesture of appreciation. This fallback protocol was piloted to ensure natural exit and avoid upsetting sellers.

Rapid-order test. In a second stage, surveyors sent an image from a randomized set of "priority products" and asked if the seller could find the item within 24 hours. This assessed the firm's ability to respond quickly to buyer demand or to test new suppliers.

Secondary Outcomes. Surveyors recorded whether the merchant added them to WhatsApp status, if delivery date was confirmed, and if the merchant mentioned quality or reputation. They also tracked total time elapsed between contact and order.

Exit. Depending on the scenario (purchase/no purchase), shoppers used an exit script, including a fallback offer of 1,000 FCFA to compensate for the merchant’s time in cases where no purchase occurred because of the 20% random draw.

Vertical quality scoring. Upon receipt, each good was evaluated by two professional tailors (and a shoemaker for footwear) using a structured 50-point scorecard developed for the study.

Each criterion was scored on a 0–2 or 0–3 scale (with a few exceptions), and we computed both the raw score and a binary high-quality indicator (equal to 1 if the item’s score was above the median for its product type). As each product was independently assessed by both experts, we average the two assessments.

Origin recording. In addition to the quality scores, surveyors recorded whether the item was labeled as “Made in Türkiye,” a strong signal of quality in this setting. When labels were absent, tailors independently evaluated origin based on design and construction cues. The resulting variable is a binary indicator equal to 1 if the label—or both experts—indicated Turkish origin.

F.4 Vertical Quality Scoring — Scorecard

Upon receipt, each good was evaluated by two professional tailors (and a shoemaker for footwear) using a structured 50-point scorecard developed for the study. We designed this scorecard with our hired experts. We also drew inspiration from [Vitali \(2024\)](#), who uses a similar approach to assess garment quality in Kampala.

Table F1: Garment Quality Scorecard (50-point scale; higher = better)

Category	Question	Polo	Dress
Fabric	Is the fabric soft?	3	3
	Are there pills?	2	2
	Is it transparent under light?	4	4
	Are the threads consistent?	3	3
Sewing	Are the seams straight and neat?	3	3
	Do the seams come apart upon gentle pulling?	3	3
	Are there stray threads or loose stitches?	2	2
	Do the seams lie flat without puckering?	2	2
	Is the hem seam reinforced?	2	2
Shoulders	Do the shoulders fall correctly?	3	3
Collar and Buttons	Is the collar well-attached and flat?	3	–
	Are the buttons securely attached?	3	–
Darts	Are there darts on the bust and waist?	–	3
	Are the darts neatly sewn?	–	3
Colourfastness	Does it lose colour upon soaking?	7	7
Pattern	Does the pattern line up with the seams?	2	2
Hem	Is the hem well done?	4	4
Sleeves	Are the sleeves the same length and width?	4	4
Absence of Other Defects	Are there any other defects not covered above?	-10	-10
Total Score		50	50